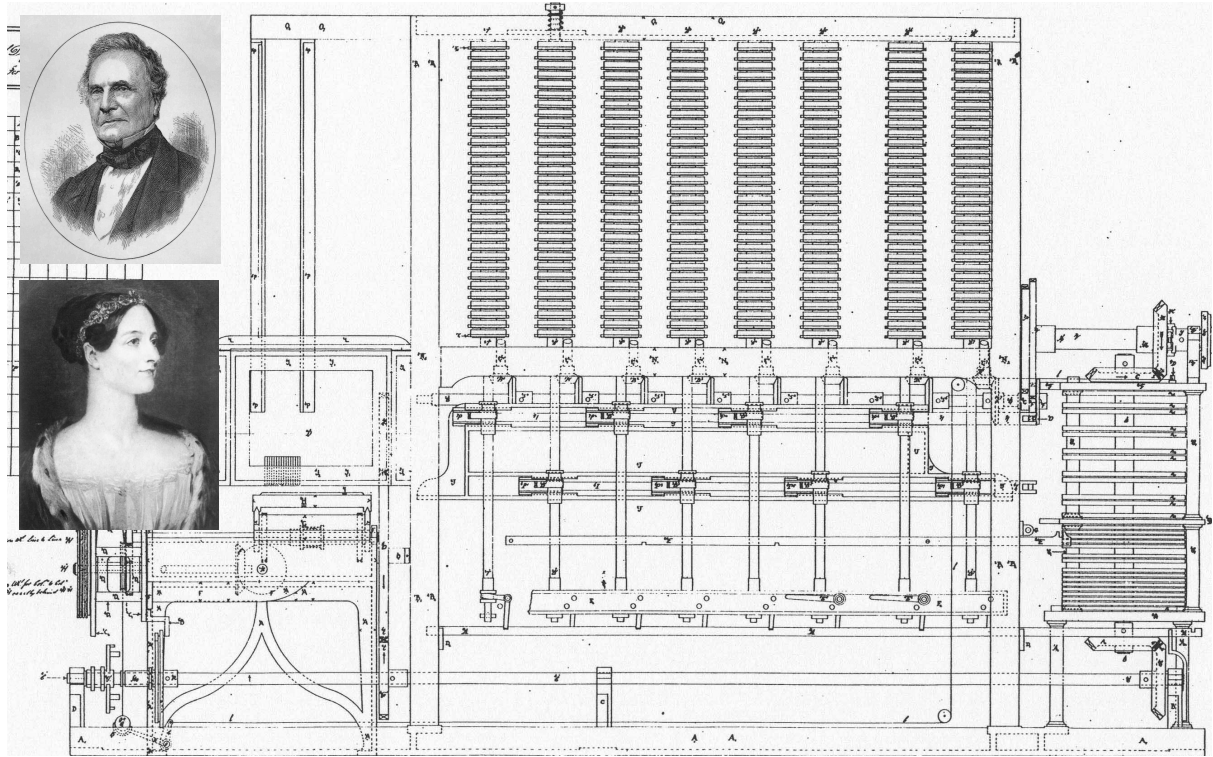


Charles Babbage, Ada Lovelace, and the Dawn of Computing

Version 3.1



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Mark M. Meysenburg

Mark M. Meysenburg, 2016

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Mark M. Meysenburg
Doane University
Department of Information Science and Technology
1014 Boswell Avenue
Crete, NE 68333

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Chapter 1

Introduction

Welcome to *Charles Babbage, Ada Lovelace, and the Dawn of Computing*, a Reacting to the Past role-playing game centered on Charles Babbage and Ada Lovelace, pioneers of the modern computer who lived in Great Britain in the nineteenth century.

1.1 Brief overview of the game

Charles Babbage, Ada Lovelace, and the Dawn of Computing takes place in early nineteenth-century Britain, focusing on the calculating engines designed by Charles Babbage. The central intellectual collisions in the game concern the conflict between imagination and reason, the nature of science and scientists (are they talented, wealthy amateurs, or is science a profession?) and whether and to what degree science and engineering projects should be subsidized by the government. The main question in the game is whether or not Charles Babbage should be awarded funds from the British government for the development of his Difference Engine (an automated calculator capable of automatically creating, typesetting, and printing mathematical tables) and / or Analytical Engine (a true proto-computer), during the early to mid-1800s.

The Engines, at a minimum, would have “mechanized number,” allowing rapid, automatic, error-free production of mathematical tables to be used in navigation, astronomy, and other applications. Ada Lovelace, regarded as the world’s first computer programmer, could envision applications of the Analytical Engine beyond mere mathematics, however. In the notes to her English translation of Luigi Federico Menabrea’s article *Sketch of the Analytical Engine Invented By Charles Babbage, Esq.* [Men43], Lovelace realized that the Engine could operate on symbols in an abstract manner, that it “might act upon other things besides number. . . the Engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.” She also foresaw applications in business, drawing graphics, and more. In addition, the precision mechanical engineering work done on the engines had profound impacts on the state of engineering in the UK and around the world. The mind boggles at the prospects had an Analytical Engine ever actually been built. What would the consequences of a steam-powered, mechanical computer have been in Victorian

Table 1.1: Overview of game sessions

Session (year) / location	Presiding character	Theme
GS 1 (1828) / 1 Dorset Street	Babbage	Babbage's party
Lab 1	n/a	Finite differences
GS 2 (1828) / Mechanic's Institute	Lardner	Public lectures 1
Lab 2	n/a	Difference Engine
GS 3 (1828) / Royal Society	Gilbert	Difference Engine
GS 4 (1830) / Royal Society	Gilbert	Celebrating the RS RS presidency
Lab 3	n/a	Analytical Engine
GS 5 (1846) / Mechanic's Institute	Brunel	Public lectures 2
GS 6 (1846) / Royal Society	RS president	Analytical Engine

England? One can easily get lost in the “steam-punk” possibilities of a 100-year head start in computing technology.

Players in the game may belong to one of two factions, or may be indeterminate. The pro-Babbage faction wants the government to provide funding adequate for the construction of one or both of Babbage's Engines. The anti-Babbage faction wants the opposite – no funding for either Engine, either because they do not feel the Engines are that important, or because they do not feel the government should provide funding for such projects. Indeterminates have no set position on this issue, and are open to persuasion by the factions.

The standard version of the game covers a broad span of time, from 1828 through 1846. There are six game sessions and three labs, briefly described in Table 1.1. The labs are optional; your gamemaster (GM) will tell you which, if any, of the labs will be included in your version of the game.

1.2 Map



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version of this map please contact: info@mapsoftheworld.com

Figure 1.1: The United Kingdom.

1.3 Prologue

You depart your lodgings at the Bull and Mouth on St. Martins le Grand in London, half a mile north of the Thames and just west of the newly-opened Guildhall Library. Your destination, Charles Babbage's house at 1 Dorset Street, is three miles to the west. The

year 1828 has been unusually wet, even for London, but on this pleasant Wednesday evening in June, the sky is clear and the sun is setting; even the stench of the Thames seems less than normal. So, you decide to walk instead of hiring a coach for the journey.

You can hardly believe that you have been invited to one of Babbage's Dorset Street parties, as they are *the* premier social events in summertime London. As Mrs. Cornelia A. H. Crosse, wife of natural philosopher Andrew Crosse, puts it, "One of three qualifications were necessary for those who sought to be invited – intellect, beauty, or rank – without one of these you might be rich as Croesus and yet be told you cannot enter here." Although you do not consider yourself in the same league as Babbage – for very, very few are – you think that it must be your intellect that garnered your invitation. Nevertheless, you are invited, and you quicken your steps as you walk past the London Stock Exchange at Capel Court.

Saint Paul's Cathedral, the dominant fixture of the London skyline, is at your back as you make your way through the teeming streets. London is the largest city in the world, recently surpassing Peking. Disraeli has quipped that "London is a nation, not a city." It seems like a good proportion of the population is out in the streets with you, all attending to some business on this fine evening. Many, like you, are members of the upper classes, dressed in finery and bound for a social event, although certainly none of these are attending anything of the magnitude of a Dorset Street party. You watch them as they walk along or depart from their carriages, making a show of their elegance, knowing that they would be envious if they knew where you were headed, and all seems sunny and bright with Britain and the world.

Then you walk past a blacking factory and the world dims slightly. The building is a ramshackle contrivance, seemly ready to tumble down or burst into a smoky inferno at a moment's notice. You can see children crowded into the factory's few windows, desperately clinging to the fading sunlight so they can see their work: pasting labels on cans of bootblack. You know a little about what these children – some as young as five – go through, working ten to twelve hours a day, six days a week, for a wage of only five or six shillings a week. Only the most callous could fail to notice the poverty and overcrowding in London's slums, where the working people live. As if to complete the vision, the breeze shifts and suddenly you can fully smell the Thames; the stench of the city's raw sewage nearly causes your knees to buckle. Passing by the Soho Square Gardens on the south of the street does nothing to lighten your mood, as you can see the decrepit state of the statue of King Charles II; the inscription at the base is now all but illegible. The less said about the Manor House on the square, and its now-closed but still infamous brothel, the better.

Leaving the blacking factory and Soho Square Gardens behind, you refocus your thoughts on the progress London has seen in the last few years: public amenities such as the London Zoo, the Guildhall Library, and horse-drawn omnibuses for transportation; and civil engineering projects such as the Hammersmith, Kingston, and London bridges, the St. Katherine docks, as well as the reconstruction of Buckingham Palace. Surely a culture capable of such accomplishments must be a just and blessed society. The spring returns to your step and your pace quickens as you near Dorset Street.

Nearing the huge house, you begin to imagine what the night will be like. Which luminaries of British society will be in attendance tonight, and what will be the topics of conversation? Perhaps the man of science John Herschel, recent winner of the prestigious Copley Medal from the Royal Society of London, will be there and regale listeners about his eighteen-inch reflecting telescope and his research on double stars.

Or maybe someone emblematic of a new kind of man of science, someone who studies the wonders of creation *professionally*? Michael Faraday, the experimentalist, perhaps? Even though he is not formally considered a gentleman, you know that he has attended Babbage's parties in the past, and you would be delighted to learn more of his experiments with electricity. Another man in this category, the engineer I. K. Brunel, might be in attendance. You imagine yourself asking him about the stalled Thames Tunnel, and his narrow brush with death during a breach and flooding of the partially completed excavation in January.

Or, even further afield, you might meet Mary Somerville, the woman who published the paper "The magnetic properties of the violet rays of the solar spectrum" in the *Proceedings of the Royal Society* two years ago. A woman publishing work about astronomy! Mrs. Somerville is "mathematician of the very first rank with all the gentleness of a woman," as Sir David Brewster describes her; how fascinating it would be to make her acquaintance!

You almost dare not imagine it, but the Prime Minister, Viscount Goderich, could be inside. You have heard that even the Duke of Wellington might be in attendance, mesmerizing the crowd with tales of his exploits during the Napoleonic Wars.

Of course, one certainty is that Babbage will be there, gregarious, engaging, and able to talk to anyone: educated commoner, man of science, or Prime Minister. He might talk about mathematics – his work on the theory of functions or how the Analytical Society modernized mathematics in Britain, bringing the kingdom into alignment with continental Europe's calculus notation. He will certainly demonstrate his Silver Lady, a clockwork-mechanized metallic statue of a young woman with a bird resting on her extended finger. When set in motion, the bird's beak opens and closes, and its wings flap! Babbage restored the workings of the Silver Lady himself, and you have heard that every new attendee to one of his parties is introduced to the mechanical wonder.

The other mechanical device that you are sure to see as a first-time guest at 1 Dorset Street is the working fragment of Babbage's Difference Engine, a beautiful assemblage of brass cogs built under Babbage's instruction by the engineer Joseph Clement. A machine that mechanizes the calculation, tabulation, and printing of numbers! Babbage is immensely proud of the machine, and he clearly considers the Engine his life's work. He is known to talk voluminously about the Engine at his parties to anyone who shows interest. You hope you are able to understand the principles and operations of the machine.

Finally you arrive at the house and are ushered inside. Many of the guests are already gathered in the parlor, and you enter excitedly, ready to participate in such a momentous social occasion.

1.4 How to react

Reacting to the Past is a series of historical role-playing games. Students are given elaborate game books which place them in moments of historical controversy and intellectual ferment. The class becomes a public body of some sort; students, in role, become particular persons from the period, often as members of a faction. Their purpose is to advance a policy agenda and achieve their victory objectives. To do so, they will undertake research and write speeches and position papers; and they will also give formal speeches, participate in informal debates and negotiations, and otherwise work to win the game. After a few preparatory lectures, the game begins and the players are in charge; the instructor serves as adviser or “gamemaster.” Outcomes sometimes differ from the actual history; a post-mortem session at the end of the game sets the record straight.

The following is an outline of what you will encounter in Reacting and what you will be expected to do. While these elements are typical of every Reacting game, it is important to remember that every game has its own special quirks.

1.4.1 Game set-up

Your instructor will spend some time before the beginning of the game helping you to understand the historical background. During the set-up period, you will read several different kinds of material:

- The game book (from which you are reading now), which includes historical information, rules and elements of the game, and essential documents; and
- Your role, which describes the historical person you will play in the game.

You may also be required to read primary and secondary sources outside the game book (perhaps including one or more accompanying books), which provide additional information and arguments for use during the game. Often you will be expected to conduct research to bolster your papers and speeches.

Read all of this contextual material and all of these documents and sources before the game begins. And just as important, go back and reread these materials throughout the game. A second reading while in role will deepen your understanding and alter your perspective: ideas take on a different aspect when seen through the eyes of a partisan actor.

Players who have carefully read the materials and who know the rules of the game will invariably do better than those who rely on general impressions and uncertain recollections.

1.4.2 Game play

Once the game begins, certain players preside over the class sessions. These presiding officers may be elected or appointed. Your instructor then becomes the gamemaster (GM)

and takes a seat in the back of the room. While not in control, the GM may do any of the following:

- Pass notes to spur players to action;
- Announce the effects of actions taken inside the game on outside parties (e.g., neighboring countries) or the effects of outside events on game actions (e.g., a declaration of war); and
- Interrupt and redirect proceedings that have gone off track.

Presiding officers may act in a partisan fashion, speaking in support of particular interests, but they must observe basic standards of fairness. As a failsafe device, most Reacting games employ the “Podium Rule,” which allows a player who has not been recognized to approach the podium and wait for a chance to speak. Once at the podium, the player has the floor and must be heard.

In order to achieve your objectives, outlined in your role sheet, you must persuade others to support you.

You must speak with others, because never will a role contain all that you need to know, and never will one faction have the strength to prevail without allies. Collaboration and coalition-building are at the heart of every game.

Most role descriptions contain secret information which you are expected to guard. Exercise caution when discussing your role with others. You may be a member of a faction, which gives you allies who are generally safe and reliable, but even they may not always be in total agreement with you.

In games where factions are tight-knit groups with fixed objectives, finding a persuadable ally can be difficult. Fortunately, every game includes roles that are undecided (or “indeterminate”) about certain issues. Everyone is predisposed on certain issues, but most players can be persuaded to support particular positions. Cultivating these players is in your interest. (By contrast, if you are assigned an “indeterminate” role, you will likely have considerable freedom to choose one or another side in the game; but often, too, indeterminates have special interests of their own.)

Cultivate friends and supporters. Before you speak at the podium, arrange to have at least one supporter second your proposal, come to your defense, or admonish those in the body not paying attention. Feel free to ask the presiding officer to assist you, but appeal to the GM only as a last resort.

Immerse yourself in the game. Regard it as a way to escape imaginatively from your usual “self” and your customary perspective as a college student in the 21st century. At first, this may cause discomfort because you may be advocating ideas that are incompatible with your own beliefs. You may also need to take actions which you would find reprehensible in real life. Remember that a Reacting game is only a game and that you and the other players are merely playing roles. When they offer criticisms, they are not criticizing you as a person. Similarly, you must never criticize another person in the game. But you will likely be obliged to criticize their persona. (For example, never say, “Sally’s argument is

ridiculous.” But feel free to say, “Governor Winthrop’s argument is ridiculous” – though you would do well to explain exactly why!) Always assume, when spoken to by a fellow player – whether in class or out of class – that that person is speaking to you in role.

Help to create this world by avoiding the colloquialisms and familiarities of today’s college life. Never should the presiding officer, for example, open a session with the salutation, “Hi guys.” Similarly, remember that it is inappropriate to trade on out-of-class relationships when asking for support within the game. (“Hey, you can’t vote against me. We’re both on the tennis team!”) Reacting to the Past seeks to approximate of the complexity of the past. Because some people in history were not who they seemed to be, so, too, some roles in Reacting may include elements of conspiracy or deceit. (For example, Brutus did not announce to the Roman Senate his plans to assassinate Caesar.) If you are assigned such a role, you must make it clear to everyone that you are merely playing a role. If, however, you find yourself in a situation where you find your role and actions to be stressful or uncomfortable, tell the GM.

1.4.3 Game requirements

Your instructor will explain the specific requirements for your class. In general, a Reacting game will require you to perform several distinct but interrelated activities:

- **Reading:** This standard academic work is carried on more purposefully in a Reacting course, since what you read is put to immediate use.
- **Research and Writing:** The exact writing requirements depend on your instructor, but in most cases you will be writing to persuade others. Most of your writing will take the form of policy statements, but you might also write autobiographies, clandestine messages, newspapers, or aftergame reflections. In most cases papers are posted on the class website for examination by others. Basic rules: Do not use big fonts or large margins. Do not simply repeat your position as outlined in your role sheets: You must base your arguments on historical facts as well as ideas drawn from assigned texts – and from independent research. (Your instructor will outline the requirements for footnoting and attribution.) Be sure to consider the weaknesses in your argument and address them; if you do not your opponents will.
- **Public Speaking and Debate:** Most players are expected to deliver at least one formal speech from the podium (the length of the game and the size of the class will affect the number of speeches). Reading papers aloud is seldom effective. Some instructors may insist that students instead speak freely from notes. After a speech, a lively and even raucous debate will likely ensue. Often the debates will culminate in a vote.
- **Strategizing:** Communication among students is a pervasive feature of Reacting games. You should find yourself writing emails, texting, and attending meetings on a fairly regular basis. If you do not, you are being outmaneuvered by your opponents.

1.4.4 Skill development

A recent Associated Press article on education and employment made the following observations:

“The world’s top employers are pickier than ever. And they want to see more than high marks and the right degree. They want graduates with so-called soft skills – those who can work well in teams, write and speak with clarity, adapt quickly to changes in technology and business conditions, and interact with colleagues from different countries and cultures.... And companies are going to ever-greater lengths to identify the students who have the right mix of skills, by observing them in role-playing exercises to see how they handle pressure and get along with others ... and [by] organizing contests that reveal how students solve problems and handle deadline pressure.”

Reacting to the Past, probably better than most elements of the curriculum, provides the opportunity for developing these “soft skills.” This is because you will be practicing persuasive writing, public speaking, critical thinking, problem-solving, and collaboration. You will also need to adapt to changing circumstances and work under pressure.

Chapter 2

Historical background

This section serves as an introduction to the history explored in the *Charles Babbage, Ada Lovelace, and the Dawn of Computing* game. This is just an introduction! You should also read the works in the Core Texts section thoroughly. Your individual role sheet may refer you to other sources as well.

2.1 Chronology

The following shows a chronology of events from the dawn of civilization through the early nineteenth century, pertaining to technology used for computing, and a few other dates relevant to our game.

Circa 2700 BCE An early form of abacus is used for calculations in Mesopotamia.

Circa 200 BCE The Chinese abacus comes into use.

Circa 150 BCE Devices such as the Antikythera mechanism, a bronze clockwork device, are used to predict eclipses and to perform other astronomical or astrological tasks.

1632 William Oughtred at the University of Oxford, England, invents the modern slide rule, allowing skilled operators to quickly perform multiplication and division operations.

1642 Blaise Pascal designs and builds his first mechanical calculator (called a *Pascaline*), capable of performing addition or subtraction of two numbers. Pascal's calculator, and others that followed, required human intervention during their calculations, thus allowing human error that could cause erroneous results.

December 26, 1791 Charles Babbage is born in Walworth, Surrey, England.

1793 - 1815 The French wars. The French Empire, led by Napoleon Bonaparte, expands to rule most of continental Europe. Rival nations, including the Austrian Empire, Prussia, and Spain, but primarily led and financed by Great Britain, finally defeat Napoleon at the Battle of Waterloo on June 18, 1815.

1801 The French weaver Joseph-Marie Jacquard invents a programmable weaving loom, controlled by punched holes in a paper tape. The pattern of holes in the tape controlled the pattern of cloth produced by the loom.

October 1810 Babbage starts at Trinity College, Cambridge.

1812 Along with fellow undergraduates John Herschel, Alexander D'Arby, Edward Ryan, Sir Edward Thomas Ffrench Bromhead, 2nd Baronet, and George Peacock, Babbage forms the Analytical Society. The Society aims to modernize British mathematics, chiefly by adopting the calculus notation used in continental Europe instead of the Newtonian notation still used in the kingdom.

December 10, 1815 Augusta Ada Byron, later the countess of Lovelace, born in London, England.

1819 On a trip to Paris with John Herschel, Babbage meets Gaspard de Prony, who had been commissioned by the French government to produce a table of logarithms and trigonometric functions in celebration of the creation of the metric system. De Prony's scheme to partition the work involved into three phases inspires the architecture of Babbage's Difference Engine.

1820 or 1821 Upon encountering many errors in the printed mathematical tables used by navigators, astronomers, and other men of science, Babbage first conceives of the Difference Engine. "I wish to God the calculations had been executed by steam" is the associated, and possibly apocryphal, quotation. (Babbage's autobiography claims this happened in 1812 or 1813, but other, earlier sources say 1820 or 1821. It seems more likely for Babbage's inspiration to have happened after the visit to Paris in 1819 rather than before.)

July 3, 1822 A six-figure-wheel working model of the Difference Engine is completed, demonstrating that the design and technology is sound. Babbage announces the project in a letter to Sir Humphrey Davy, 1st Baronet, the president of the Royal Society of London. Davies Gilbert sends a copy of the letter to Sir Robert Peel, 2nd Baronet, then serving as Home Secretary.

April 1, 1823 The Lords of the Treasury ask the Royal Society for their opinion on the merits and utility of the proposed Difference Engine.

May 1, 1823 The Royal Society replies, "Mr. Babbage has displayed great talent and ingenuity in the construction of his Machine for Computation, which the Committee think fully adequate to the attainment of the objects proposed by the inventor; and they consider Mr. Babbage as highly deserving of public encouragement in the prosecution of his arduous undertaking."

June 1823 Babbage meets with Lord Frederick John Robinson, 1st Earl of Ripon, Viscount Goderich, the Chancellor of the Exchequer. Robinson promises £1,000 of government funds to begin the creation of the Difference Engine. Babbage actually receives £1,500, and his understanding from the meeting is that this was only a start, and that Robinson intended the government to provide more funding. However, no

minutes of the meeting were made.

1823 Babbage hires the engineer Joseph Clement to build the Difference Engine.

2.2 Narrative

This section provides very brief introductions to the history of technology used for calculations, the changing nature of science in the Georgian era, British society in the early 1800s, and short biographies of two of the game’s principal characters, Charles Babbage and Ada Lovelace.

2.2.1 A brief history of computing technology

Technologies to aid human beings with mathematical calculations are as old as civilizations themselves. Writing – including methods to record numerical values – was one of the technologies associated with the Urban Revolution, beginning circa 3500 BCE. All of the first “pristine” civilizations shared common elements, including higher learning. Higher learning included writing, mathematics, and the beginnings of astronomy.

Writing can be seen as the earliest of all technologies related to mathematics: “Writing and reckoning were first and foremost practical technologies with practical origins meeting the practical needs of early civilizations. . . . All early civilizations developed arithmetical systems and systems of permanent record-keeping.” [ID06] For example, the ancient Babylonians had notations not only for keeping track of commerce, taxes, etc., but also could represent algebraic calculations.

For a millennium, writing remained the only technology aiding humans with mathematics. True mechanical calculators began with the simple abacuses used by Sumerians circa 2700 BCE. These were tables – literally – with several grooved columns, with each successive column representing the next highest power in their base-60 (sexagesimal) number system. What we most likely envision when we think about an abacus, the Chinese Suanpan (counting tray), originated about 200 BCE. These abacuses had two tiers of beads separated by a wooden bar, with two beads above the bar and five below. A Suanpan can be used for counting and arithmetic in either our usual decimal (base-10) or the hexadecimal (base-16) number system. And, in addition to simple counting, trained abacists can use the machines to quickly perform division, addition, subtraction, and even square or cube roots.

Shortly thereafter, circa 150 BCE, sophisticated clockwork computers, such as the Antikythera mechanism, were in use. Built of thirty or more inter-meshed bronze gears, this surprisingly sophisticated astronomical computer could be used to predict eclipses, among other astronomical calculations. Unfortunately, the knowledge of how to create and use such sophisticated devices was lost. When the Antikythera device was first discovered in a Greek shipwreck in 1901, archaeologists did not think it possible that the machine could be 2000 years old. Nothing as technologically precise was produced again until the 1300s.

Another kind of technology that was used for calculations was the astrolabe, in use from as early as 220 BCE through the present day. By measuring the angle from the horizon to a known star or planet, someone skilled in the use of an astrolabe can determine their latitude, thus aiding in navigation.

Around the year 500 CE, the Arabic numeral system, with place-value representation, became complete when Hindu mathematicians added the symbol representing zero to the other nine digits. The shapes of the written digits we are familiar with did not become common for centuries after that, however.

During the dark ages, as in many scientific and technological areas, there was little progress with computational technologies, whether written or mechanical.

Circa 1600, Scottish landowner and amateur mathematician John Napier, 8th Laird of Merchiston, discovered logarithms. A logarithm is a quantity representing the power to which a fixed number (the base) must be raised to produce a given number. He also discovered that large numbers can be multiplied or divided by adding or subtracting the logarithms of the numbers. Students that have learned basic arithmetic can attest that pencil-and-paper long multiplication or division is a laborious process, especially for large numbers. On the other hand, pencil-and-paper addition or subtraction can be performed more easily and with higher accuracy than multiplication or division. So, if one had a way to quickly calculate or look up the logarithms for large numbers, multiplication and division of such numbers could be done much more quickly via addition of the logarithms. This led to the need for tables of logarithms, which could be thus used to speed up calculations.

Napier's discovery also led to the invention of the first slide rule in the 1620s. The modern form of the slide rule was invented in 1632 by William Oughtred at the University of Oxford in England.

Ten years later in 1642, Isaac Newton was born in Woolsthorpe, England – and God said, “Let Newton be!” according to Alexander Pope. Newton's work in science and mathematics – not the least being co-inventor of calculus with Gottfried Leibniz – helped to cement Great Britain as the leading nation of the world in these fields.

In 1642, the French child prodigy Blaise Pascal invented a mechanical calculator that came to be known as a *Pascaline*. Pascal intended the machine to aid his father, a tax supervisor. The machine was able to add or subtract two numbers, and do multiplication or division by repeated applications of addition or subtraction. Eventually Pascal received a patent on the machine, and produced twenty or more of the machines over several years. Pascal's machines featured automatic carry operations – that is, the machines could add a ‘1’ to the next leftmost column when the sum of the current two digits was ten or more. Almost 200 years later, Charles Babbage spent much intellectual energy in designing the automatic carry mechanism for his first Difference Engine.

In 1671 Gottfried Leibniz, of calculus fame, tried to improve the Pascaline, so that it could perform multiplications automatically. His device was called the Stepped Reckoner, and the key feature in Leibniz's design was a stepped-drum gear which came to be known as the Leibniz wheel.

Although the Reckoner was not commercially successful, the Leibniz wheel was to become very important. In 1820, the French inventor Thomas de Colmar invented the Arithmometer, an invention using Leibniz wheels. The Arithmometer was the first calculator to be used in daily business use. Machines similar to Colmar's Arithmometer remained in use until the 1970s.

As we have seen, writing calculations down (e.g., performing long division) is in fact a technology for performing calculations. Many who learned to do pencil-and-paper multiplication or division in grade school will attest that this technology is fraught with the possibility of human error. In fact, the possibility of human error ruining a calculation is a problem shared by all of the calculating technologies described above. These errors could occur not only during the setup of the calculation, but during the process of the calculation itself, since human intervention was required during the process. Eliminating human error is one of the driving forces Charles Babbage felt when designing his calculating engines.

Calculating technologies began with the dawn of civilizations during the Urban Revolution. Our game takes place during the next major change in humankind's mode of being, the Industrial Revolution. The Industrial Revolution began in England around the year 1760, and was well into its transformation of the nation and the world by the 1820s.

2.2.2 The changing nature of science

First, it is important to discuss terminology. The word science itself did not come into common use until the early nineteenth century (the time period of our game). Before that, the task of explaining of the natural world, without resorting to supernatural causes, was considered a branch of philosophy, specifically, natural philosophy. The term *scientist* was coined by William Whewell in 1834, in a review of one of Mary Somerville's articles, and did not come into common use until later in the century. Therefore, it is inappropriate to refer to any of the characters in our game as scientists – certainly never while the game is underway! Instead, those that study science are called “men of science.”

A commonly held misconception is that government support of scientific research is a relatively new concept, perhaps marked most clearly by the 1940s Manhattan Project in the United States. Even the earliest civilizations performed works of science, and in these cases, the rulers or governments of the time supported the scientific activities.

In these first civilizations, in Mesopotamia, Egypt, China, the Indus valley, and Central and South America, governments supported science for practical purposes. Mathematics was required for record keeping, tax collection, and trade, and so the government required trained mathematicians. Astronomy was required for religious purposes, so that the dates of festivals and other important days could be determined. Astronomy was also useful in agriculture, as the changing seasons or flood patterns determined when to plant or harvest. Further, astronomy was useful for astrological predictions: when is it a good time to hold a royal wedding? When is it the right time to go to war? The answers were in the stars, and trained astrologists could find them. Note that the *perceived* usefulness

of scientific knowledge is what is important, not our current-day knowledge of the actual utility. Likewise, the proto-chemistry of alchemy was deemed useful, and so alchemists were required and trained.

In ancient Greece, the character and utility of natural philosophy changed. First, science was no longer required to be immediately practical. In fact, many of the philosophers of the time rejected any thought of practical application. Science became an abstract study of the natural world, without regard for any real-world usefulness. Secondly, science moved from the realm of government support into a discipline pursued by wealthy philosophers who were able to support themselves and their own intellectual pursuit with their own funds, or with funds earned by teaching students. Some of these philosophers, such as Plato and Aristotle, formed the foundations of the modern scientific traditions.

Western science faded after the fall of the Roman Empire, not to be rediscovered until the Renaissance of the 1300s. During the Dark Ages, the scientific traditions of the Greeks were kept alive and expanded by Muslim scholars. Arabic scholarship of this time had somewhat of a more practical character: much progress was made in medicine and mathematics, for example. Europeans only discovered Greek philosophy when Arabic texts were captured during the Crusades. It has been said that more ancient Greek works were found in Arabic translations than in the original language.

After the Renaissance, European science once again was a philosophical discipline mostly pursued as a hobby by the wealthy. In the time period of our game, this was just on the verge of changing. By the end of the nineteenth century, science was a discipline one could pursue as a profession instead of as a hobby; one could aspire to be a scientist, and one did not have to be born wealthy to pursue the subject.

It is also important to know that the discipline of engineering was until recently not connected to natural philosophy. Engineers did not need an abstract understanding of natural phenomena to build roads, buildings, or bridges; instead, they relied on rules of thumb derived from centuries of experience. In fact, the engineering usually preceded the science. For example, engineers knew how to build bridges before philosophers understood the abstract field of materials science. Also, engineers were working men, with little formal education outside of their apprenticeships, while natural philosophers were educated, wealthy intellectuals who had the luxury of thinking about the natural world without regard to any practical considerations. In our time, we often think of engineering as applied science, but that has only been true since the early twentieth century, marked by the Manhattan Project to develop the first atomic bomb.

2.2.3 The United Kingdom in the 1820s

The United Kingdom in the early part of the nineteenth century was a nation in transition. The American Revolutionary War and the French wars were over. The industrial revolution was well underway; gathering steam, one might be inclined to say.

King George III – familiar in the United States as the king during the American Revolution – died in 1820, succeeded by his son King George IV. George III had been blind, deaf, and

mad at the time of his death, and George IV gained a reputation for overindulgence and lax morality. The monarchy's reputation suffered in the view of the people during these years.

But, the Victorian age had yet to begin. Landed gentlemen with ties to the Church of England were still by far the dominant force in British politics. The poor, most of the growing middle class, nonconformists¹, Catholics, and women had virtually no political voice on the national scale. As people left the countryside to take factory jobs in the growing cities, slums, poverty, child labor, and dismal working conditions became major problems.

In certain ways, middle-class standards of living were on the rise, however. Those in the middle class might be able to take an occasional vacation by the sea, and a culture of consumption was starting to emerge. But the actual living conditions in large cities were worsening at the same time. The average life expectancy in the biggest cities dropped from 35 in 1820 to only 29 in 1830.

This was an era where “self-improvement” was popular. Public lectures on various scientific or other intellectual topics were popular and well-attended. One such venue was the London Mechanic's Institute, established in 1823 by Sir George Birkbeck. Mechanic's Institutes were established during this time all over the United Kingdom to provide adult education to working men.

Britain was in the midst of the industrial revolution, but the concepts of mass production were almost a century away. Most industrial concerns were relatively small, and owned by families or small partnerships rather than by large corporations.

Women were still excluded from many parts of British society during this time. Suffrage was a century away. Poor and middle-class women worked outside the home, and the changing demographics of the time saw many of them move to the cities to pursue work in the factories. By the mid-nineteenth century, the need for women and children in factory work was diminishing, but in the time of our game it was very much a fact of life.

Upper-class women, while they did not have to work in the factories, were still excluded from the boardroom. They did have important social roles, though; “...the talk of the town was almost as important as the smoke of the factory chimneys in creating a prosperous industrial economy.” [Mat00] Many men saw no need for suffrage, as they felt women were already fully involved in the system, via the role they played as homemakers and society organizers.

During this time, the infrastructure of Britain was rapidly forming, with roads, railways, canals, and steamships connecting the nation as never before. The railways, starting in the early 1830s, are illustrative, since they were created by engineers and industrialists, and unplanned by the government. Nevertheless, by the end of the century the railways provided an excellent transportation system for passengers and goods.

Politically, the Tories – the conservative party of landowners and the Church of England –

¹ In the U.K., a *nonconformist* was a Protestant Christian who did not belong to, or disagreed with the teachings of, the Church of England.

were convinced of the superiority of their approach. The fact that Britain was victorious over the French, and the rising power of the economy, served as further proof that the British system of government – the *unreformed* system of government – was a superior system, particularly when compared to the more liberal traditions that had appeared in France after its revolution. A related effect on society was the rise of a large and powerful class of businessmen, who had helped finance the wars and supply the armies.

Not everyone was so satisfied with the British system of government, however. The Whigs wished to expand franchises ², to allow underrepresented classes more political power. Radicals stirred up trouble in the cities, leading to riots, which in turn led to more repressive measures by the government. Many dissidents and criminals – hundreds of thousands by the middle of the century – were shipped to Australia.

Largely, this was an era of free trade and limited government spending. A large amount of money – nearly half of the total budget each year in the early part of the century – was spent on paying off French war debt. One-third of the government’s remaining spending was defense-related. Despite this, the income tax was abolished in 1816, increasing the amount of money that had to be raised from indirect taxation, such as taxes on food.

The British government did support a limited amount of scientific work, as long as the science was clearly practical. Such projects usually had to do with the maintenance of the British Empire, “...to map the earth and its geological and botanical resources, or to produce charts of the stars and of the earth’s magnetic field to aid safe navigation of oceans.” [Mat00]

2.2.4 Charles Babbage

Charles Babbage was born on December 26, 1791, in Walworth, Surrey. In the present day, Walworth is part of London. He was, as biographer Anthony Hyman puts it, “... the child of two revolutions: industrial revolution in Britain and political and social revolution in France.” [Hym85] The industrial revolution was changing the way of living in Britain, Europe, and the eventually the rest of the world, while the ideals of the French Revolution were starting to change the political structure of Britain as well.

Babbage’s father, Benjamin, was a banker. His family was wealthy, although not exceptionally rich. Benjamin was proud that his wealth was largely earned, rather than inherited.

Even as a schoolboy, Babbage was very curious, and also creative in the ways in which he tried to satisfy his curiosities. For example, when he was attending his first boarding school, Babbage wanted to prove or disprove the existence of ghosts and demons. He tried to summon the Devil, so he could interview him and thus confirm or deny Satan’s existence. Luckily, he was not successful in his attempts. Later in his life, Babbage produced a unique

² In the United Kingdom, *franchise* is the right to vote in public political elections. At the turn of the nineteenth century, less than five percent of the population – mostly wealthy landowners – could vote.

blending of the scientific method, his ideas about computation, and theology, when he wrote his *Ninth Bridgewater Treatise*, published in 1837.

Later, at a different boarding school, Babbage and a friend were so interested in mathematics that they awoke daily at 3:00 a.m. and went to the classroom so they could clandestinely further their study of the subject. They were discovered by some of their classmates, who insisted on joining the study sessions. Willingly awaking early in the morning to secretly study mathematics is likely difficult for many readers in our time to imagine. As the number of students grew, the amount of studying declined. Eventually the group made enough noise to waken the headmaster, causing them to be found out, thus ending the practice.

In 1810, Babbage began his studies at Trinity College, Cambridge. University life at the time was quite different from what students experience today. There were no set class schedules, assigned homework, or frequent examinations. Rather, students attended chapel and lectures, and studied their subjects from books recommended by the fellows of the college. At Cambridge, exams taken early in the second semester of the fourth year determined graduation and honors status. Some things, however, were perhaps not so different at Cambridge than at university today: Lord Byron, who graduated Cambridge in 1808, described the university as “. . . a villainous chaos of din and drunkenness, nothing but hazard and burgundy.” [BM82]

Babbage was very much looking forward to learning advanced mathematics from the professors there, but he was disappointed, partially because of the state of mathematics at the university and partially because of his own stubborn nature. The professors recommended that their students study the mathematics they would need to pass the standard graduation exam or the Tripos exam, which was taken by a select few students to determine who would graduate with honors. The mathematics questions for these exams had not been updated since 1747. In a sense, Cambridge was still resting on the laurels of Newton, while the world (particularly continental Europe) had moved forward. British mathematicians continued to use Newton’s notation in calculus problems, while everyone else had adopted Leibniz’s notation, the notation familiar to mathematics students today. In addition, Newtonian calculus approached problems with graphs, lines, and charts, while analytical calculus instead relied on the manipulation of symbolic equations to solve problems.

Babbage had purchased a French calculus textbook before starting at Cambridge, and wanted to study the analytical mathematics he learned from the text. Babbage would later claim that no one at Cambridge understood this type of mathematics, but that was not the case. At least one Cambridge scholar, Robert Woodhouse, had written extensively on the subject. So, it is not the case that no one understood Babbage, rather, the professors recommended to him that he should study what was going to be on the exam rather than pursue analytical mathematics. He did not follow their advice.

Babbage’s mathematical salvation came from two sources: first, he had access to the library, through which he could read the mathematical journals from the continent; and second, he met several fellow undergraduates who were also interested in modernizing British mathematics.

Along with friends and future luminaries John Herschel, Alexander D’Arbly, Edward Ryan, Sir Edward Thomas Ffrench Bromhead, 2nd Baronet, and George Peacock, Babbage formed the Analytical Society, dedicated to advancing the state of mathematics in the United Kingdom. They wanted to promote the use of analytical mathematics in Britain, and also to change kingdom’s usage of calculus notation from the style favored by Newton to the style of Leibniz, the notation used on the continent and by other modern mathematicians. Eventually the Society published a compilation of papers on analytical mathematics written by Babbage and Herschel entitled *Memoirs of the Analytical Society*. Although the Society was only in existence from 1812 until 1814, it was successful in its primary goal; the Leibniz notation was adopted in Britain, and the state of mathematics in the kingdom began to recover.

The Society also had a lasting impact on Babbage. From his early days, Babbage was a firm proponent of mathematical science, of collecting copious amounts of data and then analyzing it – with the tools of calculus or other mathematics – to form conclusions. The like-minded compatriots of the Analytical Society helped to firmly cement that characteristic; this Baconian scientific mindset helps explain many of Babbage’s prominent activities for the rest of his life.

Babbage graduated from Cambridge in 1814, barely. He had not prepared himself for the standard graduation exam, much less the Tripos examinations for honors students. In contrast, Herschel, who graduated in 1813, spent the entire summer before his fourth year studying for the Tripos. Once the grueling five-day examination was over, Herschel graduated as the Senior Wrangler, the highest scoring honors student.

Babbage’s graduation examination started off in a grim fashion, as he was unable on the first several days to answer question after question dealing with Newtonian mathematics. Eventually, one of the examiners (perhaps Woodhouse) asked Babbage a question based on one of his papers in the *Memoirs of the Analytical Society*, and with one correct mathematics answer, Babbage passed the exam and graduated from Cambridge.

Also in 1814, Charles married Georgina Whitmore. Charles kept his engagement secret from his friends, and his father did not approve of the marriage, leading to a strained relationship with his son. Charles and Georgina, however, were happy in the union.

On March 14th, 1816, Babbage was elected as a Fellow of the Royal Society of London. The Royal Society was officially established in 1663 by King Charles II, to promote the study of natural philosophy in the kingdom. “The Royal,” as it is sometimes called, served as a forum for men of science, a place for them to meet and discuss the subjects of the day, and also published important papers. Not everyone who was an important man of science was a member of the Society, and the Royal’s membership at the time of our game was not restricted to men of science. Politicians and titled gentlemen were also members, having been elected based on their political or social importance rather than on their scientific merits. Prominent Fellows who were not men of science included Arthur Wellesley, 1st Duke of Wellington, and Lord George Gordon Byron, 6th Baron Byron, father of Ada Lovelace.

In 1823, Babbage was elected as a member of the Royal Society Dining Club. The Dining

Club met once a week for a sumptuous, seven-course meal, with elaborate printed menus. The President of the Royal Society usually presided, and visitors were also invited to attend. The discussions at the weekly dinners were lively and entertaining, ranging over the gamut of the scientific topics of the day.

In the five years between 1815 and 1820, Babbage was a productive mathematician, specializing in the theory of functions³. Charles also pursued employment during this time, in order to be able to support his growing family without relying on his father. He sought a mathematics position at the East India Company College in Hailey, Hertfordshire, just north of London. Later, he sought the chair of mathematics at the University of Edinburgh, Scotland. Babbage did not win either post, however, likely due to his liberal political leanings and his family's nonconformist past. Benjamin Babbage missed no opportunity to remind Charles that he had not found permanent employment. Charles was still dependent on monthly allowances from his father.

In 1824, Babbage was invited to take part in the founding of a new life insurance company, The Protector. The opportunity would have offered a significant salary: £1,500 per year, which converts to approximately \$170,000 in 2016 U.S. dollars. He created the required actuarial tables, but the company never came to fruition, and so another opportunity to earn his own living was lost.

In 1819, Babbage visited Paris and observed the work of Gaspard de Prony, who had been commissioned by the French government to produce new, comprehensive tables of logarithms and trigonometric functions. The scope of the work required made it virtually impossible to produce the table via normal means, and de Prony had devised a three-stage procedure to make the project practicable. A few highly-trained mathematicians organized the work and determined how particular calculations would be carried out. A larger number of perhaps less-visionary mathematicians would figure out how these calculations could be done using only addition and subtraction, via the method of finite differences⁴. In the final stage, a much larger number of human computers – for that was a job description rather than the name of a mechanical computing device – who only knew how to perform addition and subtraction, would do the actual calculations. De Prony's division of labor for the calculation of mathematical tables a major influence on Babbage's design for the Difference Engine.

According to Babbage's own recollection, it was in 1820 or 1821 that he first had the idea to automate calculations with his Difference Engine. The thought struck while Babbage and Herschel were working on calculations that involved the use of tables of logarithms.

³ In mathematics, a *function* is a mapping from a set of inputs to a set of outputs, such that each input maps to only one output. A function transforms the input into an output. The theory of functions is an inquiry in pure mathematics, where functions and their properties are studied abstractly, without regard for practical application.

⁴ The *method of finite differences* is a way to produce each successive value of a polynomial $f(x) = c_k x^k + c_{k-1} x^{k-1} + \dots + c_1 x^1 + c_0 x^0$, using only addition and subtraction, without having to do multiplication. Other functions, such as logarithms, can be acceptably approximated via polynomials, and thus the method of finite differences can be used, if the polynomial is carefully chosen, to produce a table of logarithms (or any other useful function) using only addition and subtraction. Lab 1, if it is a part of your game, will introduce and explore the method of finite differences.

As we have seen, these extensive tomes, pages upon pages of columns of numbers, were necessary for performing more efficient computations. However, the tables had a seemingly unavoidable problem: the logarithms in the table had to be originally computed by a human at some point, and therefore the tables were subject to human error. Upon encountering a plethora of errors in the tables, Babbage allegedly said to Herschel, “I wish to God the calculations had been executed by steam,” leading to the idea for the Difference Engine.

Later, Babbage sat down and sketched out his initial ideas for the Difference Engine, convincing himself that such a device was in fact possible. Babbage later claimed that the idea was so impactful that it made him physically ill.

In February 1827, Benjamin Babbage, Charles’ father, died. Benjamin’s sizable estate was divided between Charles and his mother. Charles received approximately £100,000, relieving his financial woes. He was now able to support his family in a luxurious manner, and also fund his scientific investigations.

Later in 1827, Babbage was devastated when his beloved wife Georgina became ill and died. His youngest son, Alexander, also died, aged less than one year. Babbage took a year-long tour of continental Europe, returning to Britain in 1828, the year our game begins. In 1828, while still in Rome, Babbage read in a newspaper that he had been elected as the Lucasian Professor of Mathematics at Cambridge. The Lucasian Chair is perhaps the most prestigious academic position that one can hold anywhere in the world; Isaac Newton held the chair for 33 years in the seventeenth century, and Stephen Hawking held it for 30 years in the twentieth. Babbage replaced Sir George Biddell Airy, who had bested Charles for the chair just two years prior in 1826. Babbage held the chair for more than a decade, although he never resided at Cambridge nor did he lecture there.

Babbage was charming and energetic, a prominent fixture in British society. “He was equally at ease with intelligent working men, country clergymen, men of science, at court, or at the dining tables of the aristocracy.” [Hym85] His weekly parties at his residence at 1 Dorset Street became famous in London society, and invitations were much sought after.

During the time period covered by this game, Babbage was a liberal, bordering on radical, although later in life he became more conservative. In keeping with his broad social circles, he knew the liberal political reformers of the time.

Despite his inauspicious years at Cambridge, by the time of our game Babbage was a leading man of science, well known and respected by his colleagues.

2.2.5 Augusta Ada King-Noel, Countess of Lovelace

Augusta Ada Byron was the only legitimate child of her “mad, bad, and dangerous to know” father, poet and immense celebrity Lord Byron, and her mother, Anne Isabella Milbanke Byron. She was born on December 10th, 1815 in London. Although named Augusta for Byron’s half-sister, everyone, including Byron himself, referred to her as “Ada.” Ada’s

parents divorced in 1816, and she never knew her father. Due to Byron's notorious fame in Britain, Ada was something of a celebrity from the time of her birth.

Lord Byron died in April 1824 in Missolonghi, Greece, where he was fighting in the war for Greece's independence from the Ottoman Empire. Byron did not die while fighting, but rather of complications from medical care, specifically bloodletting.

Ada's mother passed care of her young daughter on to her mother, Lady Milbank. A quotation from one of Anne's letters to her mother, regarding Ada, may give us some insight into her (lack of) regard for her daughter:

“I talk to *it* for your satisfaction, not my own, and shall be very glad when you have *it* under your own” [Woo99] (Emphasis added.)

Anne was also controlling, and determined that Ada should not follow in her father's hedonist ways. In fact, Lady Byron considered her former husband to be insane, and so to help prevent Ada following in Byron's footsteps, Anne hired tutors to teach Ada mathematics at an early age, lest she become “poetical.” One of these tutors, William Frend, was a mathematician so conservative that he did not believe in negative numbers. Another tutor, and later friend, was Mary Somerville, a Scottish woman who published in the subjects of mathematics and astronomy, an almost unheard of accomplishment for woman of the time. As it turned out, Ada had a natural gift for mathematics. Despite her mother's best intentions, Ada later in life described her approach to natural philosophy and mathematics as “poetical science.”

Ada was a sickly child, often being assigned to bed rest, with one spell abed lasting for a year. She did not attend any formal schooling, but continued to work with tutors throughout her teenage years.

Lovelace met Charles Babbage at one of his Dorset Street parties on June 5th 1833 (see the Counterfactuals section below for a reconciliation of this with the game's timeline). After that, she learned more about the Difference Engine by attending one of Dionysius Lardner's lectures at the Mechanics Institute.

Chapter 3

The game

This section provides a general sketch of how the *Charles Babbage, Ada Lovelace, and the Dawn of Computing* game works. This information just scratches the surface, however! You will find much more specific information about your character, and what is expected of your character during the game, in your role sheet.

3.1 Major issues for debate

The central intellectual collisions in the game concern the conflict between imagination and reason, the nature of science and scientists (are they talented, wealthy amateurs, or is science a profession?) and whether and to what degree science and engineering projects should be subsidized by the government. The main question in the game is whether or not Charles Babbage should be awarded additional funds from the British government for the development of his Difference Engine (an automated calculator capable of automatically creating, typesetting, and printing mathematical tables) and / or Analytical Engine (a true proto-computer), during the early to mid-1800s. Other issues concerning politics, society, the state of mathematics in Britain, and so on, may also arise during the game.

3.2 Rules and procedures

This section briefly summarizes the rules and procedures of the game. More specifics for your character in particular may be found in your role sheets.

3.2.1 When and how to speak

Your role sheet will provide details on when you should make a formal speech to the class, e.g., John Herschel will make a speech briefly introducing the Difference Engine in game session (GS) 1, after the Georgian party games, Prime Minister's speech, and other

Table 3.1: Power cards

Card	Power	Card	Power	Card	Power	Card	Power
A ♣	Laissez faire	A ◇	Intervention	A ♥	Prescience	A ♠	Muddle
K ♣	Swing Riot	K ◇	Invigorate	K ♥	Volte-Face	K ♠	Volte-Face
Q ♣	Cholera	Q ◇	Tar & Vinegar	Q ♥	Shove	Q ♠	Shove
J ♣	Excess	J ◇	Excess	J ♥	Exflunctify	J ♠	Topsy
10 ♣	Nudge	10 ◇	Nudge	10 ♥	Nudge	10 ♠	Nudge
9 ♣	Nudge	9 ◇	Nudge	9 ♥	Nudge	9 ♠	Nudge
8 ♣	Nudge	8 ◇	Nudge	8 ♥	Nudge	8 ♠	Nudge
7 ♣	Nudge	7 ◇	Nudge	7 ♥	Nudge	7 ♠	Nudge
6 ♣	Nudge	6 ◇	Nudge	6 ♥	Nudge	6 ♠	Nudge
5 ♣	Nudge	5 ◇	Nudge	5 ♥	Nudge	5 ♠	Nudge
4 ♣	Nudge	4 ◇	Nudge	4 ♥	Nudge	4 ♠	Nudge
3 ♣	Nudge	3 ◇	Nudge	3 ♥	Nudge	3 ♠	Nudge
2 ♣	Nudge	2 ◇	Nudge	2 ♥	Nudge	2 ♠	Nudge

presentations. The basic game outline below gives an outline showing when many of the scheduled speeches will take place.

You are also most likely – depending on your role – expected to arise on occasion to speak spontaneously, based on the characteristics of the debate in the game so far. Each game session will have designated opportunities for this; when you wish to speak, simply arise and make your way to the podium. The *podium rule* says that whenever someone approaches the podium, they must be allowed to speak, eventually, by the presiding official of the game session.

3.2.2 Basic game mechanics

In this game, your fundamental goal as an individual character, and as a member of a faction (if you happen to be part of a faction), is to make persuasive arguments and excellent speeches, so that you may accumulate and strategically use *power cards*, or PCs. There are two types of PCs: *nudges* and *powers*. In the game, PCs are realized by applying meanings to the cards of a standard, 54-card deck, as shown in Table 3.1. PCs are the currency of the game: the more you can accumulate the better, and if you use them wisely you can win!

During the game, individuals and factions can earn PCs in various ways, including, but not limited to, the following:

- By scoring the highest on the faction quiz at the end of the game setup sessions
- By wearing the best period-appropriate costume during a game session, as judged by the GM

- By winning the Georgian party game that takes place during Babbage's party, in GS 1
- By making the best presentation during a game session, as judged by the GM
- Based on the outcome of the Royal Society vote in GS 3, to recommend or not recommend funding the Difference Engine
- For the anti-Babbage faction only, passing any Royal Society resolution that is unfavorable to Babbage
- As a result of a Persuasiveness Metric vote at the end of a game session (see below)
- By convincing Countess Lovelace to allow you to draw a PC from the deck (see below)
- At the beginning of each game session, one of the characters in the game will be randomly selected; this character gets to draw a PC

When a character earns a PC, they draw their card from the top of the PC deck. The GM maintains control of the deck, and will shuffle the remaining cards at the beginning of each game session. Generally speaking, once a PC is played, it will not enter back into the game. This means that PCs are a finite resource that must be spent wisely to maximize your chances of winning the game.

3.2.2.1 Nudges

Nudges are used to exert influence in two ways. First, they can influence the Prime Minister (PM) when he makes a decision regarding Babbage's machines. If you play the game well, you can accumulate enough nudges to effectively force the PM to make a decision favorable to you. Secondly, nudges can be used to help steer the selection of the indeterminate who will serve as PM during GS 6; so, you can try to "elect" a PM favorable to your position.

The presiding PM makes decisions regarding Babbage's engines at two points in the game: during GS 3 and during GS 6. For each decision, the opposing factions can use nudges to try to force the PM to make a decision favorable to their side. The nudges are placed in one of two piles: one in favor of funding Babbage, and one not in favor of funding. Just before the PM's decision, individuals, factions, and the PM may choose to place any or all of their nudges on the stacks. Individuals place their nudges first, then factions, and finally the PM. Each party in this process may place as many or as few of the nudges they control as they choose, on either or both of the stacks. Once the nudging is complete, the stack that has the most nudges in it is the decision the PM must make. If there are equal nudges on both stacks, then the PM can decide to fund or not to fund Babbage at his discretion.

The first PM decision occurs in 1828, at the end of GS 3. There will be two stacks: fund the Difference Engine and do not fund the Difference Engine. For this decision, two nudges will be placed by the GM in the fund stack.

Table 3.2: GS 6 PM selection table

PM for GS 6 & 7	Die roll
Grey	1-2
Peel	3-4
Robinson	5-6
Russell	7-8
Wellington	9-10

Table 3.3: Sample modified PM selection table

PM for GS 6 & 7	Die roll
Grey	1
Peel	3-4
Robinson	5-6
Russell	2, 7-8
Wellington	9-10

The second PM decision occurs in 1846, at the end of GS 6. The stacks for this decision are fund the Analytical Engine and do not fund the Analytical Engine. For this decision, there are no pre-allocated nudges in either of the stacks.

For the first five game sessions, the indeterminate who serves as PM is chosen randomly, so that each will serve as PM exactly one time. The indeterminate that will serve as PM during GS 6 will be chosen by rolling a ten-sided die, according to Table 3.2.

Before the die is cast, factions or individuals can play nudges to influence the outcome. Each nudge adds one to the range on one indeterminate while subtracting from another. For example, a faction might play one nudge to boost Russel while hindering Grey's chances. The modified table in this case is shown in Table 3.3. Of course, the other faction could reverse the modifications by playing their own nudge. The pro-Babbage faction will play their nudges to influence PM selection first, followed by the anti-Babbage faction, and then any indeterminates who wish to influence the PM selection. Everyone should keep in mind that nudges will be played later on in GS 6 to influence the PM's decision, and so election nudges should be played conservatively.

3.2.2.2 Powers

A significant percentage of PCs are powers rather than nudges. Powers, when played, have a particular effect on the game. Like nudges, once a power is played, that card is not returned to the PC deck. Cards that are powers are shown above in Table 3.1, and each of the powers is described below.

Cholera This PC can only be used starting in 1830 (GS 4, 5, or 6). When character *A* plays Cholera, character *A* can choose any other character (call them character *B*)

in the game to contract Cholera. Character *B* then dies (unless someone, including character *B*, plays Tar & Vinegar). When someone dies of cholera, they lose any PCs they have earned; the cards are returned to the GM, who will place them in the discard pile. The GM will provide the “dead” character with another role for the duration of the game.

Excess Excess allows a character to draw twice from the PC deck rather than once. If a character has earned a draw from the PC deck, they can use Excess to draw twice instead of once. Excess can also be applied to another character, allowing them to draw twice instead of once.

Exflunctify When character *A* plays Exflunctify, character *A* may ask all the other characters to divulge how many PCs they have, and if asked, the other characters must answer truthfully. Then, character *A* chooses another character (character *B*), who will then lose all of their PCs. The GM will collect character *B*’s PCs and place them in the discard pile. The cards are not revealed, however.

Intervention This PC nullifies the power of Laissez Faire. If Laissez Faire has been played, playing the Intervention card removes the restriction from playing other cards. Intervention only applies to the game session it is played in.

Invigorate When a character plays Invigorate, the character may, under the supervision of the GM, search through the discarded PC pile, and take one of these cards. In this way, one discarded PC is returned to the game.

Laissez Faire Once this PC is played, no further PCs (except for the Intervention card) may be played during the game session. This includes the placing of nudges to influence PM decisions or the GS 6 PM election.

Muddle When this PC is played, the GM will shuffle the PC deck.

Prescience If a character plays the Prescience card, the GM will show that character the top five cards on the PC deck. If there are fewer than five cards in the deck, Prescience allows the playing character to see all of the remaining cards. After the player has seen the cards, the GM will place them back on top of the PC deck in the same order they were before Prescience was played.

Shove Shove can be played to double a single nudge card into two nudges. Shoves can only be used when nudges are being spent (end of GS 3 or beginning or end of GS 6), but may be applied to any nudge. I.e., a character does not need to play Shove on one of their own nudges.

Swing Riot When Swing Riot is played, one of the characters in the session is chosen at random by the GM. The chosen character must control one or more PCs; if they do not, the GM repeats the random selection until someone with PCs is chosen. The chosen player places their PCs face down on a table. Captain Swing then selects one of these cards, without looking at it. The GM removes this card from the game without revealing which card was removed.

Tar & Vinegar If someone has played the Cholera card, another character can play the

Tar & Vinegar card to attempt to cure the cholera. When Tar & Vinegar is played, the GM will roll a 10-sided die; if the roll is 1, 2, 3, 4, or 5, the targeted character survives the cholera. Otherwise, the character dies, forfeits their PCs to the GM, and is given a new role for the duration of the game by the GM.

Topsy Playing Topsy changes the object of a card. Topsy only applies in certain circumstances, and the exact effect is slightly different in each case, as outlined below. For the descriptions below, character *A* has just played one of the listed PCs, and, if applicable, the card is targeting character *B*. Character *C* (which may, in fact, be character *B*) can then play Topsy to change the behavior of the PC as follows:

- **Cholera** Instead of character *B* getting cholera, character *A* does. Tar & Vinegar can still be played to save character *A*, if someone wishes to use it.
- **Excess** Instead of character *B* getting to draw from the PC deck twice, character *B* draws once and character *C* draws once.
- **Exflunctify** Instead of character *B* losing their cards, the GM confiscates character *A*'s cards.
- **Invigorate** Instead of searching through the discard pile to get another PC, character *A* loses a PC. Character *A* places their cards on a table face down, and the GM selects one and places it in the discard pile without revealing which card it is.
- **Muddle** Topsy cancels this card, so the GM does not shuffle the PC deck.
- **Prescience** Instead of character *A* looking at the top five cards of the PC deck, character *C* gets to look.
- **Swing Riot** Swing places his cards face down on the table, and the GM removes one of the cards and places it in the discard pile, without revealing the card.

Volte-Face Volte-Face reverses the effect of the last PC played. Volte-Face applies to both nudges and powers, and so Volte-Face can be used to remove one nudge from a PM decision stack or from the PM election. Note that the reversed PCs have been played, and therefore are no longer in play, even though they have been nullified. Playing a Volte-Face immediately after a Volte-Face restores the effect of the previous card.

To play a power card, a character should approach the GM and say, “I invoke the Power of *X*,” where *X* is the power the character wishes to play. The GM will announce the play, and then facilitate the card's effect.

3.2.2.3 Forfeits

Babbage will serve as the “master of the revels” during the party game at the beginning of GS 1; that is, he will run the party game. He is able to assign forfeits during the festivities, which will ultimately determine who earns an PC for the party game. Forfeits are assigned if a player cheats, cannot correctly complete an aspect of the game, commits a social *faux*

pas, and so on. For example, if in the game *I Love my Love with an A*, Dionysius Lardner cannot think of a food that begins with the letter ‘K,’ Babbage would assign him a forfeit. Or, if Mary Somerville addresses Lord Charles Grey as Charles, rather than Lord Grey or Earl Grey, Babbage would assign her a forfeit. The GM will help Babbage keep track of the forfeits he assigns during the party game.

The practice of assigning forfeits was a common practice at parties during the era in which our game takes place. At the end of the party, party goers in the early 1800s would have to pay a small amount of money for each forfeit they received during the party. In our game, the stakes are PCs. The player with the fewest forfeits (other than Babbage, of course) will draw from the PC deck at the beginning of GS 2. If more than one player is tied for the fewest number of forfeits, then the GM will break the tie with a die roll or other randomized method.

3.2.2.4 Persuasiveness metric

At or near the end of each game session, the indeterminates (including the current PM) will be polled, via secret ballot, as to who they think made the most persuasive presentation during the game session. The GM will count the votes, and the individual who receives the most votes will receive an PC. In the case of a tie, all of the tied individuals will receive an PC.

3.2.2.5 Countess Lovelace’s celebrity

The Countess Ada Lovelace is an immense celebrity in the society our game occurs in, and by association, her celebrity can be parlayed into influence by other characters in the game. To attempt this gambit, a character should first notify the GM, who will accompany them to Lovelace. The character must then recite, from memory, one of the following poem stanzas from Lord Byron, Lovelace’s father [Byr69]:

To Caroline

Think’st thou I saw thy beauteous eyes,
Suffus’d in tears, implore to stay;
And heard unmov’d thy plenteous sighs,
Which said far more than words can say?

Damætas

In law an infant, and in years a boy,
In mind a slave to every vicious joy;
From every sense of shame and virtue wean’d;
In lies an adept, in deceit a fiend;
Versed in hypocrisy, while yet a child;
Fickle as wind, of inclinations wild;

Woman his dupe, his heedless friend a tool;
 Old in the world, though scarcely broke from school;

A Fragment

When, to their airy hall, my fathers' voice
 Shall call my spirit, joyful in their choice;
 When poised upon the gale, my form shall ride,
 Or, dark in mist, descend the mountain's side;
 Oh! may my shade behold no sculptured urns
 To mark the spot where earth to earth returns!
 No lengthen'd scroll, no praise-encumber'd stone;
 My epitaph shall be my name alone;
 If *that* with honour fail to crown my clay,
 Oh! may no other fame my deeds repay;
That, only *that*, shall single out the spot;
 By that remember'd, or with that forgot.

If that High World

If that high world, which lies beyond
 Our own, surviving Love endears;
 If there the cherish'd heart be fond,
 The eye the same, except in tears –
 How welcome those untrodden spheres
 How sweet this very hour to die!
 To soar from earth and find all fears,
 Lost in thy light – Eternity!

On Jordan's Banks

On Jordan's banks the Arab's camels stray,
 On Sion's hill the False One's votaries pray,
 The Baal-adorer bows on Sinai's steep –
 Yet there – even there – O God! Thy thunders sleep:

She Walks in Beauty

She walks in beauty, like the night
 Of cloudless climes and starry skies;
 And all that's best of dark and bright
 Meet in her aspect and her eyes;
 Thus mellowed to that tender light
 Which heaven to gaudy day denies.

The Fury (or, the GM, if the Fury is not a character in your version of the game) and Lovelace will listen to the poem stanza. If both agree that the character has faithfully recited the stanza, the character will be immediately rewarded with an extra draw from the PC deck.

Table 3.4: FRS status by game year

	1828	1830	1846
Adams			
Airy			X
Babbage	X	X	X
Brunel		X	X
Buckland	X	X	X
Clement			
Crosse			
Faraday	X	X	X
Frend			
Gilbert	X	X	X
Grey			
Herschel	X	X	X
Lardner	X	X	X
Lovelace			
Peel	X	X	X
Robinson	X	X	X
Russell			
Sabine	X	X	X
Somerville			
Sussex	X	X	X
Swing			
Wellington			

The Countess Lovelace will accept at most one recitation per game session, and once a stanza has been successfully recited, it may not be reused. Consult the GM for a list of which stanzas are still available for recitation.

3.2.2.6 Royal Society membership

In game sessions where the Royal Society debates, votes, and makes recommendations, only Fellows of the Royal Society (FRS) are allowed to participate. Not every character in the game is a FRS, and further, some characters do not hold FRS status early in the game. Table 3.4 shows the characters in the game and when they possess FRS status.

3.3 Objectives and victory conditions

Generally speaking, if you are a member of the pro-Babbage or anti-Babbage faction, you win the game if your faction wins. That is to say, if the British government refuses to fund Babbage's machines in both of the opportunities for funding, then the anti-Babbage faction

Table 3.5: Franchise status in 1846

Character	Franchise
Adams	Y
Airy	Y
Babbage	Y
Brunel	Y
Buckland	Y
Byron's Ghost	
Clement	
Crosse	Y
Faraday	Y
Frend	Y
Fury	
Gilbert	Y
Grey	Y
Ornamental Hermit	
Herschel	Y
King	
Lardner	
Lovelace	
Peel	Y
Robinson	Y
Russell	Y
Sabine	Y
Somerville	
Sussex	Y
Swing	
Wellington	Y

wins, and so all the members of that faction win. On the other hand, if the government does fund Babbage's machines in both of the funding opportunities, then the pro-Babbage faction wins, and therefore all the members of that faction win. If there is a split decision (one decision to fund and one not to fund), then the faction with the largest number of remaining nudges wins. If there is a split decision and the factions have an equal number of remaining nudges, then the GM will roll a die to decide which faction wins.

If your character is an indeterminate – i.e., if you are going to serve as PM at least once, or you are one of the indeterminates who cannot serve as PM – your victory conditions are not quite so clear cut. At the end of GS 6, all the characters who have a franchise will vote for the PM who made the most eloquent and convincing PM speech; the winner of this vote will be the PM that meets their victory conditions. The characters that have franchise for the GS 6 vote are shown in Table 3.3.

The indeterminates that cannot serve as PM have different victory objectives, explained in their respective role sheets.

It is very important for you to remember that you may have different, character-specific victory conditions, even if you are a member of a faction! Refer to your role sheet for details, and make sure to keep the specifics secret from other players in the game.

3.4 Basic outline of the game

The game covers a broad span of time, from 1828 through 1846. There are three phases to the game: setup lecture(s), the six game sessions and three optional labs, and then a postmortem session to set the record straight. The phases of the game are summarized in the following sections.

3.4.1 Phase I: Setup

During the setup phase, the instructor will spend some time introducing the relevant history (briefly summarized in the Narrative section of Part 2, above). Then the instructor will explain the game, in terms of the factions, roles, game sessions, and mechanics involved.

At the end of the setup phase, the instructor will assign the members of the class to roles, and will distribute individual role sheets. **Note:** *it is likely that your role sheets will contain secret information, meant for your eyes only. You should not share your role sheet with anyone, and you should not discuss any of your secret elements with anyone else, except the GM.*

Also at the end of the setup phase, your instructor will also likely give you a faction quiz to assess your knowledge of the portions of the core texts that have been assigned before the game begins. The quiz will be taken in groups, by faction; the indeterminates will work together as a faction for the purposes of this quiz. The faction that scores the highest on the faction quiz will earn a PC. If the indeterminates win the faction quiz, a die roll will determine which indeterminate receives the PC.

3.4.2 Phase II: Game play

The game play phase consists of six game sessions and three labs, in the sequence described below. Your instructor may omit the labs, and / or include more or fewer game sessions than those outlined here. Your instructor will solidify the number and order of sessions in your version of the game during the setup phase.

The game sessions fall into three categories: one party session, two public lecture sessions, and three sessions at the Royal Society. The character and feel of a game session should mirror the category it falls in; festive, boisterous yet educational, and reserved and serious, respectively.

Each character in the game has at least one formal speech or presentation to make during the game. Several of the characters in the game have the added responsibility of presiding over one or more of the game sessions. The sections below show when each character is expected to make his or her primary speech, and which character presides over each game session. For the first five game sessions, the indeterminate selected to be PM for that session will make his PM speech at the beginning of the session.

The specific order of the presentations may vary from the outline below. The presiding character will visit with the GM before his or her session(s) to determine the actual order. If you have a presentation scheduled for a particular game session, it would be in your best interests to be prepared to make your presentation at the beginning of the game session, or at any time after the session starts.

3.4.2.1 GS 1: Babbage's party (1828)

Location: Babbage's residence, 1 Dorset Street, London

Presiding character: Babbage

Draft schedule:

- The Fury on etiquette
- Georgian party game
- PM speech
- Frend and Somerville on mathematics
- William King on the supremacy of reason
- Herschel on Babbage's Difference Engine fragment

3.4.2.2 Lab 1 (Optional)

During this lab you will learn more about the method of finite differences, which Babbage's Difference Engine was designed to employ.

3.4.2.3 GS 2: Public lectures 1 (1828)

Location: The London Mechanic's Institute

Presiding character: Lardner

Draft schedule:

- Social time
- PM speech

- Sussex on laissez-faire economics
- Sabine and Lardner on the mathematics of navigation
- Captain Swing on the plight of farmers and the dangers of automation
- Open floor for debate

3.4.2.4 Lab 2 (Optional)

During this lab, you will use a modern, Web-based simulator to explore how the Difference Engine was designed to work.

3.4.2.5 GS 3: Difference Engine (1828)

Location: The Royal Society of London

Presiding character: Gilbert

Draft schedule:

- Social time
- PM speech
- Buckland introduces a resolution against funding the Difference Engine
- Faraday (or other pro-Babbage faction member) introduces a resolution supporting funding the Difference Engine
- Open floor for debate
- FRS voting on the resolutions
- Nudge spending
- PM decision

3.4.2.6 GS 4: Celebrating the Royal Society / Royal Society presidency (1830)

Location: The Royal Society of London

Presiding character: Gilbert

Draft schedule:

- Social time
- PM speech
- Gilbert on the Royal Society

- Babbage on the Royal Society
- Nominations for Royal Society president
- Open floor for debate
- Vote on Royal Society president

3.4.2.7 Lab 3 (Optional)

During this lab, you will use a modern, Web-based simulator to explore how the Analytical Engine was designed to work.

3.4.2.8 GS 5: Public lectures II (1846)

Location: The London Mechanic's Institute

Presiding character: Brunel

Draft schedule:

- Social time
- PM speech
- Clement on the Difference Engine
- Brunel on the Thames Tunnel
- Lovelace on the Analytical Engine
- Crosse on interchangeable parts
- Open floor for debate

3.4.2.9 GS 6: Analytical Engine (1846)

Location: The Royal Society of London

Presiding character: Royal Society President

Draft schedule:

- PM selection
- Social time
- Adams on the discovery of Neptune
- Airy on the discovery of Neptune
- Presentation of resolutions

- Open floor for debate
- Royal Society vote on resolution
- Franchise voting for best PM speech (winner not revealed until end of the session)
- Nudge spending
- PM decision

3.4.3 Phase III: Postmortem

The class session(s) scheduled for the postmortem discussion take place after the final game session is complete. During the postmortem session(s), the instructor will lead discussions about the game, to reveal the winners of the game, and to uncover secrets that characters had during the game. After this, the instructor will discuss what really happened regarding Charles Babbage and his computing engines, and highlight how the outcome of the game compared to the actual history. Finally, the instructor will lead a discussion where students share their views on the central concepts and intellectual collisions that were key aspects of the game.

3.5 Assignments

Generally speaking, you will have a writing assignment for each of your formal speeches, and if you are not an indeterminate, you will write a few paragraphs summarizing your take on each PM speech. Your individual role sheet will have more detail regarding your assignments, so check there for more information.

3.6 Counterfactuals

Lardner did lecture on the Difference Engine, and Lovelace did see him speak on the subject. However, his lectures on the subject did not take place as early as 1828. In the game, his lecture on the engine takes place before the second lab and before characters need to make decisions regarding the engine.

Similarly, Ada Lovelace did meet Charles Babbage at one of his Dorset Street parties, but not until June 5th, 1833. Babbage did not pick up residence at 1 Dorset Street until 1831. In our game, Babbage is already at Dorset Street in 1828, and Ada attends the GS 1 party in 1828.

The Swing Riots did not take place until 1830, but in our game they take place in 1828; this allows Captain Swing to credibly speak during GS 2, since he cannot speak during sessions held at the Royal Society.

The period of time the game covers featured a “musical chairs” of Prime Ministers of the U.K. The PM indeterminates in the game all served as PM during this time period, but in our game the order will be random, rather than following the historical record. Not all of the men who were PM during this period are included as characters in the game. In real life, some of these men were not, in fact, indeterminate on Babbage’s machines; in the game, however, each indeterminate is instructed to be truly indeterminate.

In reality, the following characters died before our game ends in 1846: Clement (1844), Frend (1841), Gilbert (1839), Grey (1845), and Sussex (1843). For our purposes, however, these characters remain alive throughout the game.

With the exception of The Fury and Captain Swing, each of the characters in this game is a real person who lived during the era of the game, and the character biographies in the role sheets are historically accurate. However, in many cases, characters in the game perform actions that are “historical fiction” rather than “historic.” In these cases, the characters in the game serve the purposes of the game, so that students can more deeply understand the key issues involved.

Chapter 4

Roles and factions

This section briefly summarizes the two factions in *Charles Babbage, Ada Lovelace, and the Dawn of Computing* game. Each of the roles in each faction, plus each indeterminate role, is briefly described.

4.1 Pro-Babbage faction

The pro-Babbage faction is a collection of men of science, engineers, and personal friends of Charles Babbage. This group is committed to the development of Babbage’s computing engines, recognizing how important they will be. The faction members also recognize that developing the Difference Engine or the Analytical Engine will be expensive, beyond the capacity of private financing.

John Couch Adams John Couch (pronounced “cooch”) Adams is a British astronomer and mathematician, best known for his co-discovery with Urbain Le Verrier of the planet Neptune in 1846.

Charles Babbage FRS Charles Babbage is a British polymath – a person whose expertise spans a significant number of different subject areas – and the inventor of two types of calculating machines. First, Babbage invented the Difference Engine, a device capable of automatically calculating and printing certain types of mathematical tables, and then the Analytical Engine, a true proto-computer.

Isambard Kingdom (I. K.) Brunel FRS I. K. Brunel is perhaps the most accomplished mechanical and civil engineer in Britain during the time of our game. Brunel, a close personal friend of Charles Babbage, is responsible for completing the Thames Tunnel project, for building the Great Western Railway, and for the construction of the steamship S. S. Great Western.

Joseph Clement Joseph Clement is the extremely capable toolmaker, draftsman, engineer, and industrialist hired by Charles Babbage to build his Difference Engine.

Andrew Crosse Andrew Crosse is an amateur man of science, best known for his experiments with electricity. Mary Shelley, the author of the novel *Frankenstein*, is an acquaintance, and her husband Percy Bysshe Shelley attended one of Crosse's lectures on atmospheric electricity in 1814; this may or may not have helped to influence later editions of the famous novel.

Michael Faraday Michael Faraday is an extremely influential English man of science; of all the characters in this game, his influence is the greatest in our 21st-century world. His work focuses on chemistry and electricity.

John Frederick William Herschel FRS John Frederick William Herschel is another British polymath, and close friend of Charles Babbage. He and Babbage, among others, formed the Analytical Society while undergraduate students at Cambridge. Herschel studied mathematics, astronomy, chemistry, botany, and more. Herschel is the leader of the pro-Babbage faction.

Dionysius Lardner FRS Dionysius Lardner is a member of the Royal Society of London and a professor at the University of London. Lardner is a scientific populist and the first scientific journalist.

Augusta Ada King-Noel, Countess of Lovelace Augusta Ada King-Noel, Countess of Lovelace, is the only legitimate daughter of the “mad, bad, and dangerous to know” romantic poet, Lord Byron. Lovelace is a close friend of Charles Babbage, and in 1843, she translates an article on Babbage's Analytical Engine by Luigi Federico Menabrea, and adds her own voluminous notes to the translation. The notes contain what is now considered the world's first computer program.

Mary Somerville Mary Fairfax Greig Somerville is a Scottish polymath. She is very unusual for the time, having published several very influential papers and books. Somerville served as one of Ada Lovelace's tutors, and introduced her to Charles Babbage at one of his Dorset Street parties.

4.2 Anti-Babbage faction

The anti-Babbage faction is a collection of men of science, clergy, and political figures. This group is against the British government paying any more money for the development of Babbage's computing engines. Anti-Babbage faction members have different reasons for opposing the engines: some do not think the engines will be valuable or necessary for the advancement of science in Britain; some think the engines are potentially valuable, but do not think that projects such as this should be supported by the government; and some are generally against the increasing mechanization of British society, as it forces working people from their jobs.

George Biddell Airy FRS George Biddell Airy is an English mathematician and, as of 1835, the Astronomer Royal. Airy, holder of the Lucasian Professorship of Mathematics at Cambridge before Charles Babbage assumed the chair in 1828, is the leader

of the anti-Babbage faction.

Reverend William Buckland DD FRS The Reverend Doctor William Buckland, DD FRS, is an English clergyman, theologian, geologist, and paleontologist. Buckland published the first full account of a fossilized dinosaur, which he termed *Megalosaurus*.

William Frend William Frend is an English unitarian clergyman, political radical, and author of the algebra textbook, *Principles of Algebra*. Frend, along with Mary Somerville, served as one of Ada Lovelace's mathematics tutors.

Davies Gilbert PRS Davies Gilbert is a British botanist, geologist, mathematician, civil servant, politician, author, and from 1827 through 1830, president of the Royal Society of London.

Edward Sabine FRS Edward Sabine is an Irish military man, geologist, and astronomer. He served as the on-board astronomer on expeditions to attempt to discover the Northwest Passage under Sir John Ross in 1818 and again under Sir William Edward Parry in 1819 and 1820.

Prince Augustus Frederick, Duke of Sussex, KG KT GCB FRS FRSA Prince Augustus Frederick, Duke of Sussex, is the sixth son of “mad” King George III, who reigned from 1760 until 1820. Although not a man of science, Sussex does have intellectual pursuits, chiefly in biblical studies and the Hebrew language.

Captain Swing Captain Swing represents the spirit and ideas of the farmers who participated in the “Swing Riots” across England in 1830 and 1831, starting fires and breaking threshing machines, protesting for increased wages, against restrictive Poor Laws, and against the increasing use of threshing machines in agriculture. Several of the threatening letters sent by the protesters were signed with the *nom de guerre* “Captain Swing.”

4.3 Indeterminates

In this game, the characters that are Members of Parliament (MPs) will serve in the role of Prime Minister. They are indeterminate. That means that these characters have no pre-decided position regarding Babbage's engines, although each character does have some general political leanings that may inform his opinions on the subject. Other indeterminates (Byron's Ghost, the Fury, and Doctor William King) will not serve as PM, but will have things to say about other issues in the game.

The Ghost of Lord Byron Lord Byron (1788 - 1824) was an immensely famous and influential British poet, described as “mad, bad, and dangerous to know.” Ada Lovelace is Byron's only legitimate child. Byron's Ghost will work behind the scenes on behalf of the Romantic Movement, especially in regard to its veneration of nature, beauty, and imagination over industrialization, organized religion, rationalism, and social convention. Byron's Ghost is not eligible to serve as PM.

The Fury The Fury is an amalgam of the women hired by Ada Lovelace's mother, Lady Byron, Anne Isabella Noel Byron, to help to raise Ada. The Fury's primary purpose in the game is to ensure that everyone comports themselves in a manner befitting ladies and gentlemen of early Victorian Britain. The Fury is not eligible to serve as PM.

Lord Charles Grey, 2nd Earl Grey, KG PC Lord Charles Grey, 2nd Earl Grey is a Whig MP, and the namesake of Earl Grey tea.

Doctor William King Doctor William William King (not William King-Noel, 1st Earl of Lovelace, Ada Lovelace's husband) was Lovelace's physician and mentor. In this game, King will speak on the supremacy of reason over imagination. King is not eligible to serve as PM.

Sir Robert Peel, 2nd Baronet, FRS PC Sir Robert Peel, 2nd Baronet is a Tory MP.

Lord Frederick John Robinson, 1st Earl of Ripon, FRS PC Lord Frederick John Robinson, 1st Earl of Ripon is a Tory MP.

Lord John Russell, 1st Earl Russell, KG GCMG PC Lord John Russell, 1st Earl Russell is a Whig MP.

Field Marshal Arthur Wellesley, 1st Duke of Wellington, KG GCB GCH PC Field Marshal Arthur Wellesley, 1st Duke of Wellington is the biggest hero of the French Wars and Tory MP.

Chapter 5

Core texts

5.1 Passages from the Life of a Philosopher

CHAPTER II. CHILDHOOD.

FROM my earliest years I had a great desire to inquire into the causes of all those little things and events which astonish the childish mind. At a later period I commenced the still more important inquiry into those laws of thought and those aids which assist the human mind in passing from received knowledge to that other knowledge then unknown to our race. I now think it fit to record some of those views to which, at various periods of my life, my reasoning has led me. Truth only has been the object of my search, and I am not conscious of ever having turned aside in my inquiries from any fear of the conclusions to which they might lead.

As it may be interesting to some of those who will hereafter read these lines, I shall briefly mention a few events of my earliest, and even of my childish years. My parents being born at a certain period of history, and in a certain latitude and longitude, of course followed the religion of their country. They brought me up in the Protestant form of the Christian faith. My excellent mother taught me the usual forms of my daily and nightly prayer; and neither in my father nor my mother was there any mixture of bigotry and intolerance on the one hand, nor on the other of that unbecoming and familiar mode of addressing the Almighty which afterwards so much disgusted me in my youthful years.

Earliest Recollections.

Two events which impressed themselves forcibly on my memory happened, I think, previously to my eighth year.

When about five years old, I was walking with my nurse, who had in her arms an infant brother of mine, across London Bridge, holding, as I thought, by her apron. I was looking at the ships in the river. On turning round to speak to her, I found that my nurse was not there, and that I was alone upon London Bridge. My mother had always impressed upon me the necessity of great caution in passing any street-crossing: I went on therefore quietly

until I reached Tooley Street, where I remained watching the passing vehicles, in order to find a safe opportunity of crossing that very busy street.

In the mean time the nurse, having lost one of her charges, had gone to the crier, who proceeded immediately to call, by the ringing of his bell, the attention of the public to the fact that a young philosopher was lost, and to the still more important fact that five shillings would be the reward of his fortunate discoverer. I well remember sitting on the steps of the door of the linendraper's shop on the opposite corner of Tooley Street, when the gold-laced crier was making proclamation of my loss; but I was too much occupied with eating some pears to attend to what he was saying.

The fact was, that one of the men in the linendraper's shop, observing a little child by itself, went over to it, and asked what it wanted. Finding that it had lost its nurse, he brought it across the street, gave it some pears, and placed it on the steps at the door: having asked my name, the shopkeeper found it to be that of one of his own customers. He accordingly sent off a messenger, who announced to my mother the finding of young Pickle before she was aware of his loss.

Those who delight in observing coincidences may perhaps account for the following singular one. Several years ago when the houses in Tooley Street were being pulled down, I believe to make room for the new railway terminus, I happened to pass along the very spot on which I had been lost in my infancy. A slate of the largest size, called a Duchess ¹ was thrown from the roof of one of the houses, and penetrated into the earth close to my feet.

The other event, which I believe happened some time after the one just related, is as follows. I give it from memory, as I have always repeated it.

I was walking with my nurse and my brother in a public garden, called Montpelier Gardens, in Walworth. On returning through the private road leading to the gardens, I gathered and swallowed some dark berries very like black currants: – these were poisonous.

On my return home, I recollect being placed between my father's knees, and his giving me a glass of castor oil, which I took from his hand.

My father at that time possessed a collection of pictures. He sat on a chair on the right hand side of the chimney-piece in the breakfast room, under a fine picture of our Saviour taken down from the cross. On the opposite wall was a still celebrated "Interior of Antwerp Cathedral."

In after life I several times mentioned the subject both to my father and to my mother; but neither of them had the slightest recollection of the matter.

Having suffered in health at the age of five years, and again at that of ten by violent fevers, from which I was with difficulty saved, I was sent into Devonshire and placed under the care of a clergyman (who kept a school at Alphington, near Exeter) with instructions to attend to my health; but, not to press too much knowledge upon me: a mission which

¹There exists an aristocracy even amongst slates, perhaps from their occupying the most *elevated* position in every house. Small ones are called Ladies, a larger size Countesses, and the biggest of all are Duchesses.

he faithfully accomplished. Perhaps great idleness may have led to some of my childish reasonings.

Relations of ghost stories often circulate amongst children, and also of visitations from the devil in a *personal* form. Of course I shared the belief of my comrades, but still had some doubts of the existence of these personages, although I greatly feared their appearance. Once, in conjunction with a companion, I frightened another boy, bigger than myself, with some pretended ghost; how prepared or how represented by natural objects I do not now remember: I believe it was by the accidental passing shadows of some external objects upon the walls of our common bedroom.

The effect of this on my playfellow was painful; he was much frightened for several days; and it naturally occurred to me, after some time, that as I had deluded him with ghosts, I might myself have been deluded by older persons, and that, after all, it might be a doubtful point whether ghost or devil ever really existed. I gathered all the information I could on the subject from the other boys, and was soon informed that there was a peculiar process by which the devil might be raised and become personally visible. I carefully collected from the traditions of different boys the visible forms in which the Prince of Darkness had been recorded to have appeared. Amongst them were –

A rabbit,
An owl,
A black cat, very frequently,
A raven,
A man with a cloven foot, also frequent.

After long thinking over the subject, although checked by a belief that the inquiry was wicked, my curiosity at length over-balanced my fears, and I resolved to attempt to raise the devil. Naughty people, I was told, had made written compacts with the devil, and had signed them with their names written in their own blood. These had become very rich and great men during their life, a fact which might be well known. But, after death, they were described as having suffered and continuing to suffer physical torments throughout eternity, another fact which, to my uninstructed mind, it seemed difficult to prove.

As I only desired an interview with the gentleman in black simply to convince my senses of his existence, I declined adopting the legal forms of a bond, and preferred one more resembling that of leaving a visiting card, when, if not at home, I might expect the satisfaction of a return of the visit by the devil in person.

Accordingly, having selected a promising locality, I went one evening towards dusk up into a deserted garret. Having closed the door, and I believe opened the window, I proceeded to cut my finger and draw a circle on the floor with the blood which flowed from the incision.

I then placed myself in the centre of the circle, and either said or read the Lord's Prayer backwards. This I accomplished at first with some trepidation and in great fear towards the close of the scene. I then stood still in the centre of that magic and superstitious circle, looking with intense anxiety in all directions, especially at the window and at the chimney.

Fortunately for myself, and for the reader also, if he is interested in this narrative, no owl or black cat or unlucky raven came into the room.

In either case my then weakened frame might have expiated this foolish experiment by its own extinction, or by the alienation of that too curious spirit which controlled its feeble powers.

After waiting some time for my expected but dreaded visitor, I, in some degree, recovered my self-possession, and leaving the circle of my incantation, I gradually opened the door and gently closing it, descended the stairs, at first slowly, and by degrees much more quickly. I then rejoined my companions, but said nothing whatever of my recent attempt. After supper the boys retired to bed. When we were in bed and the candle removed, I proceeded as usual to repeat my prayers silently to myself. After the few first sentences of the Lord's Prayer, I found that I had forgotten a sentence, and could not go on to the conclusion. This alarmed me very much, and having repeated another prayer or hymn, I remained long awake, and very unhappy. I thought that this forgetfulness was a punishment inflicted upon me by the Almighty, and that I was a wicked little boy for having attempted to satisfy myself about the existence of a devil. The next night my memory was more faithful, and my prayers went on as usual. Still, however, I was unhappy, and continued to brood over the inquiry. My uninstructed faculties led me from doubts of the existence of a devil to doubts of the book and the religion which asserted him to be a living being. My sense of justice (whether it be innate or acquired) led me to believe that it was impossible that an almighty and all-merciful God could punish me, a poor little boy, with eternal torments because I had anxiously taken the only means I knew of to verify the truth or falsehood of the religion I had been taught. I thought over these things for a long time, and, in my own childish mind, wished and prayed that God would tell me what was true. After long meditation, I resolved to make an experiment to settle the question. I thought, if it was really of such immense importance to me here and hereafter to believe rightly, that the Almighty would not consign me to eternal misery because, after trying all means that I could devise, I was unable to know the truth. I took an odd mode of making the experiment; I resolved that at a certain hour of a certain day I would go to a certain room in the house, and that if I found the door open, I would believe the Bible; but that if it were closed, I should conclude that it was not true. I remember well that the observation was made, but I have no recollection as to the state of the door. I presume it was found open from the circumstance that, for many years after, I was no longer troubled by doubts, and indeed went through the usual religious forms with very little thought about their origin.

[...]

CHAPTER III. BOYHOOD.

DURING my boyhood my mother took me to several exhibitions of machinery. I well remember one of them in Hanover Square. by a man who called himself Merlin. I was so greatly interested in it, that the Exhibitor remarked the circumstance, and after explaining some of the objects to which the public had access, proposed to my mother to take me up to his workshop, where I should see still more wonderful automata. We accordingly ascended to the attic. There were two uncovered female figures of silver, about twelve

inches high.

One of these walked or rather glided along a space of about four feet, when she turned round and went back to her original place. She used an eye-glass occasionally, and bowed frequently, as if recognizing her acquaintances. The motions of her limbs were singularly graceful.

The other silver figure was an admirable *danseuse*, with a bird on the fore finger of her right hand, which wagged its tail, flapped its wings, and opened its beak. This lady attitudinized in a most fascinating manner. Her eyes were full of imagination, and irresistible.

These silver figures were the chef-d'œuvres of the artist: they had cost him years of unwearyed labour, and were not even then finished.

After I left Devonshire, I was placed at a school in the neighbourhood of London, in which there were about thirty boys.

My first experience was unfortunate, and probably gave an unfavourable turn to my whole career during my residence of three years.

After I had been at school a few weeks, I went with one of my companions into the play-ground in the dusk of the evening. We heard a noise, as of people talking in an orchard at some distance, which belonged to our master. As the orchard had recently been robbed, we thought that thieves were again at work. We accordingly climbed over the boundary wall, ran across the field, and saw in the orchard beyond a couple of fellows evidently running away. We pursued as fast as our legs could carry us, and just got up to the supposed thieves at the ditch on the opposite side of the orchard.

A roar of laughter then greeted us from two of our own companions, who had entered the orchard for the purpose of getting some manure for their flowers out of a rotten mulberry tree. These boys were aware of our mistake, and had humoured it.

We now returned all together towards the play-ground, when we met our master, who immediately pronounced that we were each fined one shilling for being out of bounds. We two boys who had gone out of bounds to protect our master's property, and who if thieves had really been there would probably have been half-killed by them, attempted to remonstrate and explain the case; but all remonstrance was vain, and we were accordingly fined. I never forgot that injustice.

The school-room adjoined the house, but was not directly connected with it. It contained a library of about three hundred volumes on various subjects, generally very well selected; it also contained one or two works on subjects which do not usually attract at that period of life. I derived much advantage from this library; and I now mention it because I think it of great importance that a library should exist in every school-room.

Amongst the books was a treatise on Algebra, called "Ward's Young Mathematician's Guide." I was always partial to my arithmetical lessons, but this book attracted my particular attention. After I had been at this school for about a twelvemonth, I proposed to one of my school-fellows, who was of a studious habit, that we should get up every morning at three o'clock, light a fire in the school-room, and work until five or half-past

five. We accomplished this pretty regularly for several months. Our plan had, however, become partially known to a few of our companions. One of these, a tall boy, bigger than ourselves, having heard of it, asked me to allow him to get up with us, urging that his sole object was to study, and that it would be of great importance to him in after-life. I had the cruelty to refuse this very reasonable request. The subject has often recurred to my memory, but never without regret.

Another of my young companions. Frederick Marryat,² made the same request, but not with the same motive. I told him we got up in order to work; that he would only play, and that we should then be found out. After some time, having exhausted all his arguments, Marryat told me he was determined to get up, and would do it whether I liked it not.

Marryat slept in the same room as myself: it contained five beds. Our room opened upon a landing, and its door was exactly opposite that of the master. A flight of stairs led up to a passage just over the room in which the master and mistress slept. Passing along this passage, another flight of stairs led down, on the other side of the master's bed-room, to another landing, from which another flight of stairs led down to the external door of the house, leading by a long passage to the school-room.

Through this devious course I had cautiously threaded my way, calling up my companion in his room at the top of the last flight of stairs, almost every night for several months.

One night on trying to open the door of my own bed-room, I found Marryat's bed projecting a little before the door, so that I could not open it. I perceived that this was done purposely, in order that I might awaken him. I therefore cautiously, and by degrees, pushed his bed back without awaking him, and went as usual to my work. This occurred two or three nights successively.

One night, however, I found a piece of pack-thread tied to the door lock, which I traced to Marryat's bed, and concluded it was tied to his arm or hand. I merely untied the cord from the lock, and passed on.

A few nights after I found it impossible to untie the cord, so I cut it with my pocket-knife. The cord then became thicker and thicker for several nights, but still my pen-knife did its work.

One night I found a small chain fixed to the lock, and passing thence into Marryat's bed. This defeated my efforts for that night, and I retired to my own bed. The next night I was provided with a pair of plyers, and unbent one of the links, leaving the two portions attached to Marryat's arm and to the lock of the door. This occurred several times, varying by stouter chains, and by having a padlock which I could not pick in the dark.

At last one morning I found a chain too strong for the tools I possessed; so I retired to my own bed, defeated. The next night, however, I provided myself with a ball of pack-thread. As soon as I heard by his breathing that Marryat was asleep, I crept over to the door, drew one end of my ball of packthread through a link of the too-powerful chain, and bringing it back with me to bed, gave it a sudden jerk by pulling both ends of the packthread passing through the link of the chain.

²Afterwards Captain Marryat.

Marryat jumped up, put out his hand to the door, found his chain all right, and then lay down. As soon as he was asleep again, I repeated the operation. Having awakened him for the third time, I let go one end of the string, and drew it back by the other, so that he was unable at daylight to detect the cause.

At last, however, I found it expedient to enter into a treaty of peace, the basis of which was that I should allow Marryat to join the night party; but that nobody else should be admitted. This continued for a short time; but, one by one, three or four other boys, friends of Marryat, joined our party, and, as I had anticipated, no work was done. We all got to play; we let off fire-works in the play-ground, and were of course discovered.

Our master read us a very grave lecture at breakfast upon the impropriety of this irregular system of turning night into day, and pointed out its injurious effects upon the health. This, he said, was so remarkable that he could distinguish by their pallid countenances those who had taken part in it. Now he certainly did point out every boy who had been up on the night we were detected. But it appeared to me very odd that the same means of judging had not enabled him long before to discover the two boys who had for several months habitually practised this system of turning night into day.

Another of our pranks never received its solution in our master's mind: indeed I myself scarcely know its early history. Somehow or other, a Russian young gentleman, who was a parlour-boarder, had I believe, expatiated to Marryat on the virtues of Cognac.

One evening my friend came to me with a quart bottle of what he called excellent stuff. A council was held amongst a few of us boys to decide how we should dispose of this treasure. I did not myself much admire the liquid, but suggested that it might be very good when mixed up with a lot of treacle. This thought was unanimously adopted, and a subscription made to purchase the treacle. Having no vessel sufficiently large to hold the intended mixture, I proposed to take one of our garden-pots, stopping up the hole in its bottom with a cork.

A good big earthen vessel, thus extemporised, was then filled with this wonderful mixture. A spoon or two, an oyster-shell and various other contrivances delivered it to its numerous consumers, and all the boys got a greater or less share, according to their taste for this extraordinary liqueur.

The feast was over, the garden-pot was restored to its owner, and the treacled lips of the boys had been wiped with their handkerchiefs or on their coat-sleeves, when the bell announced that it was prayer-time. We all knelt in silence at our respective desks. As soon as the prayers were over, one of the oddest scenes occurred.

Many boys rose up from their knees – but some fell down again. Some turned round several times, and then fell. Some turned round so often that they resembled spinning dervishes. Others were only more stupid than usual, some complained of being sick; many were very sleepy; others were sound asleep, and had to be carried to bed; some talked fast and heroically, two attempted psalmody, but none listened.

All investigation at the time was useless: we were sent off to bed as quickly as possible. It was only known that Count Cognac had married the sweet Miss Treacle, whom all the

boys knew and loved, and who lodged at the grocer's, in the neighbouring village. But I believe neither the pedigree of the bridegroom nor his domicile were ever discovered. It is probable that he was of French origin, and dwelt in a cellar.

[...]

CHAPTER IV. CAMBRIDGE.

[...]

Being passionately fond of algebra, I had instructed myself by means of Ward's "Young Mathematician's Guide," which had casually fallen into my hands at school. I now employed all my leisure in studying such mathematical works as accident brought to my knowledge. Amongst these were Humphrey Ditton's "Fluxions," of which I could make nothing; Madame Agnesi's "Analytical Institutions," from which I acquired some knowledge; Woodhouse's "Principles of Analytical Calculation," from which I learned the notation of Leibnitz; and Lagrange's "Théorie des Fonctions." I possessed also the Fluxions of Maclaurin and of Simpson.

Thus it happened that when I went to Cambridge I could work out such questions as the very moderate amount of mathematics which I then possessed admitted, with equal facility, in the dots of Newton the d's of Leibnitz, or the dashes of Lagrange. I had, however, met with many difficulties, and looked forward with intense delight to the certainty of having them all removed on my arrival at Cambridge. I had in my imagination formed a plan for the institution amongst my future friends of a chess club, and also of another club for the discussion of mathematical subjects.

In 1811, during the war, it was very difficult to procure foreign books. I had heard of the great work of Lacroix, on the "Differential and Integral Calculus," which I longed to possess, and being misinformed that its price was two guineas, I resolved to purchase it in London on my passage to Cambridge. As soon as I arrived I went to the French bookseller, Dulau, and to my great surprise found that the price of the book was seven guineas. After much thought I made the costly purchase, went on immediately to Cambridge, saw my tutor Hudson, got lodgings, and then spent the greater part of the night in turning over the pages of my newly-acquired purchase. After a few days, I went to my public tutor Hudson, to ask the explanation of one of my mathematical difficulties. He listened to my question, said it would not be asked in the Senate House, and was of no sort of consequence, and advised me to get up the earlier subjects of the university studies.

After some little while I went to ask the explanation of another difficulty from one of the lecturers. He treated the question just in the same way. I made a third effort to be enlightened about what was really a doubtful question, and felt satisfied that the person I addressed knew nothing of the matter, although he took some pains to disguise his ignorance.

I thus acquired a distaste for the routine of the studies of the place, and devoured the papers of Euler and other mathematicians, scattered through innumerable volumes of the academies of Petersburg, Berlin, and Paris, which the libraries I had recourse to contained.

Under these circumstances it was not surprising that I should perceive and be penetrated with the superior power of the notation of Leibnitz.

At an early period, probably at the commencement of the second year of my residence at Cambridge, a friend of mine, Michael Slegg, of Trinity, was taking wine with me, discussing mathematical subjects, to which he also was enthusiastically attached. Hearing the chapel bell ring, he took leave of me, promising to return for a cup of coffee.

At this period Cambridge was agitated by a fierce controversy. Societies had been formed for printing and circulating the Bible. One party proposed to circulate it with notes, in order to make it intelligible; whilst the other scornfully rejected all explanations of the word of God as profane attempts to mend that which was perfect.

The walls of the town were placarded with broadsides, and posters were sent from house to house. One of the latter form of advertisement was lying upon my table when Slegg left me. Taking up the paper, and looking through it, I thought it, from its exaggerated tone, a good subject for a parody.

I then drew up the sketch of a society to be instituted for translating the small work of Lacroix on the Differential and Integral Lacroix. It proposed that we should have periodical meetings for the propagation of d's; and consigned to perdition all who supported the heresy of dots. It maintained that the work of Lacroix was so perfect that any comment was unnecessary.

On Slegg's return from chapel I put the parody into his hands. My friend enjoyed the joke heartily, and at parting asked my permission to show the parody to a mathematical friend of his, Mr. Bromhead³.

The next day Slegg called on me, and said that he had put the joke into the hand of his friend, who after laughing heartily, remarked that it was too good a joke to be lost, and proposed seriously that we should form a society for the cultivation of mathematics.

The next day Bromhead called on me. We talked the subject over, and agreed to hold a meeting at his lodgings for the purpose of forming a society for the promotion of analysis.

At that meeting, besides the projectors, there were present Herschel, Peacock, D'Arblay⁴, Ryan⁵, Robinson⁶, Frederick Maule⁷, and several others. We constituted ourselves "The Analytical Society;" hired a meeting-room, open daily; held meetings, read papers, and discussed them. Of course we were much ridiculed by the Dons; and, not being put down, it was darkly hinted that we were young infidels, and that no good would come of us.

In the meantime we quietly pursued our course, and at last resolved to publish a volume of our Transactions. Owing to the illness of one of the number, and to various other

³Afterwards Sir Edward Ffrench Bromhead, Bart., the author of an interesting paper in the Transactions of the Royal Society.

⁴The only son of Madame D'Arblay.

⁵Now the Right Honourable Sir Edward Ryan.

⁶The Rev. Dr. Robinson, Master of the Temple.

⁷A younger brother of the late Mr. Justice Maule.

circumstances, the volume which was published was entirely contributed by Herschel and myself.

At last our work was printed, and it became necessary to decide upon a title. Recalling the slight imputation which had been made upon our faith, I suggested that the most appropriate title would be –

The Principles of pure D-ism in opposition to the Dot-age of the University⁸.

In thus reviving this wicked pun, I ought at the same time to record an instance of forgiveness unparalleled in history. Fourteen years after, being then at Rome, I accidentally read in Galignani's newspaper the following paragraph, dated Cambridge: – "Yesterday the bells of St. Mary rang on the election of Mr. Babbage as Lucasian Professor of Mathematics."

[...]

My views respecting the notation of Leibnitz now (1812) received confirmation from an extensive course of reading. I became convinced that the notation of fluxions must ultimately prove a strong impediment to the progress of English science. But I knew, also, that it was hopeless for any young and unknown author to attempt to introduce the notation of Leibnitz into an elementary work. This opinion naturally suggested to me the idea of translating the smaller work of Lacroix. It is possible, although I have no recollection of it, that the same idea may have occurred to several of my colleagues of the Analytical Society, but most of them were so occupied, first with their degree, and then with their examination for fellowships, that no steps were at that time taken by any of them on that subject.

Unencumbered by these distractions, I commenced the task, but at what period of time I do not exactly recollect. I had finished a portion of the translation, and laid it aside, when, some years afterwards, Peacock called on me in Devonshire Street, and stated that both Herschel and himself were convinced that the change from the dots to the d's would not be accomplished until some foreign work of eminence should be translated into English. Peacock then proposed that I should either finish the translation which I had commenced, or that Herschel and himself should complete the remainder of my translation. I suggested that we should toss up which alternative to take. It was determined by lot that we should make a joint translation. Some months after, the translation of the small work of Lacroix was published.

For several years after, the progress of the notation of Leibnitz at Cambridge was slow. It is true that the tutors of the two largest colleges had adopted it, but it was taught at none of the other colleges.

It is always difficult to think and reason in a new language, and this difficulty discouraged all but men of energetic minds. I saw, however, that by making it their interest to do so, the change might be accomplished. I therefore proposed to make a large collection of examples of the differential and integral calculus, consisting merely of the statement of each problem and its final solution. I foresaw that if such a publication existed, all those tutors who did

⁸Leibnitz indicated fluxions by a *d*, Newton by a dot.

not approve of the change of the Newtonian notation would yet, in order to save their own time and trouble, go to this collection of examples to find problems to set to their pupils. After a short time the use of the new signs would become familiar, and I anticipated their general adoption at Cambridge as a matter of course.

I commenced by copying out a large portion of the work of Hirsch. I then communicated to Peacock and Herschel my view, and proposed that they should each contribute a portion.

Peacock considerably modified my plan by giving the process of solution to a large number of the questions. Herschel prepared the questions in finite differences, and I supplied the examples to the calculus of functions. In a very few years the change was completely established; and thus at last the English cultivators of mathematical science, untrammelled by a limited and imperfect system of signs, entered on equal terms into competition with their continental rivals.

CHAPTER V. DIFFERENCE ENGINE NO. 1.

CALCULATING Machines comprise various pieces of mechanism for assisting the human mind in executing the operations of arithmetic. Some few of these perform the whole operation without any mental attention when once the given numbers have been put into the machine.

Others require a moderate portion of mental attention: these latter are generally of much simpler construction than the former, and it may also be added, are less useful.

The simplest way of deciding to which of these two classes any calculating machine belongs is to ask its maker – Whether, when the numbers on which it is to operate are placed in the instrument, it is capable of arriving at its result by the mere motion of a spring, a descending weight, or any other constant force? If the answer be in the affirmative, the machine is really automatic; if otherwise, it is not self acting.

Of the various machines I have had occasion to examine, many of those for Addition and Subtraction have been found to be automatic. Of machines for Multiplication and Division, which have fully come under my examination, I cannot at present recall one to my memory as absolutely fulfilling this condition.

The earliest idea that I can trace in my own mind of calculating arithmetical Tables by machinery arose in this manner: –

One evening I was sitting in the rooms of the Analytical Society, at Cambridge, my head leaning forward on the Table in a kind of dreamy mood, with a Table of logarithms lying open before me. Another member, coming into the room, and seeing me half asleep, called out, “Well, Babbage, what are you dreaming about?” to which I replied, “I am thinking that all these Tables (pointing to the logarithms) might be calculated by machinery.”

I am indebted to my friend, the Rev. Dr. Robinson, the Master of the Temple, for this anecdote. The event must have happened either in 1812 or 1813.

About 1819 I was occupied with devising means for accurately dividing astronomical in-

struments, and had arrived at a plan which I thought was likely to succeed perfectly. I had also at that time been speculating about making machinery to compute arithmetical Tables.

One morning I called upon the late Dr. Wollaston, to consult him about my plan for dividing instruments. On talking over the matter, it turned out that my system was exactly that which had been described by the Duke de Chaulnes, in the Memoirs of the French Academy of Sciences, about fifty or sixty years before. I then mentioned my other idea of computing Tables by machinery, which Dr. Wollaston thought a more promising subject.

I considered that a machine to execute the mere isolated operations of arithmetic would be comparatively of little value, unless it were very easily set to do its work, and unless it executed not only accurately, but with great rapidity, whatever it was required to do.

On the other hand, the method of differences supplied a general principle by which *all* Tables might be computed through limited intervals, by one uniform process. Again, the method of differences required the use of mechanism for Addition only. In order, however, to insure accuracy in the printed Tables, it was necessary that the machine which computed Tables should also set them up in type, or else supply a mould in which stereotype plates of those Tables could be cast.

I now began to sketch out arrangements for accomplishing the several partial processes which were required. The arithmetical part must consist of two distinct processes – the power of adding one digit to another, and also of carrying the tens to the next digit, if it should be necessary.

The first idea was, naturally, to add each digit successively. This, however, would occupy much time if the numbers added together consisted of many places of figures.

The next step was to add all the digits of the two numbers each to each at the same instant, but reserving a certain mechanical memorandum, wherever a carriage became due. These carriages were then to be executed successively.

Having made various drawings, I now began to make models of some portions of the machine, to see how they would act. Each number was to be expressed upon wheels placed upon an axis; there being one wheel for each figure in the number operated upon.

Having arrived at a certain point in my progress, it became necessary to have teeth of a peculiar form cut upon these wheels. As my own lathe was not fit for this job, I took the wheels to a wheel-cutter at Lambeth, to whom I carefully conveyed my instructions, leaving with him a drawing as his guide.

These wheels arrived late one night, and the next morning I began putting them in action with my other mechanism, when, to my utter astonishment, I found they were quite unfit for their task. I examined the shape of their teeth, compared them with those in the drawings, and found they agreed perfectly; yet they could not perform their intended work. I had been so certain of the truth of my previous reasoning, that I now began to be somewhat uneasy. I reflected that, if the reasoning about which I had been so certain should prove

to have been really fallacious, I could then no longer trust the power of my own reason. I therefore went over with my wheels to the artist who had formed the teeth, in order that I might arrive at some explanation of this extraordinary contradiction.

On conferring with him, it turned out that, when he had understood fully the peculiar form of the teeth of wheels, he discovered that his wheel-cutting engine had not got amongst its divisions that precise number which I had required. He therefore had asked me whether another number, which his machine possessed, would not equally answer my object. I had inadvertently replied in the affirmative. He then made arrangements for the precise number of teeth I required; and the new wheels performed their expected duty perfectly.

The next step was to devise means for printing the tables to be computed by this machine. My first plan was to make it put together moveable type. I proposed to make metal boxes, each containing 3,000 types of one of the ten digits. These types were to be made to pass out one by one from the bottom of their boxes when required by the computing part of the machine.

But here a new difficulty arose. The attendant who put the types into the boxes might, by mistake, put a wrong type in one or more of them. This cause of error I removed in the following manner: – There are usually certain notches in the side of the type. I caused these notches to be so placed that all the types of any given digit possessed the same characteristic notches, which no other type had. Thus, when the boxes were filled, by passing a small wire down these peculiar notches, it would be impeded in its passage, if there were included in the row a single wrong figure. Also, if any digit were accidentally turned upside down, it would be indicated by the stoppage of the testing wire.

One notch was reserved as common to every species of type. The object of this was that, before the types which the Difference Engine had used for its computation were removed from the iron platform on which they were placed, a steel wire should be passed through this common notch, and remain there. The tables, composed of moveable types, thus interlocked, could never have any of their figures drawn out by adhesion to the inking roller, and then by possibility be restored in an inverted order. A small block of such figures tied together by a bit of string, remained unbroken for several years although it was rather roughly used as a plaything by my children. One such box was finished, and delivered its type satisfactorily.

Another plan for printing the tables, was to place the ordinary printing type round the edges of wheels. Then, as each successive number was produced by the arithmetical part, the type-wheels would move down upon a plate of soft composition, upon which the tabular number would be impressed. This mould was formed of a mixture of plaste- of-Paris with other materials, so as to become hard in the course of a few hours.

The first difficulty arose from the impression of one number on the mould being distorted by the succeeding one.

I was not then aware that a very slight depth of impression from the type would be quite sufficient. I surmounted the difficulty by previously passing a roller, having longitudinal wedge-shaped projections, over the plastic material. This formed a series of small depres-

sions in the matrix between each line. Thus the expansion arising from the impression of one line partially filled up the small depression or ditch which occurred between each successive line.

The various minute difficulties of this kind were successively overcome; but subsequent experience has proved that the depth necessary for stereotype moulds is very small, and that even thick paper, prepared in a peculiar manner, is quite sufficient for the purpose.

Another series of experiments were, however, made for the purpose of punching the computed numbers upon copper plate. A special machine was contrived and constructed, which might be called a co-ordinate machine, because it moved the copper plate and steel punches in the direction three rectangular co-ordinates. This machine was afterwards found very useful for many other purposes. It was, in fact, a general shaping machine, upon which many parts of the Difference Engine were formed.

[...]

I have proposed and drawn various machines for the purpose of calculating a series of numbers forming Tables by means of a certain system called "The Method of Differences," which it is the object of this sketch to explain.

The first Difference Engine with which I am acquainted comprised a few figures, and was made by myself, between 1820 and June 1822. It consisted of from six to eight figures. A much larger and more perfect engine was subsequently commenced in 1823 for the Government.

[...]

Explanation of the Difference Engine.

Those who are only familiar with ordinary arithmetic may, by following out with the pen some of the examples which will be given, easily make themselves acquainted with the simple principles on which the Difference Engine acts.

It is necessary to state distinctly at the outset, that the Difference Engine is not intended to answer special questions. Its object is to calculate and print a *series* of results formed according to given laws. These are called Tables – many such are in use in various trades. For example – there are collections of Tables of the amount of any number of pounds from 1 to 100 lbs. of butchers' meat at various prices per lb. Let us examine one of these Tables: viz. – the price of meat 5*d* per lb we find

Number.	Table.
Lbs.	Price.
	s. d.
1	0 5
2	0 10
3	1 3
4	1 8
5	2 1

There are two ways of computing this Table: –

1st. We might have multiplied the number of lbs. in each line by 5, the price per lb., and have put down the result in *l. s. d.* as in the 2nd column: or,

2nd. We might have put down the price of 1 lb., which is *5d.*, and have added five pence for each succeeding lb.

Let us now examine the relative advantages of each plan. We shall find that if we had multiplied each number of lbs. in the Table by 5, and put down the resulting amount, then every number in the Table would have been computed independently. If, therefore, an error had been committed, it would not have affected any but the single tabular number at which it had been made. On the other hand, if a single error had occurred in the system of computing by adding five at each step, any such error would have rendered the whole of the rest of the Table untrue.

Thus the system of calculating by differences, which is the easiest, is much more liable to error. It has, on the other hand, this great advantage: viz., that when the Table has been so computed, if we calculate its last term directly, and if it agree with the last term found by the continual addition of 5, we shall then be quite certain that every term throughout is correct. In the system of computing each term directly, we possess no such check upon our accuracy.

Now the Table we have been considering is, in fact, merely a Table whose first difference is constant and equal to five. If we express it in pence it becomes –

	Table.	1st Difference.
1	5	5
2	10	5
3	15	5
4	20	5
5	25	

Any machine, therefore, which could add one number to another, and at the same time retain the original number called the first difference for the next operation, would be able to compute all such Tables.

Let us now consider another form of Table which might readily occur to a boy playing with his marbles, or to a young lady with the balls of her solitaire board.

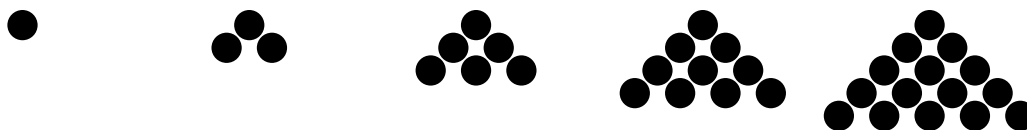
The boy may place a row of his marbles on the sand, equal distances from each other, thus –



He might then, beginning with the second, place two other marbles under each, thus –



He might then, beginning with the third, place three other marbles under each group, and so on; commencing always one group later, and making the addition one marble more each time. The several groups would stand thus arranged –



He will not fail to observe that he has thus formed a series of triangular groups, every group having an equal number of marbles in each of its three sides. Also that the side of each successive group contains one more marble than that of its preceding group.

Now an inquisitive boy would naturally count the numbers in each group and he would find them thus—

1 3 6 10 15 21

He might also want to know how many marbles the thirtieth or any other distant group might contain. Perhaps he might go to papa to obtain this information; but I much fear papa would snub him, and would tell him that it was nonsense — that it was useless — that nobody knew the number, and so forth. If the boy is told by papa, that he is not able to answer the question, then I recommend him to pay careful attention to whatever that father may at any time say, for he has overcome two of the greatest obstacles to the acquisition of knowledge — inasmuch as he possesses the consciousness that he does not know — and he has the moral courage to avow it⁹.

If papa fail to inform him, let him go to mamma, who will not fail to find means to satisfy her darling's curiosity. In the meantime the author of this sketch will endeavour to lead young friend to make use of his own common sense for the purpose of becoming better acquainted with the figures he has formed with his marbles.

In the case of the Table of the price of butchers meat, it was obvious that it could be formed by adding the same *constant* difference continually to the first term. Now suppose we place the numbers of our groups of marbles in a column, as we did our prices of various weights of meat. Instead of adding a certain difference, as we did in the former case, let us subtract the figures representing each group of marbles from the figures of the succeeding group in the Table. The process will stand thus: —

	Table.	1st Difference.	2nd Difference.
Number of the Group.	Number of Marbles in each Group.	Difference between the number of Marbles in each Group and that in the next.	
1	1	1	1
2	3	2	1
3	6	3	1
4	10	4	1
5	15	5	1
6	21	6	
7	28	7	

⁹The most remarkable instance I ever met with of the distinctness with which any individual perceived the exact boundary of his own knowledge, was that of the late Dr Wollaston.

It is usual to call the third column thus formed *the column of first differences*. It is evident in the present instance that that column represents the natural numbers. But we already know that the first difference of the natural numbers is constant and equal to unity. It appears, therefore, that a Table of these numbers, representing the group of marbles, might be constructed to any extent by mere addition – using the number 1 as the first number of the Table, the number 1 as the first Difference, and also the number 1 as the second Difference, which last always remains constant.

Now as we could find the value of any given number of pounds of meat directly, without going through all the previous part of the Table, so by a somewhat different rule we can find at once the value of any group whose number is given.

Thus, if we require the number of marbles in the fifth group, proceed thus: –

Take the number of the group	.	.	.	.	5
Add 1 to this number, it becomes	.	.	.	.	6
Multiply these numbers together					2) 30
Divide the product by 2					15

This gives 15, the number of marbles in the 5th group.

If the reader will take the trouble to calculate with his pencil the five groups given above, he will soon perceive the general truth of this rule.

We have now arrived at the fact that this Table – like that of the price of butchers' meat – can be calculated by two different methods. By the first, each number of the Table is calculated independently; by the second, the truth of each number depends upon the truth of all the previous numbers.

Perhaps my young friend may now ask me, What is the use of such Tables? Until he has advanced further in his arithmetical studies, he must take for granted that they are of some use. The very Table about which he has been reasoning possesses a special name – it is called a Table of Triangular Numbers. Almost every general collection of Tables hitherto published contains portions of it of more or less extent.

Above a century ago, a volume in small quarto, containing the first 20,000 triangular numbers, was published at the Hague by E. De Joncourt, A.M., and Professor of Philosophy¹⁰. I cannot resist quoting the author's enthusiastic expression of the happiness he enjoyed in composing his celebrated work:

“The Trigonal numbers here to be found, and nowhere else, are exactly elaborate. Let the candid reader make the best of these numbers, and feel (if possible) in perusing my work the pleasure I had in composing it.”

“That sweet joy may arise from such contemplations cannot be denied. Numbers and lines have many charms, unseen by vulgar eyes, and only discovered to the

¹⁰‘On the Nature and Notable Use of the most Simple Trigonal Numbers.’ By E. De Joncourt, at the Hague, 1762

unwearied and respectful sons of Art. In features the serpentine line (who starts not at the name) produces beauty and love; and in numbers, high powers, and humble roots give soft delight.”

“Lo! the raptured arithmetician! Easily satisfied, he asks no Brussels lace, nor a coach and six. To calculate, contents his liveliest desires, and obedient numbers are within his reach.”

I hope my young friend is acquainted with the fact – that the product of any number multiplied by itself is called the square of that number. Thus 36 is the product of 6 multiplied by 6, and 36 is called the square of 6. I would now recommend him to examine the series of square numbers

1, 4, 9, 16, 25, 36, 49, 64, &c.,

and to make for his own instruction, the series of their first and second differences, and then to apply to it the same reasoning which has been already applied to the Table of Triangular Numbers.

When he feels that he has mastered that Table, I shall be happy to accompany mamma’s darling to Woolwich or to Portsmouth, where he will find some practical illustrations of the use of his newly-acquired numbers. He will find scattered about in the Arsenal various heaps of cannon balls, some of them triangular, others square or oblong pyramids.

Looking on the simplest form – the triangular pyramid – he will observe that it exactly represents his own heaps of marbles placed each successively above one another until the top of the pyramid contains only a single ball.

The new series thus formed by the addition of his own triangular numbers is –

Number.	Table.	1st Difference.	2nd Difference	3rd Difference
1	1	3	3	1
2	4	6	4	1
3	10	10	5	1
4	20	15	6	
5	35	21		
6	56			

He will at once perceive that this Table of the number of cannon balls contained in a triangular pyramid can be carried to any extent by simply adding successive differences, the third of which is constant.

The next step will naturally be to inquire how any number in this Table can be calculated by itself. A little consideration will lead him to a fair guess; a little industry will enable him to confirm his conjecture.

It will be observed at p. 58 that in order to find independently any number of the Table of the price of butchers’ meat, the following rule was observed: –

Take the number whose tabular number is required. Multiply it by the first difference. This product is equal to the required tabular number.

Again, at p. 61, the rule for finding any triangular number was: –

Take the number of the group	.	.	.	.	5
Add 1 to this number, it becomes	.	.	.	.	6
Multiply these numbers together					2) 30
Divide the product by 2					15

This gives 15, the number of marbles in the 5th group.

Now let us make a bold conjecture respecting the Table of cannon balls, and try this rule: –

Take the number whose tabular number is

required, say	.	.	.	.	.	.	5
Add 1 to that number	.	.	.	.	.	.	6
Add 1 more to that number	.	.	.	.	.	.	7
Multiply all three numbers together	.	.	.	.	.	.	2) 210
Divide by 2	.	.	.	.	.	.	105

The real number in the 5th pyramid is 35. But the number 105 at which we have arrived is exactly three times as great. If, therefore, instead of dividing by 2 we had divided by 2 and also by 3, we should have arrived at a true result in this instance.

The amended rule is therefore –

Take the number whose tabular number is

required, say	.	.	.	.	.	.	n
Add 1 to it	.	.	.	.	.	.	$n + 1$
Add 1 to this	.	.	.	.	.	.	$n + 2$
Multiply these three numbers together							$n \times (n + 1) \times (n + 2)$
Divide by $1 \times 2 \times 3$							
The result is	.	.	.	.	.	.	$\frac{n(n+1)(n+2)}{6}$

This rule will, upon trial, be found to give correctly every tabular number.

By similar reasoning we might arrive at the knowledge of the number of cannon balls in square and rectangular pyramids. But it is presumed that enough has been stated to enable the reader to form some general notion of the method of calculating arithmetical Tables by differences which are constant.

It may now be stated that mathematicians have discovered that all the Tables most important for practical purposes, such as those relating to Astronomy and Navigation, can, although they may not possess any constant differences, still be calculated in detached portions by that method.

Hence the importance of having machinery to calculate differences, which, if well made, cannot err; and which, if carelessly set, presents in the last term it calculates the power verification of every antecedent term.

*Of the Mechanical Arrangements necessary for computing
Tables by the Method of Differences.*

From the preceding explanation it appears that all Tables may be calculated, to a greater or less extent, by the method of Differences. That method requires, for its successful execution, little beyond mechanical means of performing the arithmetical operation of Addition. Subtraction can, by the aid of a well-known artifice, be converted into Addition.

The process of Addition includes two distinct parts – 1st. The first consists of the addition of any one digit to another digit; 2nd. The second consists in carrying the tens to the next digit above.

Let us take the case of the addition of the two following numbers, in which no carriages occur : –

$$\begin{array}{r} 6023 \\ \underline{1970} \\ 7993 \end{array}$$

It will be observed that, in making this addition, the mind acts by successive steps. The person adding says to himself –

0 and 3 make three, 7 and 2 make nine, 9 and 0 make nine, 1 and 6 make seven.

In the following addition there are several carriages: –

$$\begin{array}{r} 2648 \\ \underline{4564} \\ 7212 \end{array}$$

The person adding says to himself –

4 and 8 make 12: put down 2 and carry one.
1 and 6 are 7 and 4 make 11: put down 1 and carry one.
1 and 5 are 6 and 6 make 12: put down 2 and carry one.
1 and 4 are 5 and 2 make 7: put down 7.

Now the length of time required for adding one number to another is mainly dependent upon the number of figures to be added. If we could tell the average time required by the mind to add two figures together, the time required for adding any given number of figures to another equal number would be found by multiplying that average time by the number of digits in either number.

When we attempt to perform such additions by machinery we might follow exactly the usual process of the human mind. In that case we might take a series of wheels, each having marked on its edges the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. These wheels might be placed above each other upon an axis. The lowest would indicate the units' figure, the next above the tens, and so on, [...]

Several such axes, with their figure wheels, might be placed around a system of central wheels, with which the wheels of any one or more axes might at times be made to gear. Thus the figures on any one axis might, by means of those central wheels, be added to the figure wheels of any other axis.

But it may fairly be expected, and it is indeed of great importance that calculations made by machinery should not merely be exact, but that they should be done in a much shorter time than those performed by the human mind. Suppose there were no tens to carry, as in the first of the two cases; then, if we possessed mechanism capable of adding any one digit to any other in the units' place of figures, a similar mechanism might be placed above it to add the tens' figures, and so on for as many figures as might be required.

But in this case, since there are no carriages, each digit might be added to its corresponding digit at the same time. Thus, the time of adding by means of mechanism, any two numbers, however many figures they might consist of, would not exceed that of adding a single digit to another digit. If this could be accomplished it would render additions and subtractions with numbers having ten, twenty, fifty, or any number of figures, as rapid as those operations are with single figures.

Let us now examine the case in which there were several carriages. Its successive stages may be better explained, thus –

	2 6 4 8
	4 5 8 4
Stages	
1 Add units' figure = 4	2 6 4 2
2 Carry	1
	2 6 5 2
3 Add tens' figure = 8	8
	2 6 3 2
4 Carry	1
	2 7 3 2
5 Add hundreds' figure = 5	5
	2 2 3 2
6 Carry	1
	3 2 3 2
7 Add thousands' figure = 4	4
	7 2 3 2
8 Carry 0. There is no carr.	

Now if, as in this case, all the carriages were known, it would then be possible to make all the additions of digits at the same time, provided we could also record each carriage as it became due. We might then complete the addition by adding, at the same instant, each carriage in its proper. place The process would then stand thus: –

	2 6 4 8	
	4 5 6 4	
Stages		
1	$\left\{ \begin{array}{l} 6 \ 1 \ 0 \ 2 \\ 1 \ 1 \ 1 \end{array} \right.$	Add each digit to the digit above. Record the carriages.
2	$\left\{ \begin{array}{l} 7 \ 2 \ 1 \ 2 \end{array} \right.$	Add the above carriages.

Now, whatever mechanism is contrived for adding any one digit to any other must, of course, be able to add the largest digit, nine, to that other digit. Supposing, therefore, one unit of number to be passed over in one second of time, it is evident that any number of pairs of digits may be added together in nine seconds, and that, when all the consequent carriages are known, as in the above case, it will cost one second more to make those carriages. Thus addition and carriage would be completed in ten seconds, even though the numbers consisted each of a hundred figures.

But, unfortunately, there are multitudes of cases which the carriages that become due are only known in successive periods of time. As an example, add together the two following

numbers: –

	8	4	7	3
	1	5	2	8
Stages				
1 Add all the digits	9	9	9	1
2 Carry on tens and warn next car.			1	
	9	9	0	1
3 Carry on hundreds, and ditto			1	
	9	0	0	1
4 Carry on thousands, and ditto			1	
	0	0	0	0
5 Carry on ten thousands				1
	1	0	0	0
				1

In this case the carriages only become known successively, and they amount to the number of figures to be added; consequently, the mere addition of two numbers, each of fifty places of figures, would require only nine seconds of time, whilst the possible carriages would consume fifty seconds.

The mechanical means I employed to make these carriages bears some slight analogy to the operation of the faculty of memory. A toothed wheel had the ten digits marked upon its edge; between the nine and the zero a projecting tooth was placed. Whenever any wheel, in receiving addition, passed from nine to zero, the projecting tooth pushed over a certain lever. Thus, as soon as the nine seconds of time required for addition were ended, every carriage which had *become due* was indicated by the altered position of its lever. An arm now went round, which was so contrived that the act of replacing that lever caused the carriage which its position indicated to be made to the next figure above. But this figure might be a nine, in which case, in passing to zero, it would put over its lever, and so on. By placing the arms spirally round an axis, these successive carriages were accomplished.

Multitudes of contrivances were designed, and almost endless drawings made, for the purpose of economizing the time and simplifying the mechanism of carriage. [...]

Description of the existing portion of Difference Engine No. 1.

[...]

It consists of three columns; each column contains six cages; each cage contains one figure-wheel.

The column on the right hand has its lowest figure-wheel covered by a shade which is never removed, and to which the reader's attention need not be directed.

The figure-wheel next above may be placed by hand at any one of the ten digits. In the woodcut it stands at zero.

The third, fourth, and fifth, cages are exactly the same as the second.

The sixth cage contains exactly the same as the four just described. It also contains two other figure-wheels, which with a similar one above the frame, may also be dismissed from the reader's attention. Those wheels are entirely unconnected with the moving part of the engine, and are only used for memoranda.

It appears, therefore, that there are in the first column on the right hand five figure-wheels, each of which may be set by hand to any of the figures 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

The lowest of these figure-wheels represents the unit's figure of any number, the next above the ten's figure, and so on. The highest figure-wheel will therefore represent tens of thousands.

Now, as each of these figure-wheels may be set by hand to any digit, it is possible to place on the first column any number up to 99999. It is on these wheels that the Table to be calculated by the engine is expressed. This column is called the Table column, and the axis of the wheels the Table axis.

The second or middle column has also six cages, in each of which a figure-wheel is placed. It will be observed that in the lowest cage, the figure on the wheel is concealed by a shade. It may therefore be dismissed from the attention. The five other figure-wheels are exactly like the figure-wheels on the Table axis, and can also represent any number up to 99999.

This column is called the First Difference column, and the axis is called the First Difference axis.

The third column, which is that on the left hand, has also six cages, in each of which is a figure-wheel capable of being set by hand to any digit.

The mechanism is so contrived that whatever may be the numbers placed respectively on the figure-wheels of each of the three columns, the following succession of operations will take place as long as the handle is moved: –

1st. Whatever number is found upon the column of first differences will be added to the number found upon the Table column.

2nd. The same first difference remaining upon its own column, the number found upon the column of second differences will be added to that first difference.

It appears, therefore, that with this small portion of the Engine any Table may be computed by the method of differences, provided neither the Table itself, nor its first and second differences, exceed five places of figures.

[...]

In the existing portion there are three bells; they can be respectively ordered to ring when the Table; its first difference and its second difference, pass from positive to negative. Several weeks after the machine had been placed in my drawing-room, a friend came by appointment to test its power of calculating Tables. After the Engine had computed several Tables, I remarked that it was evidently finding the root of a quadratic equation; I therefore

set the bells to watch it. After some time the proper bell sounded twice, indicating, and giving the two positive roots to be 28 and 30. The Table thus calculated related to the barometer and really involved a quadratic equation, although its maker had not previously observed it. I afterwards set the Engine to tabulate a formula containing impossible roots, and of course the other bell warned me when it had attained those roots. I had never before used these bells, simply because I did not think the power it thus possessed to be of any practical utility.

Again, the lowest cages of the Table, and of the first difference, have been made use of for the purpose of illustrating three important faculties of the finished engine.

1st. The portion exhibited can calculate any Table whose third difference is constant and less than 10.

2nd. It can be used to show how much more rapidly astronomical Tables can be calculated in an engine in which there is no constant difference.

3rd. It can be employed to illustrate those singular laws which might continue to be produced through ages, and yet after an enormous interval of time change into other different laws; each again to exist for ages and then to be superseded by new laws. These views were first proposed in the “Ninth Bridgewater Treatise.”

Amongst the various questions which have been asked respecting the Difference Engine, I will mention a few of the most remarkable: – One gentleman addressed me thus: “Pray, Mr Babbage, can you explain to me in two words what is the principle of this machine?” Had the querist possessed a moderate acquaintance with mathematics I might in four words have conveyed to him the required information by answering, “The method of differences.” The question might indeed have been answered with six characters thus –

$$\Delta^7 u_x = 0$$

but such information would have been unintelligible to such inquirers.

On two occasions I have been asked, – “Pray, Mr Babbage, if you put into the machine wrong figures, will the right answers come out?” In one case a member of the Upper, and in the other a member of the Lower, House put this question. I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question. I did, however, explain the following property, which might in some measure approach towards an answer to it.

It is possible to construct the [...] Engine in such a manner that after the question is once communicated to the engine, it may be stopped at any turn of the handle and set on again as often as may be desired. At each stoppage every figure wheel throughout the Engine, which is capable of being moved without breaking, may be moved on to any other digit. Yet after each of these apparent falsifications the engine will be found to make the next calculation with perfect truth.

The explanation is very simple, and the property itself useless. The whole of the mechanism ought of course to be enclosed in glass, and kept under lock and key, in which case the

mechanism necessary to give it the property alluded to would be useless.

CHAPTER VI. Statement relative to the Difference Engine, drawn up by the late Sir H. Nicolas from the Author's Papers.

THE following statement was drawn up by the late Sir Harris Nicolas, G.S.M. & G., from papers and documents in my possession relating to the Difference Engine. I believe every paper I possessed at all bearing on the subject was in his hands for several months.

For some time previous to 1822, Mr. Babbage had been engaged in contriving machinery for the execution of extensive arithmetical operations, and in devising mechanism by which the machine that made the calculations might also print the results.

On the 3rd of July, 1822, he published a letter to Sir Humphry Davy, President of the Royal Society, containing a statement of his views on that subject; and more particularly describing an Engine for calculating astronomical, nautical and other Tables, by means of the "method of differences." In that letter it is stated that a small Model, consisting of six figures, and capable of working two orders of differences, had been constructed; and that it performed its work in a satisfactory manner.

The concluding paragraph of that letter is as follows: –

"Whether I shall construct a larger Engine of this kind, and bring to perfection the others I have described, will, in a great measure, depend on the nature of the encouragement I may receive.

Induced, by a conviction of the great utility of such Engines, to withdraw, for some time, my attention from a subject on which it has been engaged during several years, and which possesses charms of a higher order, I have now arrived at a point where success is no longer doubtful. It must, however, be attained at a very considerable expense, which would not probably be replaced, by the works it might produce, for a long period of time; and which is an undertaking I should feel unwilling to commence, as altogether foreign to my habits and pursuits."

The Model alluded to had been shown to a large number of Mr. Babbage's acquaintances, and to many other persons; and copies of his letter having been given to several of his friends, it is probable that one of the copies was sent to the Treasury.

On the 1st of April, 1823, the Lords of the Treasury referred that Letter to the Royal Society, requesting –

The opinion of the Royal Society on the merits and utility of this invention.

On the 1st of May the Royal Society reported to the Treasury, that –

"Mr. Babbage has displayed great talent and ingenuity in the construction of his Machine for Computation, which the Committee think fully adequate to the attainment of the objects proposed by the inventor; and they consider Mr. Babbage as highly deserving of public encouragement, in the prosecution of his

arduous undertaking.”¹¹

On the 21st of May these papers were ordered to be printed by the House of Commons.

In July, 1823, Mr. Babbage had an interview with the Chancellor of the Exchequer (Mr. Robinson¹²), to ascertain if it was the wish of the Government that he should construct a large Engine of the kind, which would also print the results it calculated.

From the conversation which took place on that occasion, Mr. Babbage apprehended that such was the wish of the Government. The Chancellor of the Exchequer remarked that the Government were in general unwilling to make grants of money for any inventions, however meritorious; because, if they really possessed the merit claimed for them, the sale of the article produced would be the best, as well as largest reward of the inventor; but that the present case was an *exception*; it being apparent that the construction of such a Machine could not be undertaken with a view to profit from the sale of its produce; and that, as mathematical Tables were peculiarly valuable for nautical purposes, it was deemed a fit object of encouragement by the Government.

The Chancellor of the Exchequer mentioned two modes of advancing money for the construction either through the recommendation of a Committee of the House of Commons or by taking a sum from the Civil Contingencies and he observed that as the Session of Parliament was near its termination the latter course might perhaps be the most convenient.

Mr. Babbage thinks the Chancellor of the Exchequer also made some observation, indicating that the amount of money taken from the Civil Contingencies would be smaller than that which might be had by means of a Committee of the House of Commons: and he then proposed to take £1,000 as a commencement from the Civil Contingencies Fund. To this Mr Babbage replied, in words which he distinctly remembers, “Would it be too much in the first instance to take £1,500?” The Chancellor of the Exchequer immediately answered, that £1,500 should be advanced.

Mr. Babbage’s opinion at that time was, that the Engine would be completed in two, or at the most in three years: and that by having £1,500 in the first instance, he would be enabled to advance from his own private funds, the residue of the £3,000, or even £5,000, which he then imagined the Engine might possibly cost; so that he would not again have occasion to apply to Government until it was completed. Some observations were made by the Chancellor of the Exchequer about the mode of accounting for the money received, as well as about its expenditure; but it seemed to be admitted that it was not possible to prescribe any very definite system, and that much must be left to Mr. Babbage’s own judgment.

Very unfortunately, no Minute of that conversation was made at the time, nor was any sufficiently distinct understanding between the parties arrived at. Mr. Babbage’s conviction was, that whatever might be the labour and difficulty of the undertaking, the Engine itself would, of course, become the property of the Government, which had paid for its

¹¹Parliamentary Paper, No. 370, printed 22nd May, 1823.

¹²Afterwards Lord Goderich, now Earl of Ripon

construction.

Soon after this interview with the Chancellor of the Exchequer, a letter was sent from the Treasury to the Royal Society, informing that body that the Lords of the Treasury

“Had directed the issue of £1,500 to Mr Babbage to enable him to bring his invention to perfection, in the manner recommended .”

These latter words, “*in the manner recommended*,” can only refer to the previous recommendation of the Royal Society; but it does not appear, from the Report of the Royal Society, that any *plan*, *terms*, or *conditions* had been pointed out by that body.

Towards the end of July, 1823, Mr. Babbage took measures for the construction of the present *Difference Engine*,¹³ and it was regularly proceeded with for four years.

In October, 1827, the expense incurred had amounted to £3,475; and Mr. Babbage having suffered severe domestic affliction, and being in a very ill state of health, was recommended by his medical advisers to travel on the Continent. He left, however, sufficient drawings to enable the work to be continued, and gave an order to his own banker to advance £1,000 during his absence: he also received, from time to time, drawings and inquiries relating to the mechanism, and returned instructions to the engineer who was constructing it.

As it now appeared probable that the expense would much exceed what Mr. Babbage had originally anticipated, he thought it desirable to inform the Government of that fact, and to procure a further grant. As a preliminary step, he wrote from Italy to his brother-in-law, Mr. Wolryche Whitmore, to request that he would see Lord Goderich upon the subject of the interview in July, 1823, but it is probable that he did not sufficiently inform Mr. Whitmore of all the circumstances of the case.

Mr. Whitmore, having had some conversation with Lord Goderich on the subject, addressed a letter, dated on the 29th of February, 1828, to Mr. Babbage, who was then at Rome, stating that

“That interview was unsatisfactory; that Lord Goderich did not like to admit that there was any understanding, at the time the £1,500 was advanced, that more would be given by Government.”

On Mr. Babbage’s return to England, towards the end of 1828, he waited in person upon Lord Goderich, who admitted that the understanding of 1823 was not very definite. He then addressed a statement to the Duke of Wellington, as the head of the Government, explaining the previous steps in the affair; stating the reasons for his inferences from what took place at the interview with the Chancellor of the Exchequer in July, 1823; and referring

¹³Note. – It will be convenient to distinguish between –

1. The small *Model* of the original or Difference Engine.
2. The *Difference Engine* itself, belonging to the Government, a part only of which has been put together.
3. The designs for another *Engine*, which in this Statement is called the Analytical Engine.

his Grace for further information to Lord Goderich, to whom also he sent a copy of that statement.

The Duke of Wellington, in consequence of this application, requested the Royal Society to inquire –

“Whether the progress of the Machine confirms them in their former opinion, that it will ultimately prove adequate to the important object it was intended to attain.”

[...]

CHAPTER VIII OF THE ANALYTICAL ENGINE.

THE circular arrangement of the axes of the Difference Engine round large central wheels led to the most extended prospects. The whole of arithmetic now appeared within the grasp of mechanism. A vague glimpse even of an Analytical Engine at length opened out, and I pursued with enthusiasm the shadowy vision. The drawings and the experiments were of the most costly kind. Draftsmen of the highest order were necessary to economize the labour of my own head; whilst skilled workmen were required to execute the experimental machinery to which I was obliged constantly to have recourse.

In order to carry out my pursuits successfully, I had purchased a house with above a quarter of an acre of ground in a very quiet locality. My coach-house was now converted into a forge and a foundry, whilst my stables were transformed into a workshop. I built other extensive workshops myself, and had a fire-proof building for my drawings and draftsmen. Having myself worked with a variety of tools, and having studied the art of constructing each of them, I at length laid it down as a principle – that, except in rare cases, I would never do anything myself if I could afford to hire another person who could do it for me.

The complicated relations which then arose amongst the various parts of the machinery would have baffled the most tenacious memory. I overcame that difficulty by improving and extending a language of signs, the Mechanical Notation, which in 1826 I had explained in a paper printed in the “Phil. Trans.” By such means I succeeded in mastering trains of investigation so vast in extent that no length of years ever allotted to one individual could otherwise have enabled me to control. By the aid of the Mechanical Notation, the Analytical Engine became a reality: for it became susceptible of demonstration.

Such works could not be carried on without great expenditure. The fluctuations in the demand and supply of skilled labour were considerable. The railroad mania withdrew from other pursuits the most intellectual and skilful draftsmen. One who had for some years been my chief assistant was tempted by an offer so advantageous that in justice to his own family he could scarcely have declined it. Under these circumstances I took into consideration the plan of advancing his salary to one guinea per day. Whilst this was in abeyance, I consulted my venerable surviving parent. When I had fully explained the circumstances my excellent mother replied: “My dear son, you have advanced far in the accomplishment of a great object, which is worthy of your ambition. You are capable of completing it. My advice is

– pursue it, even if it should oblige you to live on bread and cheese.”

This advice entirely accorded with my own feelings. I therefore retained my chief assistant at his advanced salary.

The most important part of the Analytical Engine was undoubtedly the mechanical method of carrying the tens. On this I laboured incessantly, each succeeding improvement advancing me a step or two. The difficulty did not consist so much in the more or less complexity of the contrivance as in the reduction of the *time* required to effect the carriage. Twenty or thirty different plans and modifications had been drawn. At last I came to the conclusion that I had exhausted the principle of successive carriage. I concluded also that nothing but teaching the Engine to foresee and then to act upon that foresight could ever lead me to the object I desired, namely, to make the whole of any unlimited number of carriages in one unit of time. One morning, after I had spent many hours in the drawing-office in endeavouring to improve the system of successive carriages, I mentioned these views to my chief assistant, and added that I should retire to my library, and endeavour to work out the new principle. He gently expressed a doubt whether the plan was *possible*, to which I replied that, not being able to prove its impossibility, I should follow out a slight glimmering of light which I thought I perceived.

After about three hours’ examination, I returned to the drawing-office with much more definite ideas upon the subject. I had discovered a principle that proved the possibility, and I had contrived mechanism which, I thought, would accomplish my object.

I now commenced the explanation of my views, which I soon found were but little understood by my assistant; nor was this surprising, since in the course of my own attempt at explanation, I found several defects in my plan, and was also led by his questions to perceive others. All these I removed one after another, and ultimately terminated at a late hour my morning’s work with the conviction that *anticipating* carriage was not only within my power, but that I had devised one mechanism at least by which it might be accomplished.

Many years after, my assistant, on his return from a long residence abroad, called upon me, and we talked over the progress of the Analytical Engine. I referred back to the day on which I had made that most important step, and asked him if he recollected it. His reply was that he perfectly remembered the circumstance; for that on retiring to my library, he seriously thought that my intellect was beginning to become deranged. The reader may perhaps be curious to know how I spent the rest of that remarkable day.

After working, as I constantly did, for ten or eleven hours a day, I had arrived at this satisfactory conclusion, and was revising the rough sketches of the new contrivance, when my servant entered the drawing-office, and announced that it was seven o’clock – that I dined in Park Lane – and that it was time to dress. I usually arrived at the house of my friend about a quarter of an hour before the appointed time, in order that we might have a short conversation on subjects on which we were both much interested. Having mentioned my recent success, in which my host thoroughly sympathized, I remarked that it had produced an exhilaration of the spirits which not even his excellent champagne could rival. Having enjoyed the society of Hallam, of Rogers, and of some few others of that delightful

circle, I retired, and joined one or perhaps two much more extensive reunions. Having thus forgotten science, and enjoyed society for four or five hours, I returned home. About one o'clock I was asleep in my bed, and thus continued for the next five hours.

This new and rapid system of the tens when two numbers are added together, reduced the actual time of the addition of any number of digits, however large, to nine units of time for the addition, and one unit for the carriage. Thus in ten's units of time, any two numbers, however large, might be added together. A few more units of time, perhaps five or six, were required for making the requisite previous arrangements.

Having thus advanced as nearly as seemed possible to the minimum of time requisite for arithmetical operations, I felt renewed power and increased energy to pursue the far higher object I had in view.

To describe the successive improvements of the Analytical Engine would require many volumes. I only propose here to indicate a few of its more important functions, and to give to those whose minds are duly prepared for it some information which will remove those vague notions of wonder, and even of its impossibility, with which it is surrounded in the minds of some of the most enlightened.

To those who are acquainted with the principles of the Jacquard loom, and who are also familiar with analytical formulae, a general idea of the means by which the Engine executes its operations may be obtained without much difficulty. In the Exhibition of 1862 there were many splendid examples of such looms.

It is known as a fact that the Jacquard loom is capable of weaving any design which the imagination of man may conceive. It is also the constant practice for skilled artists to be employed by manufacturers in designing patterns. These patterns are then sent to a peculiar artist, who, by means of a certain machine, punches holes in a set of pasteboard cards in such a manner that when those cards are placed in a Jacquard loom, it will then weave upon its produce the exact pattern designed by the artist.

Now the manufacturer may use, for the warp and weft of his work, threads which are all of the same colour; let us suppose them to be unbleached or white threads. In this case the cloth will be woven all of one colour. but there will be a damask pattern upon it such as the artist designed.

But the manufacturer might use the same cards, and put into the warp threads of any other colour. Every thread might even be of a different colour, or of a different shade of colour; but in all these cases the *form* of the pattern will be precisely the same – the colours only will differ.

The analogy of the Analytical Engine with this well-known process is nearly perfect.

The Analytical Engine consists of two parts: –

1st. The store in which all the variables to be operated upon, as well as all those quantities which have arisen from the result of other operations, are placed.

2nd. The mill into which the quantities about to be operated upon are always brought.

Every formula which the Analytical Engine can be required to compute consists of certain algebraical operations to be performed upon given letters, and of certain other modifications depending on the numerical value assigned to those letters.

There are therefore two sets of cards, the first to direct the nature of the operations to be performed – these are called operation cards: the other to direct the particular variables on which those cards are required to operate – these latter are called variable cards. Now the symbol of each variable or constant, is placed at the top of a column capable of containing any required number of digits.

Under this arrangement, when any formula is required to be computed, a set of operation cards must be strung together, which contain the series of operations in the order in which they occur. Another set of cards must then be strung together, to call in the variables into the mill, the order in which they are required to be acted upon. Each operation card will require three other cards, two to represent the variables and constants and their numerical values upon which the previous operation card is to act, and one to indicate the variable on which the arithmetical result of this operation is to be placed.

But each variable has below it, on the same axis, a certain number of figure-wheels marked on their edges with the ten digits: upon these any number the machine is capable of holding can be placed. Whenever variables are ordered into the mill, these figures will be brought in, and the operation indicated by the preceding card will be performed upon them. The result of this operation will then be replaced in the store.

The Analytical Engine is therefore a machine of the most general nature. Whatever formula it is required to develop, the law of its development must be communicated to it by two sets of cards. When these have been placed, the engine is special for that particular formula. The numerical value of its constants must then be put on the columns of wheels below them, and on setting the Engine in motion it will calculate and print the numerical results of that formula.

Every set of cards made for any formula will at any future time recalculate that formula with whatever constants may be required.

Thus the Analytical Engine will possess a library of its own. Every set of cards once made will at any future time reproduce the calculations for which it was first arranged. The numerical value of its constants may then be inserted.

It is perhaps difficult to apprehend these descriptions without a familiarity both with analytical forms and mechanical structures. I will now, therefore, confine myself to the mathematical view of the Analytical Engine, and illustrate by example some of its supposed difficulties.

An excellent friend of mine, the late Professor MacGullagh, of Dublin, was discussing with me, at breakfast, the various powers of the Analytical Engine. After a long conversation on the subject, he inquired what the machine could do if, in the midst of algebraic operations, it was required to perform logarithmic or trigonometric operations.

My answer was, that whenever the Analytical Engine should exist, all the developments of

formula would be directed by this condition – that the machine should be able to compute their numerical value in the shortest possible time. I then added that if this answer were not satisfactory. I had provided means by which, with equal accuracy, it might compute by logarithmic or other Tables.

I explained that the Tables to be used must, of course, be computed and punched on cards by the machine, in which case they would undoubtedly be correct. I then added that when the machine wanted a tabular number, say the logarithm of a given number, that it would ring a bell and then stop itself. On this, the attendant would look at a certain part of the machine, and find that it wanted the logarithm of a given number, say of 2303. The attendant would then go to the drawer containing the pasteboard cards representing its table of logarithms. From amongst these he would take the required logarithmic card, and place it in the machine. Upon this the engine would first ascertain whether the assistant had or had not given him the correct logarithm of the number; if so, it would use it and continue its work. But if the engine found the attendant had given him a wrong logarithm, it would then ring a louder bell, and stop itself. On the attendant examining the engine, he would observe the words, “Wrong tabular number,” and then discover that he really had given the wrong logarithm, and of course he would have to replace it by the right one.

Upon this, Professor MacCullagh naturally asked why, if the machine could tell whether the logarithm was the right one, it should have asked the attendant at all? I told him that the means employed were so ridiculously simple that I would not at that moment explain them; but that if he would come in the course of a few days, I should be ready to explain it. Three or four days after, Bessel and Jacobi, who had just arrived in England, were sitting with me, inquiring about the Analytical Engine, when fortunately my friend MacCullagh was announced. The meeting was equally agreeable to us all, and we continued our conversation. After some time Bessel put to me the very same question which MacCullagh had previously asked. On this Jacobi remarked that he, too, was about to make the same inquiry when Bessel had asked the question. I then explained to them the following very simple means by which that verification was accomplished.

Besides the sets of cards which direct the nature of the operations to be performed, and the variables or constants which are to be operated upon, there is another class of cards called number cards. These are much less general in their uses than the others, although they are necessarily of much larger size.

Any number which the Analytical Engine is capable of using or of producing can, if required, be expressed by a card with certain holes in it; thus –

Number.				Table.						
2	3	0	3	3	6	2	2	9	3	9
●	●	○	●	●	●	●	●	●	●	●
●	●	○	●	●	●	●	●	●	●	●
○	●	○	●	●	●	○	○	●	●	●
○	○	○	○	○	●	○	○	●	○	●
○	○	○	○	○	●	○	○	●	○	●
○	○	○	○	○	●	○	○	●	○	●
○	○	○	○	○	○	○	○	●	○	●
○	○	○	○	○	○	○	○	●	○	●
○	○	○	○	○	○	○	○	●	○	●

The above card contains eleven vertical rows for holes, each row having nine or any less number of holes, In this example the tabular number is 3 6 2 2 9 3 9, whilst its number in the order of the table is 2 3 0 3. In fact, the former number is the logarithm of the latter.

The Analytical Engine will contain,

- 1st. Apparatus for printing on paper, one, or, if required, two copies of its results.
- 2nd. Means for producing a stereotype mould of the tables or results it computes.
- 3rd. Mechanism for punching on blank pasteboard cards or metal plates the numerical results of any of its computations.

Of course the Engine will compute all the Tables which it may itself be required to use. These cards will therefore be entirely free from error. Now when the Engine requires a tabular number, it will stop, ring a bell, and ask for such number. In the case we have assumed, it asks for the logarithm of 2 3 0 3.

When the attendant has placed a tabular card in the Engine, the first step taken by it will be to verify the *number* of the card given it by subtracting its number from 2 3 0 3, the number whose logarithm it asked for. If the remainder is zero, then the engine is certain that the logarithm must be the right one, since it was computed and punched by itself.

Thus the Analytical Engine first computes and punches on cards its own tabular numbers. These are brought to it by its attendant when demanded. But the engine itself takes care that the *right* card is brought to it by verifying the *number* of that card by the number of the card which it demanded. The Engine will always reject a wrong card by continually ringing a loud bell and stopping itself until supplied with the precise intellectual food it demands.

It will be an interesting question, which time only can solve, to know whether such tables of cards will ever be required for the Engine. Tables are used for saving the time of continually computing individual numbers. But the computations to be made by the Engine are so rapid that it seems most probable that it will make shorter work by computing directly from proper formulae than by having recourse even to its own Tables.

The Analytical Engine I propose will have the power of expressing every number it uses to fifty places of figures. It will multiply any two such numbers together, and then, if required, will divide the product of one hundred figures by number of fifty places of figures.

Supposing the velocity of the moving parts of the Engine to be not greater than forty feet per minute, I have no doubt that

Sixty additions or subtractions may be completed and printed in one minute.

One multiplication of two numbers, each of fifty figures, in one minute.

One division of a number having 100 places of figures by another of 50 in one minute.

In the various sets of drawings of the modifications of the mechanical structure of the Analytical Engines, already numbering upwards of thirty, two great principles were embodied to an unlimited extent.

1st. The entire control over *arithmetical* operations, however large, and whatever might be the number of their digits.

2nd. The entire control over the *combinations* of algebraic symbols, however lengthened those processes may be required. The possibility of fulfilling these two might reasonably be doubted by the most accomplished mathematician as well as by the most ingenious mechanician.

The difficulties which naturally occur to those capable of examining the question, as far as they relate to arithmetic, are these, –

- (a) The number of digits in *each constant* inserted in the Engine must be without limit.
- (b) The number of constants to be inserted in the Engine must also be without limit.
- (c) The number of operations necessary for arithmetic is only four, but these four may be repeated an *unlimited* number of times.
- (d) These operations may occur in any order, or follow an *unlimited* number of laws.

The following conditions relate to the algebraic portion of the Analytical Engine: –

- (e) The number of *litteral* constants must be *unlimited*.
- (e) The number of *variables* must be *without limit*.
- (e) The combinations of the algebraic signs must *be unlimited*.
- (e) The number of *functions* to be employed must be *without limit*.

This enumeration includes eight conditions each of which is absolutely *unlimited* as to the number of its combinations.

Now it is obvious that no *finite* machine can include infinity. It is also certain that no question *necessarily* involving infinity can ever be converted into any other in which the idea of infinity under some shape or other does not enter.

It is impossible to construct machinery occupying unlimited space; but it is possible to construct finite machinery, and to use it through unlimited time. It is this substitution of the *infinity of time* for the *infinity of space* which I have made use of, to limit the size of the engine and yet to retain its unlimited power.

(a). I shall now proceed briefly to point out the means by which I have effected this change.

Since every calculating machine must be constructed for the calculation of a definite number of figures, the first datum must be to fix upon that number. In order to be somewhat in advance of the greatest number that may ever be required, I chose fifty places of figures as the standard for the Analytical Engine. The intention being that in such a machine two numbers, each of fifty places of figures, might be multiplied together and the resultant product of one hundred places might then be divided by another number of fifty places. It seems to me probable that a long period must elapse before the demands of science will exceed this limit. To this it may be added that the addition and subtraction of numbers in an engine constructed for n places of figures would be equally rapid whether n were equal to five or five thousand digits. With respect to multiplication and division, the time required is greater: –

Thus if $a \cdot 10^{50} + b$ and $a' \cdot 10^{50} + b'$ are two numbers each of less than a hundred places of figures, then each can be expressed upon two columns of fifty figures, and a, b, a', b , are each less than fifty places of figures: they can therefore be added and subtracted upon any column holding fifty places of figures.

The product of two such numbers is –

$$aa'10^{100} + (ab' + a'b)10^{50} + bb'.$$

This expression contains four pair of factors, aa', ab', ab', bb'^{14} , each factor of which has less than fifty places of figures. Each multiplication can therefore be executed in the Engine. The time, however, of multiplying two numbers each consisting of any number of digits between fifty and one hundred, will be nearly four times as long as that of two such numbers of less than fifty places of figures.

The same reasoning will show that if the numbers of digits of each factor are between one hundred and one hundred and fifty, then the time required for the operation will be nearly nine times that of a pair of factors having only fifty digits.

Thus it appears that whatever may be the number of digits the Analytical Engine is capable of holding, if it is required to make all the computations with k times that number digits, then it can be executed by the same Engine, but in an amount of time equal to k^2 times the former. Hence the condition (a), or the unlimited number of digits contained in each constant employed, is fulfilled.

It must, however, be admitted that this advantage is gained at the expense of diminishing the number of the constants the Engine can hold. An engine of fifty digits, when used as

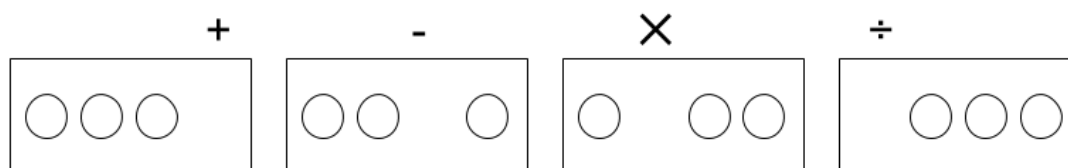
¹⁴[sic. The factors are $aa', ab', a'b$, and bb']

one of a hundred digits, can only contain half the number of variables. An engine containing m columns, each holding n digits, if used for computations requiring kn digits, can only hold $\frac{m}{k}$ constants or variables.

(b). The next step is therefore to prove (b), viz.: to show that a finite engine can be used as if it contained an unlimited number of constants. The method of punching cards for tabular numbers has already been alluded to. Each Analytical Engine will contain one or more apparatus for printing any numbers put into it, and also an apparatus for punching on pasteboard cards the holes corresponding to those numbers. At another part of the machine a series of number cards, resembling those of Jacquard, but delivered to and computed by the machine itself, can be placed. These can be called for by the Engine itself in any order in which they may be placed, or according to *any law* the Engine may be directed to use. Hence the condition (b) is fulfilled, namely an *unlimited number of constants* can be inserted in the machine in an *unlimited* time.

I propose in the Engine I am constructing to have places for only a thousand constants, because I think it will be more than sufficient. But if it were required to have ten, or even a hundred times that number, it would be quite possible to make it, such is the simplicity of its structure of that portion of the Engine.

(c). The next stage in the arithmetic is the number of times the four processes of addition, subtraction, multiplication, and division can be repeated. It is obvious that four different cards thus punched



would give the orders for the four rules of arithmetic.

Now there is no limit to the number of such cards which may be strung together according to the nature of the operations required. Consequently the condition (c) is fulfilled.

(d). The fourth arithmetical condition (d), that the order of succession in which these operations can be varied, is itself *unlimited*, follows as a matter of course.

The four remaining conditions which must be fulfilled, in order to render the Analytical Engine as general as the science of which it is the powerful executive, relate to algebraic quantities with which it operates.

The thousand columns, each capable of holding any number of less than fifty-one places of figures, may each represent a constant or a variable quantity. These quantities I have called by the comprehensive title of variables, and have denoted them by V_n , with an index below. In the machine I have designed, n may vary from 0 to 999. But after any one or more columns have been used for variables, if those variables are not required afterwards, they may be printed upon paper, and the columns themselves again used for other variables. In

such cases the variables must have a new index; thus, ${}^mV^n$. I propose to make n vary from 0 to 99. If more variables are required, these may be supplied by Variable Cards, which may follow each other in unlimited succession. Each card will cause its symbol to be printed with its proper indices.

For the sake of uniformity, I have used V with as many indices as may be required throughout the Engine. This, however, does not prevent the printed result of a development from being represented by any letters which may be thought to be more convenient. In that part in which the results are printed, type of any form may be used, according to the taste of the proposer of the question.

It thus appears that the two conditions, (e) and (f), which require that the number of constants and of variables should be unlimited, are both fulfilled.

The condition (g) requiring that the number of combinations of the four algebraic signs shall be unlimited, is easily fulfilled by placing them on cards in any order of succession the problem may require.

The last condition (h), namely that the number of functions to be employed must be without limit, might seem at first sight to be difficult to fulfil. But when it is considered that any function of any number of operations performed upon any variables is but a combination of the four simple signs of operation with various quantities, it becomes apparent that any function whatever may be represented by two groups of cards, the first being signs of operation, placed in the order in which they succeed each other, and the second group of cards representing the variables and constants placed in the order of succession in which they are acted upon by the former.

Thus it appears that the whole of the conditions which enable a *finite* machine to make calculations of *unlimited* extent are fulfilled in the Analytical Engine. The means I have adopted are uniform. I have converted the infinity of space, which was required by the conditions of the problem, into the infinity of time. The means I have employed are in daily use in the art of weaving patterns. It is accomplished by systems of cards punched with various holes strung together to any extent which may be demanded. Two large boxes, the one empty and the other filled with perforated cards, are placed before and behind a polygonal prism, which revolves at intervals upon its axis, and advances through a short space, after which it immediately returns.

A card passes over the prism just before each stroke of the shuttle; the cards that have passed hang down until they reach the empty box placed to receive them, into which they arrange themselves one over the other. When the box is full, another empty box is placed to receive the coming cards, and a new full box on the opposite side replaces the one just emptied. As the suspended cards on the entering side are exactly equal to those on the side at which the others are delivered, they are perfectly balanced, so that whether the formulas to be computed be excessively complicated or very simple, the force to be exerted always remains nearly the same.

In 1840 I received from my friend M. Plana a letter pressing me strongly to visit Turin at the then approaching meeting of Italian philosophers. In that letter M. Plana stated that

he had inquired anxiously of many of my countrymen about the power and mechanism of the Analytical Engine. He remarked that from all the information he could collect the case seemed to stand thus: –

“Hitherto the *legislative* department of our analysis has been all powerful – the *executive* all feeble.”

“Your engine seems to give us the same control over the executive which we have hitherto only possessed over the legislative department.”

Considering the exceedingly limited information which could have reached my friend respecting the Analytical Engine, I was equally surprised and delighted at his exact prevision of its powers. Even at the present moment I could not express more clearly, and in fewer terms, its real object. I collected together such of my models, drawings, and notations as I conceived to be best adapted to give an insight into the principles and mode of operating of the Analytical Engine. On mentioning my intention to my excellent friend the late Professor MacCullagh, he resolved to give up a trip to the Tyrol, and join me at Turin.

We met at Turin at the appointed time, and as soon as the first bustle of the meeting had a little abated, I had the great pleasure of receiving at my own apartments, for several mornings, Messrs. Plana, Menabrea, Mossotti, MacCullagh, Plantamour, and others of the most eminent geometers and engineers of Italy.

Around the room were hung the formula, the drawings, notations, and other illustrations which I had brought with me. I began on the first day to give a short outline of the idea. My friends asked from time to time further explanations of parts I had not made sufficiently clear. M. Plana had at first proposed to make notes, in order to write an outline of the principles of the engine. But his own laborious pursuits induced him to give up this plan, and to transfer the task to a younger friend of his, M. Menabrea, who had already established his reputation as a profound analyst.

These discussions were of great value to me in several ways. I was thus obliged to put into language the various views I had taken, and I observed the effect of my explanations on different minds. My own ideas became clearer, and I profited by many of the remarks made by my highly-gifted friends.

One day Mosotti, who had been unavoidably absent from the previous meeting, when a question of great importance had been discussed, again joined the party. Well aware of the acuteness and rapidity of my friend’s intellect, I asked my other friends to allow me five minutes to convey to Professor Mosotti the substance of the preceding sitting. After putting a few questions to Mosotti himself, he placed before me distinctly his greatest difficulty.

He remarked that he was now quite ready to admit the power of mechanism over numerical, and even over algebraical relations, to any extent. But he added that he had no conception how the machine could perform the act of judgment sometimes required during an analytical inquiry, when two or more different courses presented themselves, especially as the proper course to be adopted could not be known in many cases until all the previous portion had been gone through.

I then inquired whether the solution of a numerical equation of any degree by the usual, but very tedious proceeding of approximation would be a type of the difficulty to be explained. He at once admitted that it would be a very eminent one.

For the sake of perspicuity and brevity I shall confine my present explanation to possible roots.

I then mentioned the successive stages: –

Number of Operation

Cards used.

- | | |
|---|--|
| 1 | a. Ascertain the number of possible roots by applying Sturm's theorem to the coefficients. |
| 2 | b. Find a number greater than the greatest root. |
| 3 | c. Substitute the powers of ten (commencing with that next greater than the greatest root, and diminishing the powers by unity at each step) for the value of x in the given equation. Continue this until the sign of the resulting number changes from positive to negative. |

Number of Operation

Cards used.

- | | |
|---|--|
| 4 | d. Substitute successively for x in the original equation 0×10^n , 1×10^n , 2×10^n , 3×10^n , $\dots 9 \times 10^n$, until a change of sign occurs in the result. The digit previously substituted will be the first figure of the root sought. |
| 5 | e. Transform the original equation into another whose roots are less by the number thus found. The transformed equation will have a real root, the digit, less than 10^n . |
| 6 | f. Substitute $1 \times 10^{n-1}$, $2 \times 10^{n-1}$, $3 \times 10^{n-1}$, &c., successively for the root of this equation, until a change of sign occurs in the result, as in process 4.
This will give the second figure of the root.
This process of alternately finding a new figure in the root, and then transforming the equation into another (as in process 4 and 5), must be carried on until as many figures as are required, whether whole numbers or decimals, are arrived at. |
| 7 | g. The root thus found must now be used to reduce the original equation to one dimension lower. |
| 8 | h. This new equation of one dimension lower must now be treated by sections 3, 4, 5, 6, and 7, until the new root is found. |
| 9 | i. The repetition of sections 7 and 8 must go on until all the roots have been found. |

Now it will be observed that Professor Mosoiit was quite ready to admit at once that each

of these difference processes could be performed by the Analytical Machine through the medium of properly-arranged sets of Jacquard cards.

His real difficulty consisted in teaching the engine to know when to change from one set of cards to another, and back again repeatedly, at intervals not known to the person who gave the orders.

The dimensions of the algebraic equation being known, the number of arithmetical processes necessary for Sturm's theorem is consequently known. A set of operation cards can therefore be prepared. These must be accompanied by a corresponding set of variable cards, which will represent the columns in the store, on which the several coefficients of the given equation, and the various combinations required amongst them, are to be placed.

The next stage is to find a number greater than the greatest root of the given equation. There are various courses for arriving at such a number. Any one of these being selected, another set of operation and variable cards can be prepared to execute this operation.

Now, as this second process invariably follows the first, the second set of cards may be attached to the first set, and the engine will pass on from the first to the second process, and again from the second to the third process.

But here a difficulty arises: successive powers of ten are to be substituted for x in the equation, until a certain event happens. A set of cards may be provided to make the substitution of the highest power of ten, and similarly for the others; but on the occurrence of a certain event, namely, the change of a sign from $+$ to $-$, this stage of the calculation is to terminate.

Now at a very early period of the inquiry I had found it necessary to teach the engine to know when any numbers it might be computing passed through zero or infinity.

The passage through zero can be easily ascertained, this: Let the continually-decreasing number which is being computed be placed upon a column of wheels in connection with a carrying apparatus. After each process this number will be diminished, until at last a number is subtracted from it which is greater than the number expressed on those wheels.

$$\begin{array}{r} \text{Thus let it be .} \quad 00000,00000,00000,00423 \\ \text{Subtract . . .} \quad 00000,00000,00000,00511 \\ \hline \quad \quad \quad 99999,99999,99999,99912 \end{array}$$

Now in every case of a carriage becoming due, a certain lever is transferred from one position to another in the cage next above it.

Consequently in the highest cage of all (say the fiftieth in the Analytical Engine), an arm will be moved or not moved accordingly as the carriages do or do not run up beyond the highest wheel.

This arm can, of course, make any change which has been previously decided upon. In the instance we have been considering it would order the cards to be turned on to the next set.

If we wish to find when any number, which is increasing, exceeds in the number of its

digits the number of wheels on the columns of the machine, the same carrying arm can be employed. Hence any directions may be given which the circumstances require.

It will be remarked that this does not actually prove, even in the Analytical Engine of fifty figures, that the number computed has passed through infinity; but only that it has become greater than any number of fifty places of figures.

There are, however, methods by which any machine made for a given number of figures may be made to compute the same formulae with double or any multiple of its original number. But the nature of this work prevents me from explaining that method.

It may here be remarked that in the process, the cards employed to make the substitutions of the powers of ten are *operation* cards. They are, therefore, quite independent of the numerical values substituted. Hence the same set of operation cards which order the substitutions of 1×10^n will, if backed, order the substitution of 2×10^n , &c. We may, therefore, avail ourselves of mechanism for backing these cards, and call it into action whenever the circumstances themselves require it.

The explanation of M. Mosotti's difficulty is this:— Mechanical means have been provided for backing or advancing the operation cards to any extent. There exist means of expressing the conditions under which these various processes are required to be called into play. It is not even necessary that two courses only should be possible. Any number of course may be possible at the same time; and the choice of each may depend upon any number of conditions.

It was during these meetings that my highly valued friend, M. Menabrea, collected the materials for that lucid and admirable description which he subsequently published in the *Bibli. Univ. de Genève*, t. xli. Oct. 1842.

The elementary principles on which the Analytical Engine rests were thus in the first instance brought before the public by General Menabrea.

Some time after the appearance of his memoir on the subject in the “*Bibliothèque Universelle de Genève*,” the late Countess of Lovelace¹⁵ informed me that she had translated the memoir of Menabrea. I asked why she had not herself written an original paper on a subject with which she was so intimately acquainted? To this Lady Lovelace replied that the thought had not occurred to her. I then suggested that she should add some notes to Menabrea's memoir; an idea which was immediately adopted.

We discussed together the various illustrations that might be introduced: I suggested several, but the selection was entirely her own. So also was the algebraic working out of the different problems, except, indeed, that relating to the numbers of Bernoulli, which I had offered to do to save Lady Lovelace the trouble. This she sent back to me for an amendment, having detected a grave mistake which I had made in the process.

The notes of the Countess Lovelace extend to about three times the length of the original memoir. Their author has entered fully into almost all the very difficult and abstract questions connected with the subject.

¹⁵Ada Augusta, Countess of Lovelace, only child of the Poet Byron.

These two memoirs taken together furnish, to those who are capable of understanding the reasoning, a complete demonstration – *That the whole of the developments and operations of analysis are now capable of being executed by machinery.*

There are various methods by which these developments are arrived at: – 1. By the aid of the Differential and Integral Calculus. 2. By the Combinatorial Analysis of Hindenburg. 3. By the Calculus of Derivations of Arbogast.

Each of these systems professes to expand any function according to any laws. Theoretically each method may be admitted to be perfect; but practically the time and attention required are, in the greater number of cases, more than the human mind is able to bestow. Consequently, upon several highly interesting questions relative to the Lunar theory, some of the ablest and most indefatigable of existing analysts are at variance.

The Analytical Engine is capable of executing the laws prescribed by each of these methods. At one period I examined the Combinatorial Analysis, and also took some pains to ascertain from several of my German friends, who had had far more experience of it than myself, whether it could be used with greater facility than the Differential system. They seemed to think that it was more readily applicable to all the usual wants of analysis.

I have myself worked with the system of Arbogast, and if I were to decide from my own limited use of the three methods, I should, for the purpose of the Analytical Engine, prefer the Calcul de Derivations.

As soon as the Analytical Engine exists, it will necessarily guide the future course of the science. Whenever any result is sought by its aid, the question will then arise – By what course of calculation can these results be arrived at by the machine in the *shortest time*?

In the drawings I have prepared I proposed to have a thousand variables, upon each of which any number not having more than fifty figures can be placed. This machine would multiply 50 figures by 50, and print the product of 100 figures. Or it would divide any number having 100 figures by any other of 50 figures, and print the quotient of 50 figures. Allowing but a moderate velocity for the machine, the time occupied by either of these operations would be about one minute.

The whole of the *numerical* constants throughout the works of Laplace, Plana, Le Verrier, Hansen, and other eminent men whose indefatigable labours have brought astronomy to its present advanced state, might easily be recomputed. They are but the numerical coefficients of the various terms of functions developed according to certain series. In all cases in which these numerical constants can be calculated by more than one method, it might be desirable to compute them by several processes until frequent practice shall have confirmed our belief in the infallibility of mechanism.

The great importance of having accurate Tables is admitted by all who understand their uses; but the multitude of errors really occurring is comparatively little known. Dr. Lardner, in the “Edinburgh Review,” has made some very instructive remarks on this subject.

I shall mention two within my own experience: these are selected because they occurred in works where neither care nor expense were spared on the part of the Government to insure perfect accuracy. It is, however, but just to the eminent men who presided over the preparation of these works for the press to observe, that the fault lay not in them but in *the nature of things*.

In 1828 I lent the Government an original MS. of the table of Logarithmic Sines, Cosines, &c., computed to every second of the quadrant, in order that they might have it compared with Taylor's Logarithms, 4to., 1792, of which they possessed a considerable number of copies. Nineteen errors were thus detected, and a list of these errata was published in the Nautical Almanac for 1832: these may be called

Nineteen errata of the first order . . 1832

An error being detected in one of these errata, in the following Nautical Almanac we find an

Erratum of the errata in N. Alm. 1832 . . 1833

But in this very erratum of the second order a new mistake was introduced larger than any of the original mistakes. In the year next following there ought to have been found

Erratum in the erratum of the errata in N. Alm.
1832 1834

[...]

I am unwilling to terminate this chapter without reference to another difficulty now arising, which is calculated to impede the progress of Analytical Science. The extension of analysis is so rapid, its domain so unlimited, and so many inquirers are entering into its fields, that a variety of new symbols have been introduced, formed on no common principles. Many of these are merely new ways of expressing well-known functions. Unless some philosophical principles are generally admitted as the basis of all notation, there appears a great probability of introducing the confusion of Babel into the most accurate of all languages.

A few months ago I turned back to a paper in the Philosophical Transactions, 1844, to examine some analytical investigations of great interest by an author who has thought deeply on the subject. It related to the separation of symbols of operation from those of quantity, a question peculiarly interesting to me, since the Analytical Engine contains the embodiment of that method. There was no ready, sufficient, and simple mode of distinguishing letters which represented quantity from those which indicated operation. To understand the results the author had arrived at, it became necessary to read the whole Memoir.

Although deeply interested in the subject, I was obliged, with great regret, to give up the attempt; for it not only occupied much time, but placed too great a strain on the memory.

Whenever I am thus perplexed it has often occurred to me that the very simple plan

I have adopted in my *Mechanical Notation* for lettering drawings might be adopted in analysis.

On the geometrical drawings of machinery every piece of matter which represents framework is invariably denoted by an *upright* letter; whilst all letters indicating moveable parts are marked by *inclined* letters.

The analogous rule would be –

Let all letters indicating operations or modifications be expressed by *upright* letters;

Whilst all letters representing quantity should be represented by *inclined* letters.

The subject of the principles and laws of notation is so important that it is desirable, before it is too late, that the scientific academies of the world should each contributed the results of their own examination and conclusions, and that some congress should assemble to discuss them. Perhaps it might be still better if each academy would draw up its own views, illustrated by examples, and have a sufficient number printed to send to all other academies.

CHAPTER IX. OF THE MECHANICAL NOTATION.

Soon after I had commenced the Difference Engine, my attention was strongly directed to the imperfection of all known modes of explaining and demonstrating the construction of machinery. It soon became apparent that my progress would be seriously impeded unless I could devise more rapid means of understanding and recalling the interpretation of my own drawings.

By a new system of very simple signs I ultimately succeeded in rendering the most complicated machine capable of explanation almost without the aid of words.

In order thoroughly to understand the action of any machine, we must have full information upon the following subjects, and it is of the greatest importance that this information should be acquired in the shortest possible time.

I. The actual shape and relative position of every piece of matter of which the machine is composed.

This can be accomplished by the ordinary mechanical drawings. Such drawings usually have letters upon them for the sake of reference in the description of the machine. Hitherto such letters were chosen without any principle, and in fact gave no indication of anything except the mere spot upon the paper on which they were written.

I then laid down rules for the selection of letters. I shall only mention one or two of them:
–

1. All upright letters, as *a, c, d, e, A, B*, represent framing.
2. All inclined letters, as *aa, c, d, e, A, B*, represent moveable parts.

3. All small letters represent working points. One of the most obvious advantages of these rules is that they enable the attention to be more easily confined to the immediate object sought.

By other rules it is rendered possible, when looking at a plan of any complicated machine, to perceive the *relative order* of super-position of any number of wheels, arms, &c., without referring to the elevation or end view.

II. The actual time and duration of every motion throughout the action of any machine can be ascertained almost instantly by a system of signs called the Notations of Periods.

It possesses equal facilities for ascertaining every contemporaneous as well as for every successive system of movements.

III. The actual connection of every moveable piece of the machine with every other on which it acts. Thus, taking from any special part of the drawing the indicating letter, and looking for it on a certain diagram, called the trains, the whole course of its movements may be traced, up to the prime mover, or down to the final results.

I have called this system of signs the Mechanical Notation. By its application to geometrical drawing it has given us a new demonstrative science, namely, that of proving that any given machine can or cannot exist; and if it can exist, that it will accomplish its desired object.

It is singular that this addition to human knowledge should have been made just about the period when it was beginning to be felt by those most eminently skilled in analysis that the time has arrived when many of its conclusions rested only on provable evidence. This state of things arose chiefly from the enormous extent to which the developments were necessarily carried in the lunar and planetary theories.

After employing this language for several years, it was announced, in December 1825, that King William IV. had founded two medals of fifty guineas each, to be given annually by the Royal Society according to rules to be laid down by the Council.

On the 26th January 1826, it was resolved,

“That it is the opinion of the Council that the medals be awarded for the most important discoveries or series of investigations, completed and made known to the Royal Society in the year preceding the day of the award.”

This rule reduced the number of competitors to a very few. Although I had had some experience as to the mode in which medals were awarded, and therefore valued them accordingly, I was simple enough to expect that the Council of the Royal Society would not venture upon a fraud on the very first occasion of exercising the royal liberality. I had also another motive for taking a ticket in this philosophical lottery of medals.

In 1824, the Astronomical Society did me the honour to award to me the first gold medal they ever bestowed. It was rendered still more grateful by the address of that eminent man, the late Henry Thomas Colebrooke, the President, who in a spirit of prophecy anticipated the results of years, at that period, long future.

“It may not, therefore, be deemed too sanguine an anticipation, when I express the hope that an instrument which in its simpler form attains to the extraction of the roots of numbers, and approximates the roots of equations, may, in a more advanced state of improvement, rise to the approximate solutions of algebraic equations of elevated degrees. I refer to solutions of such equations proposed by Lagrange, and more recently by other analysts, which involve operations too tedious and intricate for use, and which must remain without efficacy, unless some mode be devised of abridging the labour or facilitating the means of performance.” ¹⁶

I felt, therefore, that the *first* Royal Medal might fairly become an object of ambition, whatever might be the worth of subsequent ones.

In order to qualify myself for this chance, I carefully drew up a paper, “On a Method of expressing by Signs the Action of Machinery,” which I otherwise should not have published at that time.

This Memoir was read at the Royal Society on the 16th March, 1826. To the system of signs which it first expounded I afterwards gave the name of “Mechanical Notation.” It had been used in England and in Ireland, although not taught in its schools. It applies to the description of a combat by sea or by land. It can assist in representing the functions of animal life; and I have had both from the Continent and from the United States, specimens of such applications. Finally, to whatever degree of simplicity I may at last have reduced the Analytical Engine, the course through which I arrived at it was the most entangled and perplexed which probably ever occupied the human mind. Through the aid of the Mechanical Notation I examined numberless plans and systems of computing, and I am sure, from the nature of its self-necessary verifications that it is impossible I can have been deceived.

On the 16th November, 1826, that very council of the Royal Society which had made the law took the earliest opportunity to violate it by awarding the two Royal Medals, the first to Dalton, whose great discovery had been made nearly twenty years before, and the other to Ivory, for a paper published in their “Transactions” three years before. The history of their proceedings will be found in the “Decline of Science in England,” p. 115, 1830.

¹⁶‘Discourse of the President on delivering the first Gold Medal of the Astronomical Society to Charles Babbage, Esq .’ ‘Memoirs of the Astronomical Society,’ vol. i. p. 509.

5.2 ON THE APPLICATION OF ...

ON THE APPLICATION OF MACHINERY TO THE PURPOSE OF CALCULATING AND PRINTING MATHEMATICAL TABLES

By CHARLES BABBAGE

A Letter to Sir Humphry Davy, Bart., President of the Royal Society, July 3, 1822

My Dear Sir,

The great interest you have expressed in the success of that system on contrivances which has lately occupied a considerable portion of my attention, induces me to adopt this channel for stating more generally the principles on which they proceed, and for pointing out the probable extent and important consequences to which they appear to lead. Acquainted as you were with this inquiry almost from its commencement, much of what I have now to say cannot fail to have occurred to your own mind: you will, however, permit me to restate it for the consideration of those with whom the principles and the machinery are less familiar.

The intolerable labour and fatiguing monotony of a continued repetition of similar arithmetical calculations, first excited the desire, and afterwards suggested the idea, of a machine, which, by the aid of gravity or any other moving power, should become a substitute for one of the lowest operations of human intellect. It is not my intention in the present Letter to trace the progress of this idea, or the means which I have adopted for its execution; but I propose stating some of their general applications, and shall commence with describing the powers of several engines which I have contrived: of that part which is already executed I shall speak more in the sequel.

The first engine of which drawings were made was one which is capable of computing any table by the aid of differences, whether they are positive or negative, or of both kinds. With respect to the number of the order of differences, the nature of the machinery did not in my own opinion, nor in that of a skillful mechanic whom I consulted, appear to be restricted to any very limited number; and I should venture to construct one with ten or a dozen orders with perfect confidence. One remarkable property of this machine is, that the greater the number of differences the more the engine will outstrip the most rapid calculator.

By the application of certain parts of not great degree of complexity, this may be converted into a machine for extracting the roots of equations, and consequently the roots of numbers: and the extent of the approximation depends on the magnitude of the machine.

Of a machine for multiplying any number of figures (m) by any other number (n), I have several sketches; but it is not yet brought to that degree of perfection which I should wish to give it before it is to be executed.

I have also certain principles by which, if it should be desirable, a table of prime numbers might be made, extending from 0 to ten millions.

Another machine, whose plans are much more advanced than several of those just named, is one for constructing tables which have no order of differences constant.

A vast variety of equations of finite differences may by its means be solved, and a variety of tables, which could be produced in successive parts by the first machine I have mentioned, could be calculated by the latter one with a still less exertion of human thought. Another and very remarkable point in this structure of this machine is, that will calculate tables governed by laws which have not been hitherto shown to be explicitly determinable, or that it will solve equations for which analytical methods of solution have not yet been contrived.

Supposing these engines executed, there would yet be wanting other means to ensure the accuracy of the printed tables to be produced by them.

The errors of the persons employed to copy the figures presented by the engines would first interfere with their correctness. To remedy this evil, I have contrived means by which the machines themselves shall take from several boxes containing type, the numbers which they calculate, and place them side by side, thus becoming at the same time a substitute for the compositor and the computer; by which means all error in copying as well as in printing is removed. There are, however, two sources of error which have not yet been guarded against. The ten boxes with which the engine is provided contain each about three thousand types; any box having of course, only those of one number in it. It may happen that the person employed in filling these boxes shall accidentally place a wrong type in some of them; as for instance, the number 2 in the box which ought only to contain 7's. When these boxes are delivered to the superintendent of the engine, I have provided a simple and effectual means by which he shall, in less than half an hour, ascertain whether, amongst these 30,000 types, there be any individual misplaced, or even inverted. The other cause of error to which I have alluded, arises from the type falling out when the page has been set up: this I have rendered impossible by means of a similar kind.

The quantity of errors from carelessness in correcting the press, even in tables of the greatest credit, will scarcely be believed, except by those who have had constant occasion for their use. A friend of mine, whose skill in practical as well as theoretical astronomy is well known, produced to me a copy of the tables published by order of the French Board of Longitude, containing those of the Sun by Delambre, and of the Moon by Burg, in which he had corrected above *five hundred errors*: most of these appear to be errors of the press; and it is somewhat remarkable, that in turning over the leaves in the fourth page I opened we observed a new error before unnoticed. These errors are so much more dangerous, because independent computers using the same tables will agree in the same errors.

To bring to perfection the various machinery which I have contrived, would require an expense of both time and money, which can be known only to those who have themselves attempted to execute mechanical inventions. Of the greater part of that which has been mentioned, I have a present contented myself with sketches on paper, accompanied by short memorandums, by which I might at any time more fully develop the contrivances; and where any new principles are introduced, I have had models executed, in order to examine their actions. For the purpose of demonstrating the practicability of these views, I have chosen the engine for differences, and have constructed one of them, which will produce any tables whose second differences are constant. Its size is the same as that which I should propose for any more extensive one of the same kind: the chief difference would be, that in one

intended for use there would be a greater repetition of the same parts in order to adapt it to the calculation of a larger number of figures. Of the action of this engine, you have yourself had opportunities of judging, and I will only at present mention a few trials which have since been made by some scientific gentlemen to whom it has been shown, in order to determine the rapidity with which it calculates. The computed table is presented to the eye at two opposite sides of the machine; and a friend having undertaken to write down the numbers as they appeared, it proceeded to make a table from the formula $x^2 + x + 41$. In the earlier numbers my friend, in writing quickly, rather more than kept pace with the engine; but as soon as four figures were required, the machine was at least equal in speed to the writer.

In another trial it was found that thirty numbers of the same table were calculated in two minutes and thirty seconds: as these contained eighty-two figures, the engine produced thirty-three every minute.

In another trial it produced figures as the rate of forty-four in a minute. As the machine may be made to move uniformly by a weight, this rate might be maintained for any length of time, and I believe few writers would be found to copy with equal speed for many hours together. Imperfect as a first machine generally is, and suffering as this particular one does from great defect in the workmanship, I have every reason to be satisfied with the accuracy of its computations; and by the few skillful mechanics to whom I have in confidence shown it, I am assured that its principles are such that it may be carried to any extent. In fact, the parts of which it consists are few but frequently repeated, resembling in this respect the arithmetic to which it is applied, which, by the aid of a few digits often repeated, produces all the wide variety of number. The wheels of which it consists are numerous, but few move at the same time; and I have employed a principle by which any small error that may arise from accident or bad workmanship is corrected as soon as it is produced, in such a manner as effectually to prevent any accumulation of small errors from producing a wrong figure in the calculation.

Of those contrivances by which the composition is to be effected, I have made many experiments and several models; the results of these leave me no reason to doubt of success, which is still further confirmed by a working model that is just finished.

As the engine for calculating tables by the method of differences is the only one yet completed, I shall, in my remarks on the utility of such instruments, confine myself to a statement of the powers which that method possesses.

I would however premise, that if any one shall be of opinion, notwithstanding all the precaution I have taken and means I have employed to guard against the occurrence of error, that it may still be possible for it to arise, the method of differences enables me to determine its existence. Thus, if proper numbers are placed at the outset in the engine, and if it has composed a page of any kind of table, then by comparing the last number it has set up with that number previously calculated, if they are found to agree, the whole page must be correct: should any disagreement occur, it would scarcely be worth the trouble of looking for its origin, as the shortest plan would be to make the engine recalculated the whole page, and nothing would be lost but a few hours' labour of the moving power.

Of the variety of tables which such an engine could calculate, I shall mention but a few. The tables of powers and products published at the expense of the Board of Longitude, and calculated by Dr. Hutton, were solely executed by the method of differences; and other tables of the roots of numbers have been calculated by the same gentleman on similar principles.

As it is not my intention in the present instance to enter into the theory of differences, a field far too wide for the limits of this letter, and which will probably be yet further extended in consequence of the machinery I have contrived, I shall content myself with describing the course pursued in one of the most stupendous monuments of arithmetical calculation which the world has yet produced, and shall point out the mode in which it was conducted, and what share of mental labour would have been saved by the employment of such an engine as I have contrived.

The tables to which I allude are those calculated under the direction of M. Prony, by order of the French Government, – a work which will ever reflect the highest credit on the nation which patronized and on the scientific men who executed it. The tables computed were the following.

1. The natural sines of each 10,000 of the quadrant calculated to twenty-five figures with seven or eight orders of differences.
2. The logarithmic sines of each 100,000 of the quadrant calculated to fourteen decimals with five orders of differences.
3. The logarithm of the ratios of the sines to their arcs of the first 5,000 of the 100,000ths of the quadrant calculated to fourteen decimals with three orders of differences.
4. The logarithmic tangents corresponding to the logarithmic sines calculated to the same extent.
5. The logarithms of the ratios of the tangents to their arcs calculated in the same manner as the logarithms of the ratios of the sines to their arcs.
6. The logarithms of numbers from 1 to 10,000 calculated to nineteen decimals.
7. The logarithms of all numbers from 10,000 to 200,000 calculated to fourteen figures with five orders of differences.

Such are the tables which have been calculated, occupying in their present state seventeen large folio volumes. It will be observed that the trigonometric tables are adapted to the decimal system, which has not been generally adopted even by the French, and which has not been at all employed in this country. But, notwithstanding this objection, such was the opinion entertained of their value, that a distinguished member of the English Board of Longitude was not long since commissioned by our Government to make a proposal to the Board of Longitude of France to print an abridgement of these tables at the joint expense of the two countries; and five thousand pounds were named as the sum our Government was willing to advance for this purpose. It is gratifying to record this disinterested offer, so far above those little jealousies which frequently interfere between nations long rivals, and manifesting so sincere a desire to render useful to mankind the best materials of science in

whatever country they might be produced. Of the reasons why this proposal was declined by our neighbors, I am at present uninformed: but, from a personal acquaintance with many of the distinguished foreigners to whom it was referred, I am convinced that it was received with the same good feelings as those who dictated it.

I will now endeavor shortly to state the manner in which this enormous mass of computation was executed; on table of which (that of the logarithms of numbers) must contain about eight millions of figures.

The calculators were divided into three sections. The first section comprised five or six mathematicians of the highest merit, amongst whom were M. Prony and M. Legendre. These were occupied entirely with the analytical part of the work; they investigated and determined on the formulae to be employed.

The second section consisted of seven or eight skillful calculators habituated both to analytical and arithmetical computations. These received the formulae from the first section, converted them into numbers, and furnished to the third section the proper differences at the stated intervals.

They also received from that section the calculated results, and compared the two sets, which were computed independently for the purpose of verification. The third section, on whom the most laborious part of the operations devolved, consisted of from sixty to eighty persons, few of them possessing a knowledge of more than the first rules of arithmetic: these received from the second class certain numbers and differences, with which, by additions and subtractions in a prescribed order, they completed the whole of the tables above mentioned.

I will now examine what portion of this labour might be dispensed with, in case it should be deemed advisable to compute these or any similar tables of equal extent by the aid of the engine I have referred to.

In the first place, the labour of the first section would be considerably reduced, because the formulae used in the great work I have been describing have already been investigated and published. On person, or at the utmost two, might therefore conduct it.

If the persons composing the second section, instead of delivering the numbers they calculate to the computers of the third section, were to deliver them to the engine, the whole of the remaining operations would be executed by machinery, and it would only be necessary to employ people to copy down as fast as they were able the figures presented to them by the engine. If, however, the contrivances for printing were brought to perfection and employed, even this labour would be unnecessary, and a few superintendents would manage the machine and receive the calculated pages set up in type. Thus the number of calculators employed, instead of amounting to ninety-six, would be reduced to twelve. This number might, however, be considerably diminished, because when an engine is used the intervals between the differences calculated by the second section may be greatly enlarged. In the tables of logarithms M. Prony caused the differences to be calculated at intervals of two hundred, in order to save the labour of the third section: but as that would not devolve on machinery, which would scarcely move the slower for its additional burden,

the intervals might properly be enlarged to three or four times that quantity. This would cause a considerable diminution of the labour of the second section. If to this diminution of mental labour we add that which arises from the whole work of the compositor being executed by the machine, and the total suppression of that most annoying of all literary labour, the correction of the errors in the press ¹⁷, I think I am justified in presuming that if engines were made purposely for this object, and were afterwards useless, the tables could be produced at a much cheaper rate; and of their superior accuracy there could be no doubt. Such engines would, however, be far from useless: containing within themselves the power of generating, to an almost unlimited extent, tables whose accuracy would be unrivalled, at an expense comparatively moderate, they would become active agents in reducing the abstract inquiries of geometry to a form and an arrangement adapted to the ordinary purposes of human society.

I should be unwilling to terminate this Letter without noticing another class of tables of the greatest importance, almost the whole of which are capable of being calculated by the method of differences. I refer to all astronomical tables for determining the positions of the sun or planets: it is scarcely necessary to observe that the constituent parts of these are of the form $a \sin \Theta$, where a is a constant quantity, and Θ is what is usually called the argument. Viewed in this light they differ but little from a table of sines, and like it may be computed by the method of differences.

I am aware that the statements contained in this Letter may perhaps be viewed as something more than Utopian, and that the philosophers of Laputa may be called up to dispute by claim to originality. Should such be the case, I hope the resemblance will be found to adhere to the nature of the subject rather than to the manner in which it has been treated. Conscious, from my own experience, of the difficulty of convincing those who are but little skilled in mathematical knowledge, of the possibility of making a machine which shall perform calculations, I was naturally anxious, in introducing it to the public, to appeal to the testimony of one so distinguished in the records of British science. Of the extent to which the machinery whose nature I have described may be carried, opinions will necessarily fluctuate, until experiment shall have finally decided their relative value: but of that engine which already exists I think I shall be supported, both by yourself and by several scientific friends who have examined it, in stating that it performs with rapidity and precision all those calculations for which it was designed.

Whether I shall construct a larger engine of this kind, and bring to perfection the others I have described, will in a great measure depend on the nature of the encouragement I may receive.

Induced, by a conviction of the great utility of such engines, to withdraw for some time my attention from a subject on which it has been engaged during several years, and which possesses charms of a higher order, I have now arrived at a point where success is no longer doubtful. It must, however, be attained at a very considerable expense, which would not probably be replaced, by the works it might produce, for a long period of time, and which

¹⁷ I have been informed that the publishers of a valuable collection of mathematical tables, now reprinting, pay to the gentlemen employed in correcting the press, at the rate of three guineas a sheet, a sum by no means too large for the faithful execution of such a laborious duty.

is an undertaking I should feel unwilling to commence, as altogether foreign to my habits and pursuits.

I remain, my dear Sir,

Faithfully yours,

C. Babbage

5.3 Sketch of the Analytical Engine Invented by Charles Babbage, Esq.

SKETCH
OF THE
ANALYTICAL ENGINE
INVENTED BY
CHARLES BABBAGE, Esq.
By L. F. MENABREA,
of Turin,
WITH NOTES BY THE TRANSLATOR.
[Extracted from the ‘Scientific Memoirs,’ vol. iii.]

[BEFORE submitting to our readers the translation of M Menabrea's memoir 'On the Mathematical Principles of the Analytical Engine' invented by Mr. Babbage, we shall present to them a list of the printed papers connected with the subject, and also of those relating to the Difference Engine by which it was preceded.

For information on Mr. Babbage's "Difference Engine," which is but slightly alluded to by M. Menabrea, we refer the reader to the following sources: –

1. Letter to Sir Humphry Davy, Bart., P.R.S., on the Application of Machinery to Calculate and Print Mathematical Tables. By Charles Babbage, Esq. F.R.S. London, July 1822. Reprinted, with a Report of the Council of the Royal Society, by order of the House of Commons. May 1823.
2. On the Application of Machinery to the Calculation of Astronomical and Mathematical Tables. By Charles Babbage, Esq. – *Memoirs of the Astronomical Society*, vol. i. part 2. London, 1822.
3. Address to the Astronomical Society by Henry Thomas Colebrooke, Esq., F.R.S., President, on presenting the first Gold Medal of the Society to Charles Babbage, Esq., for the invention of the Calculating Engine. – *Memoirs of the Astronomical Society*. London, 1822.
4. On the Determination of the General Term of a New Class of Infinite Series. By Charles Babbage, Esq. – *Transactions of the Cambridge Philosophical Society*.
5. On Mr. Babbage's New Machine for Calculating and Printing Mathematical Tables. – Letter from Francis Baily, Esq., F.R.S., to M. Schumacher. No. 46, *Astronomische Nachrichten*. Reprinted in the *Philosophical Magazine*, May 1824.
6. On a Method of expressing by Signs the Action of Machinery. By Charles Babbage, Esq. – *Philosophical Transactions*. London, 1826.
7. On Errors common to many Tables of Logarithms. By Charles Babbage, Esq. – *Memoirs of the Astronomical Society*, London, 1827.
8. Report of the Committee appointed by the Council of the Royal Society to consider the subject referred to in a communication received by them from the Treasury respecting Mr. Babbage's Calculating Engine, and to report thereon. London, 1829.
9. *Economy of Manufactures*, chap. xx 8vo. London, 1832.
10. Article on Babbage's Calculating Engine. – *Edinburgh Review*, July 1834. No. 120. vol. lix.

The present state of the Difference Engine, which has always been the property of Government, is as follows: – The drawings are nearly finished, and the mechanical notation of the whole, recording every motion of which it is susceptible, is completed. A part of that Engine, comprising sixteen figures, arranged in three orders of differences, has been put together, and has frequently been used during the last eight years. It performs its work with absolute precision. This portion of the Difference Engine, together with all the drawings, are at present deposited in the Museum of King's College, London.

Of the ANALYTICAL ENGINE, which forms the principal object of the present memoir, we are not aware that any notice has hitherto appeared, except a Letter from the Inventor to M. Quetelet, Secretary to the Royal Academy of Sciences at Brussels, by whom it was communicated to that body. We subjoin a translation of this Letter, which was itself a translation of the original, and was not intended for publication by its author.

Royal Academy of Sciences at Brussels, General Meeting of the 7th and 8th of May, 1835.

“A Letter from Mr. Babbage announces that he has for six months been engaged in making the drawings of a new calculating machine of far greater power than the first.

“ ‘I am myself astonished,’ says Mr. Babbage, ‘at the power I have been enabled to give to this machine; a year ago I should not have believed this result possible. This machine is intended to contain a hundred variables (or numbers susceptible of changing); each of these numbers may consist of twenty-five figures, $v_1, v_2, \dots v_n$ being any numbers whatever, n being less than a hundred; if $f(v_1, v_2, v_3, \dots v_n)$ be any given function which can be formed by addition, subtraction, multiplication, division, extraction of roots, or elevation to powers, the machine will calculate its numerical value; it will afterwards substitute this value in the place of v , or of any other variable, and will calculate this second function with respect to v . It will reduce to tables almost all equations of finite differences. Let us suppose that we have observed a thousand values of a, b, c, d , and that we wish to calculate them by the formula $p = \sqrt{\frac{a+b}{cd}}$, the machine must be set to calculate the formula; the first series of the values of a, b, c, d must be adjusted to it; it will then calculate them, print them, and reduce them to zero; lastly, it will ring a bell to give notice that a new set of constants must be inserted. When there exists a relation between any number of successive coefficients of a series, provided it can be expressed as has already been said, the machine will calculate them and make their terms known in succession; and it may afterwards be disposed so as to find the value of the series for all the values of the variable.’

“Mr. Babbage announces, in conclusion, ‘that the greatest of the invention have already been surmounted, and that the plans will be finished in a few months.’ ”

In the Ninth Bridgewater Treatise, Mr. Babbage has employed several arguments deduced from the Analytical Engine, which afford some idea of its powers. See Ninth Bridgewater Treatise, 8vo, second edition. London, 1834.

Some of the numerous drawings of the Analytical Engine have been engraved on wooden blocks, and from these (by a mode contrived by Mr. Babbage) various stereotype plates have been taken. They comprise –

1. Plan of the figure wheels for one method of adding numbers.
2. Elevation of the wheels and axis of ditto.
3. Elevation of framing only of ditto.
4. Section of adding wheels and framing together.
5. Section of the adding wheels, sign wheels and framing complete.
6. Impression from the original wood block.

7. Impressions from a stereotype cast of No. 6, with the letters and signs inserted. Nos. 2, 3, 4 and 5 were stereotypes taken from this.
8. Plan of adding wheels and of long and short pinions by means of which *stepping* is accomplished
 N.B. This process performs the operation of multiplying or dividing a number by any power of ten.
9. Elevation of long pinions in the position for addition.
10. Elevation of long pinions in the position for stepping.
11. Plan of mechanism for carrying the tens (by anticipation), connected with long pinions.
12. Section of the chain of wires for anticipating carriage.
13. Sections of the elevation of parts of the preceding carriage.

All these were executed about five years ago. At a later period (August 1840) Mr. Babbage caused one of his general plans (No. 25) of the whole Analytical Engine to be lithographed at Paris.

Although these illustrations have not been published, on account of the time which would be required to describe them, and the rapid succession of improvements made subsequently, yet copies have been freely given to many of Mr. Babbage's friends, and were in August 1838 presented at Newcastle to the British Association for the Advancement of Science, and in August 1840 to the Institute of France through M. Arago, as well as to the Royal Academy of Turin through M. Plana. – Editor]

Those labours which belong to the various branches of the mathematical sciences, although on first consideration they seem to be the exclusive province of intellect, may, nevertheless, be divided into two distinct sections; one of which may be called the mechanical, because it is subjected to precise and invariable laws, that are capable of being expressed by means of the operations of matter; while the other, demanding the intervention of reasoning, belongs more specially to the domain of the understanding. This admitted, we may propose to execute, by means of machinery, the mechanical branch of these labours, reserving for pure intellect that which depends on the reasoning faculties. Thus the rigid exactness of those laws which regulate numerical calculations must frequently have suggested the employment of material instruments, either for executing the whole of such calculations or for abridging them; and thence have arisen several inventions having this object in view, but which have in general but partially attained it. For instance, the much-admired machine of Pascal is now simply an object of curiosity, which, whilst it displays the powerful intellect of its inventor, is yet of little utility in itself. Its powers extended no further than the execution of the four¹⁸ first operations of arithmetic, and indeed were in reality confined to that of

¹⁸ This remark seems to require further comment, since it is in some degree calculated to strike the mind as being at variance with the subsequent passage (page 109), where it is explained that *an engine which*

the two first, since multiplication and division were the result of a series of additions and subtractions. The chief drawback hitherto on most of such machines is, that they require the continual intervention of a human agent to regulate their movements, and thence arises a source of errors; so that, if their use has not become general for large numerical calculations, it is because they have not in fact resolved the double problem which the question presents, that of *correctness* in the results united with a *æconomy* of time.

Struck with similar reflections, Mr. Babbage has devoted some years to the realization of a gigantic idea. He proposed to himself nothing less than the construction of a machine capable of executing not merely arithmetical calculations, but even all those of analysis, if their laws are known. The imagination is at first astounded at the idea of such an undertaking; but the more calm reflection we bestow on it, the less impossible does success appear, and it is felt that it may depend on the discovery of some principle so general, that if applied to machinery, the latter may be capable of mechanically translating the operations which may be indicated to it by algebraical notation. The illustrious inventor having been kind enough to communicate to me some of his views on this subject during a visit he made at Turin, I have, with his approbation, thrown together the impressions they have left on my mind. But the reader must not expect to find a description of Mr. Babbage's engine; the comprehension of this would entail studies of much length; and I shall endeavour merely to give an insight into the end proposed, and to develop the principles on which its attainment depends.

I must first premise that this engine is entirely different from that of which there is a notice in the 'Treatise on the Æconomy of Machinery,' by the same author. But as the latter gave rise¹⁹ to the idea of the engine in question, I consider it will be a useful preliminary briefly to recall what were Mr. Babbage's first essays, and also the circumstances in which they originated.

It is well known that the French government, wishing to promote the extension of the decimal system, had ordered the construction of logarithmical and trigonometrical tables

can effect these four operations can in fact effect *every species of calculation*. The apparent discrepancy is stronger too in the translation than in the original, owing to its being impossible to render precisely into the English tongue all the niceties of distinction which the French idiom happens to admit of in the phrases used for the two passages we refer to. The explanation lies in this: that in the one case the execution of these four operations is the *fundamental starting point*, and the object proposed for attainment by the machine is the *subsequent combination of these* in every possible variety; whereas in the other case the execution of some *one* of these four operations, selected at pleasure, is the *ultimatum*, the sole and utmost result that can be proposed for attainment by the machine referred to, and which result it cannot any further combine or work upon. The one *begins* where the other *ends*. Should this distinction not now appear perfectly clear, it will become so on perusing the rest of the Memoir, and the Notes that are appended to it. – Note by Translator.

¹⁹ The idea that the one engine is the offspring and has grown out of the other, is an exceedingly natural and plausible supposition, until reflection reminds us that no *necessary* sequence and connexion need exist between two such inventions, and that they *may* be wholly independent. M. Menabrea has shared this idea in common with persons who have not his profound and accurate insight into the nature of either engine. In Note A. (see the Notes at the end of the Memoir) it will be found sufficiently explained, however, that this supposition is unfounded. M. Menabrea's opportunities were by no means such as could be adequate to afford him information on a point like this, which would be naturally and almost unconsciously *assumed*, and would scarcely suggest any inquiry with reference to it. – Note by Translator.

of enormous extent. M. de Prony, who had been entrusted with the direction of this undertaking, divided it into three sections, to each of which were appointed a special class of persons. In the first section the formulae were so combined as to render them subservient to the purposes of numerical calculation; in the second, these same formulae were calculated for values of the variable, selected at certain successive distances; and under the third section, comprising about eighty individuals, who were most of them only acquainted with the two first rules of arithmetic, the values which were intermediate to those calculated by the second section were interpolated by means of simple additions and subtractions.

An undertaking similar to that just mentioned having been entered upon in England, Mr. Babbage conceived that the operations performed under the third section might be executed by a machine; and this idea he realized by means of mechanism, which has been in part put together, and to which the name Difference Engine is applicable, on account of the principle upon which its construction is founded. To give some notion of this, it will suffice to consider the series of whole square numbers, 1, 4, 9, 16, 25, 36, 49, 64, &c. By subtracting each of these from the succeeding one, we obtain a new series, which we will name the Series of First Differences, consisting of the numbers 3, 5, 7, 9, 11, 13, 15, &c. On subtracting from each of these the preceding one, we obtain the Second Differences, which are all constant and equal to 2. We may represent this succession of operations, and their results, in the following table: –

	A. Column of Square Numbers	B. First Differences	C. Second Differences
	1		
	4	3	
	9	5	2 <i>b</i>
<i>a</i>	16	7	2 <i>d</i>
	25	9	2
<i>c</i>	36	11	2

From the mode in which the two last columns B and C have been formed, it is easy to see that if, for instance, we desire to pass from the number 5 to the succeeding one 7, we must add to the former the constant difference 2; similarly, if from the square number 9 we would pass to the following one 16, we must add to the former the difference 7, which

difference is in other words the preceding difference 5, plus the constant difference 2; or again, which comes to the same thing, to obtain 16 we have only to add together the three numbers 2, 5, 9, placed obliquely in the direction *ab*. Similarly, we obtain the number 25 by summing up the three numbers placed in the oblique direction *dc*: commencing by the addition $2 + 7$, we have the first difference 9 consecutively to 7; adding 16 to the 9 we have the square 25. We see then that the three numbers 2, 5, 9 being given, the whole series of successive square numbers, and that of their first differences likewise, may be obtained by means of simple additions.

Now, to conceive how these operations may be reproduced by a machine, suppose the latter to have three dials, designated as A, B, C, on each of which are traced, say a thousand divisions, by way of example, over which a needle shall pass. The two dials C, B, shall have in addition a registering hammer, which is to give a number of strokes equal to that of the divisions indicated by the needle. For each stroke of the registering hammer of the dial C, the needle B shall advance one division; similarly, the needle A shall advance one division for every stroke of the registering hammer of the dial B. Such is the general disposition of the mechanism.

This being understood, let us at the beginning of the series of operations we wish to execute, place the needle C on the division 2, the needle B on the division 5, and the needle A on the division 9. Let us allow the hammer of the dial C to strike; it will strike twice, and at the same time the needle B will pass over two divisions. The latter will then indicate the number 7, which succeeds the number 5 in the column of first differences. If we now permit the hammer of the dial B to strike in its turn, it will strike seven times, during which the needle A will advance seven divisions; these added to the nine already marked by it, will give the number 16, which is the square number consecutive to 9. If we now recommence these operations, beginning with the needle C, which is always to be left on the division 2, we shall perceive that by repeating them indefinitely, we may successively reproduce the series of whole square numbers by means of a very simple mechanism.

The theorem on which is based the construction of the machine we have just been describing, is a particular case of the following more general theorem: that if in any polynomial whatever, the highest power of whose variable is m , this same variable be increased by equal degrees; the corresponding values of the polynomial then calculated, and the first, second, third, &c. differences of these be taken (as for the preceding series of squares); the m th differences will all be equal to each other. So that, in order to reproduce the series of values of the polynomial by means of a machine analogous to the one above described, it is sufficient that there be $(m + 1)$ dials, having the mutual relations we have indicated. As the differences may be either positive or negative, the machine will have a contrivance for either advancing or retrograding each needle, according as the number to be algebraically added may have the sign *plus* or *minus*.

If from a polynomial we pass to a series having an infinite number of terms, arranged according to the ascending powers of the variable, it would at first appear, that in order to apply the machine to the calculation of the function represented by such a series, the mechanism must include an infinite number of dials, which would in fact render the thing impossible. But in many cases the difficulty will disappear, if we observe that for a great

number of functions the series which represent them may be rendered convergent; so that, according to the degree of approximation desired, we may limit ourselves to the calculation of a certain number of terms of the series, neglecting the rest. By this method the question is reduced to the primitive case of a finite polynomial. It is thus that we can calculate the succession of the logarithms of numbers. But since, in this particular instance, the terms which had been originally neglected receive increments in a ratio so continually increasing for equal increments of the variable, that the degree of approximation required would ultimately be affected, it is necessary, at certain intervals, to calculate the value of the function by different methods, and then respectively to use the results thus obtained, as data whence to deduce, by means of the machine, the other intermediate values. We see that the machine here performs the office of the third section of calculators mentioned in describing the tables computed by order of the French government, and that the end originally proposed is thus fulfilled by it.

Such is the nature of the first machine which Mr. Babbage conceived. We see that its use is confined to cases where the numbers required are such as can be obtained by means of simple additions or subtractions: that the machine is, so to speak, merely the expression of one²⁰ particular theorem of analysis; and that, in short, its operations cannot be extended so as to embrace the solution of an infinity of other questions included within the domain of mathematical analysis. It was while contemplating the vast field which yet remained to be traversed, that Mr. Babbage, renouncing his original essays, conceived the plan of another system of mechanism whose operations should themselves possess all the generality of algebraical notation, and which, on this account, he denominates the *Analytical Engine*.

Having now explained the state of the question, it is time for me to develop the principle on which is based the construction of this latter machine. When analysis is employed for the solution of any problem, there are usually two classes of operations to execute: firstly, the numerical calculation of the various coefficients; and secondly, their distribution in relation to the quantities affected by them. If, for example, we have to obtain the product of two binomials $(a + bx)(m + nx)$, the result will be represented by $am + (an + bm)x + bnx^2$, in which expression we must first calculate am, an, bm, bn ; then take the sum of $an + bm$; and lastly, respectively distribute the coefficients thus obtained, amongst the powers of the variable. In order to reproduce these operations by means of a machine, the latter must therefore possess two distinct sets of powers: first, that of executing numerical calculations; secondly, that of rightly distributing the values so obtained.

But if human intervention were necessary for directing each of these partial operations, nothing would be gained under the heads of correctness and æconomy of time; the machine must therefore have the additional requisite of executing by itself all the successive operations required for the solution of a problem proposed to it, when once the *primitive numerical data* for this same problem have been introduced. Therefore, since from the moment that the nature of the calculation to be executed or of the problem to be resolved have been indicated to it, the machine is, by its own intrinsic power, of itself to go through all the intermediate operations which lead to the proposed result, it must exclude all methods

²⁰ See Note A.

of trial and guess-work, and can only admit the direct processes of calculation²¹.

It is necessarily thus; for the machine is not a thinking being, but simply an automaton which acts according to the laws imposed upon it. This being fundamental, one of the earliest researches its author had to undertake, was that of finding means for effecting the division of one number by another without using the method of guessing indicated by the usual rules of arithmetic. The difficulties of effecting this combination were far from being among the least; but upon it depended the success of every other. Under the impossibility of my here explaining the process through which this end is attained, we must limit ourselves to admitting that the four first operations of arithmetic, that is addition, subtraction, multiplication and division, can be performed in a direct manner through the intervention of the machine. This granted, the machine is thence capable of performing every species of numerical calculation, for all such calculations ultimately resolve themselves into the four operations we have just named. To conceive how the machine can now go through its functions according to the laws laid down, we will begin by giving an idea of the manner in which it materially represents numbers.

Let us conceive a pile or vertical column consisting of an indefinite number of circular discs, all pierced through their centres by a common axis, around which each of them can take an independent rotatory movement. If round the edge of each of these discs are written the ten figures which constitute our numerical alphabet, we may then, by arranging a series of these figures in the same vertical line, express in this manner any number whatever. It is sufficient for this purpose that the first disc represent units, the second tens, the third hundreds, and so on. When two numbers have been thus written on two distinct columns, we may propose to combine them arithmetically with each other, and to obtain the result on a third column. In general, if we have a series of columns²² consisting of discs, which columns we will designate as V_0, V_1, V_2, V_3, V_4 , &c, we may require, for instance, to divide the number written on the column V_1 by that on the column V_4 , and to obtain the result on the column V_7 . To effect this operation, we must impart to the machine two distinct arrangements; through the first it is prepared for executing a *division*, and through the second the columns it is to operate on are indicated to it, and also the column on which the result is to be represented. If this division is to be followed, for example, by the addition of two numbers taken on other columns, the two original arrangements of the machine must be simultaneously altered. If, on the contrary, a series of operations of the same nature is to be gone through, then the first of the original arrangements will remain, and the second alone must be altered. Therefore, the arrangements that may be communicated to the various parts of the machine, may be distinguished into two principal classes:

First, that relative to the *Operations*.

Secondly, that relative to the *Variables*.

By this latter we mean that which indicates the columns to be operated on. As for the operations themselves, they are executed by a special apparatus, which is designated by

²¹This must not be understood in too unqualified a manner. The engine is capable, under certain circumstances, of feeling about to discover which of two or more possible contingencies has occurred, and of then shaping its future course accordingly. – Note by Translator.

²² See Note B.

the name of *mill*, and which itself contains a certain number of columns, similar to those of the Variables. When two numbers are to be combined together, the machine commences by effacing them from the columns where they are written, that is it places *zero*²³ on every disc of the two vertical lines on which the numbers were represented; and it transfers the numbers to the mill. There, the apparatus having been disposed suitably for the required operation, this latter is effected, and, when completed, the result itself is transferred to the column of Variables which shall have been indicated. Thus the mill is that portion of the machine which works, and the columns of Variables constitute that where the results are represented and arranged. After the preceding explanations, we may perceive that all fractional and irrational results will be represented in decimal fractions. Supposing each column to have forty discs, this extension will be sufficient for all degrees of approximation generally required.

It will now be inquired how the machine can of itself, and without having recourse to the hand of man, assume the successive dispositions suited to the operations. The solution of this problem has been taken from Jacquard's apparatus²⁴, used for the manufacture of brocaded stuffs, in the following manner: –

Two species of threads are usually distinguished in woven stuffs; one is the *warp* or longitudinal thread, the other the *woof* or transverse thread, which is conveyed by the instrument called the shuttle, and which crosses the longitudinal thread or warp. When a brocaded stuff is required, it is necessary in turn to prevent certain threads from crossing the woof, and this according to a succession which is determined by the nature of the design that is to be reproduced. Formerly this process was lengthy and difficult, and it was requisite that the workman, by attending to the design which he was to copy, should himself regulate the movements the threads were to take. Thence arose the high price of this description of stuffs, especially if threads of various colours entered into the fabric. To simplify this manufacture, Jacquard devised the plan of connecting each group of threads that were to act together, with a distinct lever belonging exclusively to that group. All these levers terminate in rods, which are united together in one bundle, having usually the form of a parallelopiped with a rectangular base. The rods are cylindrical, and are separated from each other by small intervals. The process of raising the threads is thus resolved into that of moving these various lever-arms in the requisite order. To effect this, a rectangular sheet of pasteboard is taken, somewhat larger in size than a section of the bundle of lever-arms. If this sheet be applied to the base of the bundle, and an advancing motion be then communicated to the pasteboard, this latter will move with it all the rods of the bundle, and consequently the threads that are connected with each of them. But if the pasteboard, instead of being plain, were pierced with holes corresponding to the extremities of the levers which meet it, then, since each of the levers would pass through the pasteboard during the motion of the latter, they would all remain in their places. We thus see that it is easy so to determine the position of the holes in the pasteboard, that, at any given moment, there shall be a certain number of levers, and consequently of parcels of threads, raised, while the rest remain where they were. Supposing this process is successively repeated according

²³ Zero is not *always* substituted when a number is transferred to the mill. This is explained further on in the memoir, and still more fully in Note D. – Note by Translator.

²⁴ See Note C.

to a law indicated by the pattern to be executed, we perceive that this pattern may be reproduced on the stuff. For this purpose we need merely compose a series of cards according to the law required, and arrange them in suitable order one after the other; then, by causing them to pass over a polygonal beam which is so connected as to turn a new face for every stroke of the shuttle, which face shall then be impelled parallelly to itself against the bundle of lever-arms, the operation of raising the threads will be regularly performed. Thus we see that brocaded tissues may be manufactured with a precision and rapidity formerly difficult to obtain.

Arrangements analogous to those just described have been introduced into the Analytical Engine. It contains two principal species of cards: first, Operation cards, by means of which the parts of the machine are so disposed as to execute any determinate series of operations, such as additions, subtractions, multiplications, and divisions; secondly, cards of the Variables, which indicate to the machine the columns on which the results are to be represented. The cards, when put in motion, successively arrange the various portions of the machine according to the nature of the processes that are to be effected, and the machine at the same time executes these processes by means of the various pieces of mechanism of which it is constituted.

In order more perfectly to conceive the thing, let us select as an example the resolution of two equations of the first degree with two unknown quantities. Let the following be the two equations, in which x and y are the unknown quantities: –

$$\begin{cases} mx + ny &= d \\ m'x + n'y &= d' \end{cases}$$

We deduce $x = \frac{dn' - d'n}{n'm - nm'}$, and for y an analogous expression.

Let us continue to represent V_0, V_1, V_2 , &c. the different columns which contain the numbers, and let us suppose that the first eight columns have been chosen for expressing on them the numbers represented by m, n, d, m', n', d', n and n' , which implies that $V_0 = m, V_1 = n, V_2 = d, V_3 = m', V_4 = n', V_5 = d', V_6 = n, V_7 = n'$.

The series of operations commanded by the cards, and the results obtained, may be represented in the following table: –

Number of the operations	Operation-cards	Cards of the variables		Progress of the operations
	Symbols indicating the nature of the operations	Columns on which operations are to be performed	Columns which receive results of operations	
1	$\times$	$V_2 \times V_4 =$	$V_8 \dots\dots$	$= dn'$
2	$\times$	$V_5 \times V_1 =$	$V_9 \dots\dots$	$= d'n$
3	$\times$	$V_4 \times V_0 =$	$V_{10} \dots\dots$	$= n'm$
4	$\times$	$V_1 \times V_3 =$	$V_{11} \dots\dots$	$= nm'$
5	$-$	$V_8 - V_9 =$	$V_{12} \dots\dots$	$= dn' - d'n$
6	$-$	$V_{10} - V_{11} =$	$V_{13} \dots\dots$	$= n'm - nm'$
7	$\div$	$\frac{V_{12}}{V_{13}} =$	$V_{14} \dots\dots$	$= x = \frac{dn' - d'n}{n'm - nm'}$

Since the cards do nothing but indicate in what manner and on what columns the machine shall act, it is clear that we must still, in every particular case, introduce the numerical data for the calculation. Thus, in the example we have selected, we must previously inscribe the numerical values of m, n, d, m', n', d' , in the order and on the columns indicated, after which the machine when put in action will give the value of the unknown quantity x for this particular case. To obtain the value of y , another series of operations analogous to the preceding must be performed. But we see that they will be only four in number, since the denominator of the expression for y , excepting the sign, is the same as that for x , and equal to $n'm - nm'$. In the preceding table it will be remarked that the column for operations indicates four successive *multiplications*, two *subtractions*, and one *division*. Therefore, if desired, we need only use three operation cards; to manage which, it is sufficient to introduce into the machine an apparatus which shall, after the first multiplication, for instance, retain the card which relates to this operation, and not allow it to advance so as to be replaced by another one, until after this same operation shall have been four times repeated. In the preceding example we have seen, that to find the value of x we must begin by writing the coefficients m, n, d, m', n', d' , upon eight columns, thus repeating n and n' twice. According to the same method, if it were required to calculate y likewise, these coefficients must be written on twelve different columns. But it is possible to simplify this process, and thus to diminish the chances of errors, which chances are greater, the larger the number of the quantities that have to be inscribed previous to setting the machine in action. To understand this simplification, we must remember that every number written on a column must, in order to be arithmetically combined with another number, be effaced from the column on which it is, and transferred to the *mill*. Thus, in the example we have discussed, we will take the two coefficients m and n' , which are each of them to enter into *two* different products, that is m into mn' and md' , n' into mn' and $n'd$. These coefficients will be inscribed on the columns V_0 and V_4 . If we commence the series of operations by the product of m into n' , these numbers will be effaced from the columns V_0 and V_4 , that they may be transferred to the mill, which will multiply them into each other, and will then command the machine to represent the result, say on the column V_6 . But as these numbers are each to be used again in another operation, they must again be inscribed somewhere; therefore, while the mill is working out their product, the machine will inscribe them anew on any two columns that may be indicated to it through the cards; and, as in the actual case, there is no reason why they should not resume their former places, we will suppose

them again inscribed on V_0 and V_4 , whence in short they would not finally disappear, to be reproduced no more, until they should have gone through all the combinations in which they might have to be used.

We see, then, that the whole assemblage of operations requisite for resolving the two²⁵ above equations of the first degree, may be definitively represented in the following table:

—

²⁵ See Note D.

Columns on which are inscribed the primitive data	Number of the operations	Cards of the operations		Variable cards			Statement of results
		No. of the Oper'n-cards	Nature of each oper'n	Columns acted on by each operation	Columns that require the result of each operation	Indication of change of value on any column	
${}^1V_0 = m$	1	1	$\times$	${}^1V_0 \times {}^1V_4 =$	${}^1V_6, \dots$	$\begin{cases} {}^1V_0 = {}^1V_0 \\ {}^1V_4 = {}^1V_4 \end{cases}$	${}^1V_6 = mn'$
${}^1V_1 = n$	2	"	$\times$	${}^1V_3 \times {}^1V_1 =$	${}^1V_7, \dots$	$\begin{cases} {}^1V_3 = {}^1V_3 \\ {}^1V_1 = {}^1V_1 \end{cases}$	${}^1V_7 = m'n$
${}^1V_2 = d$	3	"	$\times$	${}^1V_2 \times {}^1V_4 =$	${}^1V_8, \dots$	$\begin{cases} {}^1V_2 = {}^1V_2 \\ {}^1V_4 = {}^0V_4 \end{cases}$	${}^1V_8 = dn'$
${}^1V_3 = m'$	4	"	$\times$	${}^1V_5 \times {}^1V_1 =$	${}^1V_9, \dots$	$\begin{cases} {}^1V_5 = {}^1V_5 \\ {}^1V_1 = {}^0V_1 \end{cases}$	${}^1V_9 = d'n$
${}^1V_4 = n'$	5	"	$\times$	${}^1V_0 \times {}^1V_5 =$	${}^1V_{10}, \dots$	$\begin{cases} {}^1V_0 = {}^0V_0 \\ {}^1V_5 = {}^0V_5 \end{cases}$	${}^1V_{10} = d'm$
${}^1V_5 = d'$	6	"	$\times$	${}^1V_2 \times {}^1V_3 =$	${}^1V_{11}, \dots$	$\begin{cases} {}^1V_2 = {}^0V_2 \\ {}^1V_3 = {}^0V_3 \end{cases}$	${}^1V_{11} = dm'$
	7	2	$-$	${}^1V_6 - {}^1V_7 =$	${}^1V_{12}, \dots$	$\begin{cases} {}^1V_6 = {}^0V_6 \\ {}^1V_7 = {}^0V_7 \end{cases}$	${}^1V_{12} = mn' - m'n$
	8	"	$-$	${}^1V_8 - {}^1V_9 =$	${}^1V_{13}, \dots$	$\begin{cases} {}^1V_8 = {}^0V_8 \\ {}^1V_9 = {}^0V_9 \end{cases}$	${}^1V_{13} = dn' - d'n$
	9	"	$-$	${}^1V_{10} - {}^1V_{11} =$	${}^1V_{14}, \dots$	$\begin{cases} {}^1V_{10} = {}^0V_{10} \\ {}^1V_{11} = {}^0V_{11} \end{cases}$	${}^1V_{14} = d'm - dm'$
	10	3	$\div$	${}^1V_{13} \div {}^1V_{12} =$	${}^1V_{15}, \dots$	$\begin{cases} {}^1V_{13} = {}^0V_{13} \\ {}^1V_{12} = {}^1V_{12} \end{cases}$	${}^1V_{15} = \frac{dn' - d'n}{mn' - m'n} = x$
	11	"	$\div$	${}^1V_{14} \div {}^1V_{12} =$	${}^1V_{16}, \dots$	$\begin{cases} {}^1V_{14} = {}^0V_{14} \\ {}^1V_{12} = {}^0V_{14} \end{cases}$	${}^1V_{16} = \frac{d'm - dm'}{mn' - m'n} = y$
1	2	3	4	5	6	7	8

In order to diminish to the utmost the chances of error in inscribing the numerical data of the problem, they are successively placed on one of the columns of the mill; then, by means of cards arranged for this purpose, these same numbers are caused to arrange themselves on the requisite columns, without the operator having to give his attention to it; so that his undivided mind may be applied to the simple inscription of these same numbers.

According to what has now been explained, we see that the collection of columns of Variables may be regarded as a *store* of numbers, accumulated there by the mill, and which, obeying the orders transmitted to the machine by means of the cards, pass alternately from the mill to the store, and from the store to the mill, that they may undergo the transformations demanded by the nature of the calculation to be performed.

Hitherto no mention has been made of the *signs* in the results, and the machine would be far from perfect were it incapable of expressing and combining amongst each other positive and negative quantities. To accomplish this end, there is, above every column, both of the mill and of the store, a disc, similar to the discs of which the columns themselves consist. According as the digit on this disc is even or uneven, the number inscribed on the corresponding column below it will be considered as positive or negative. This granted, we may, in the following manner, conceive how the signs can be algebraically combined in the machine. When a number is to be transferred from the store to the mill and *vice versa*, it

will always be transferred with its sign, which will be effected by means of the cards, as has been explained in what precedes. Let any two numbers then, on which we are to operate arithmetically, be placed in the mill with their respective signs. Suppose that we are first to add them together; the operation-cards will command the addition: if the two numbers be of the same sign, one of the two will be entirely effaced from where it was inscribed, and will go to add itself on the column which contains the other number; the machine will, during this operation, be able, by means of a certain apparatus, to prevent any movement in the disc of signs which belongs to the column on which the addition is made, and thus the result will remain with the sign which the two given numbers originally had. When two numbers have two different signs, the addition commanded by the card will be changed into a subtraction through the intervention of mechanisms which are brought into play by this very difference of sign. Since the subtraction can only be effected on the larger of the two numbers, it must be arranged that the disc of signs of the larger number shall not move while the smaller of the two numbers is being effaced from its column and subtracted from the other, whence the result will have the sign of this latter, just as in fact it ought to be. The combinations to which algebraical subtraction give rise, are analogous to the preceding. Let us pass on to multiplication. When two numbers to be multiplied are of the same sign, the result is positive; if the signs are different, the product must be negative. In order that the machine may act conformably to this law, we have but to conceive that on the column containing the product of the two given numbers, the digit which indicates the sign of that product, has been formed by the mutual addition of the two digits that respectively indicated the signs of the two given numbers; it is then obvious that if the digits of the signs are both even, or both odd, their sum will be an even number, and consequently will express a positive number; but that if, on the contrary, the two digits of the signs are one even and the other odd, their sum will be an odd number, and will consequently express a negative number. In the case of division, instead of adding the digits of the discs, they must be subtracted one from the other, which will produce results analogous to the preceding; that is to say, that if these figures are both even or both uneven, the remainder of this subtraction will be even; and it will be uneven in the contrary case. When I speak of mutually adding or subtracting the numbers expressed by the digits of the signs, I merely mean that one of the sign-discs is made to advance or retrograde a number of divisions equal to that which is expressed by the digit on the other sign-disc. We see, then, from the preceding explanation, that it is possible mechanically to combine the signs of quantities so as to obtain results conformable to those indicated by algebra²⁶.

The machine is not only capable of executing those numerical calculations which depend on a given algebraical formula, but it is also fitted for analytical calculations in which there are one or several variables to be considered. It must be assumed that the analytical expression to be operated on can be developed according to powers of the variable, or according to determinate functions of this same variable, such as circular functions, for instance; and similarly for the result that is to be attained. If we then suppose that above

²⁶ Not having had leisure to discuss with Mr Babbage the manner of introducing into his machine the combination of algebraical signs, I do not pretend here to expose the method he uses for this purpose; but I considered that I ought myself to supply the deficiency, conceiving that this paper would have been imperfect if I had omitted to point out one means that might be employed for resolving this essential part of the problem in question.

the columns of the store, we have inscribed the powers or the functions of the variable, arranged according to whatever is the prescribed law of development, the coefficients of these several terms may be respectively placed on the corresponding column below each. In this manner we shall have a representation of an analytical development; and, supposing the position of the several terms composing it to be invariable, the problem will be reduced to that of calculating their coefficients according to the laws demanded by the nature of the question. In order to make this more clear, we shall take the following²⁷ very simple example, in which we are to multiply $(a + bx')$ by $(A + B \cos^1 x)$. We shall begin by writing $x^0, x^1, \cos^0 x, \cos^1 x$, above the columns V_0, V_1, V_2, V_3 ; then, since from the form of the two functions to be combined, the terms which are to compose the products will be of the following nature, $x^0 \cdot \cos^0 x, x^0 \cdot \cos^1 x, x^1 \cdot \cos^0 x, x^1 \cdot \cos^1 x$; these will be inscribed above the columns V_4, V_5, V_6, V_7 . The coefficients of a $x^0, x^1, \cos^0 x, \cos^1 x$ being given, they will, by means of the mill, be passed to the columns V_0, V_1, V_2 and V_3 . Such are the primitive data of the problem. It is now the business of the machine to work out its solution, that is to find the coefficients which are to be inscribed on V_4, V_5, V_6, V_7 . To attain this object, the law of formation of these same coefficients being known, the machine will act through the intervention of the cards, in the manner indicated by the following table: —²⁸

(?) Columns above which are written the functions of the variable.	Coefficients.		Cards of the operations.		Cards of the variables.			
	Given.	To be formed.	Number of the operations.	Nature of the operation.	Columns on which operations are to be performed.	Columns on which are to be inserted the results of the operations.	Indication of change of value on any column submitted to an operation.	Results of the operations.
$x^0 \dots \dots \dots {}^1V_0$	a	"	"	"	"	"	"	" "
$x^1 \dots \dots \dots {}^1V_1$	b	"	"	"	"	"	"	" "
$\cos^0 x \dots \dots \dots {}^1V_2$	A	"	"	"	"	"	"	" "
$\cos^1 x \dots \dots \dots {}^1V_3$	B	"	"	"	"	"	"	" "
$x^0 \cos^0 x \dots \dots {}^0V_4$	...	aA	1	$\times$	${}^1V_0 \times {}^1V_2 =$	${}^1V_4 \dots \dots$	$\begin{Bmatrix} {}^1V_0 = {}^1V_0 \\ {}^1V_2 = {}^1V_2 \end{Bmatrix}$	${}^1V_4 = aA$ coefficients of $x^0 \cos^0 x$
$x^0 \cos^1 x \dots \dots {}^0V_5$	...	aB	2	$\times$	${}^1V_0 \times {}^1V_3 =$	${}^1V_5 \dots \dots$	$\begin{Bmatrix} {}^1V_0 = {}^0V_0 \\ {}^1V_3 = {}^1V_3 \end{Bmatrix}$	${}^1V_5 = aB \dots \dots x^0 \cos^1 x$
$x^1 \cos^0 x \dots \dots {}^0V_6$	...	bA	3	$\times$	${}^1V_1 \times {}^1V_2 =$	${}^1V_6 \dots \dots$	$\begin{Bmatrix} {}^1V_1 = {}^1V_1 \\ {}^1V_2 = {}^0V_2 \end{Bmatrix}$	${}^1V_6 = bA \dots \dots x^1 \cos^0 x$
$x^1 \cos^1 x \dots \dots {}^0V_7$	...	bB	4	$\times$	${}^1V_1 \times {}^1V_3 =$	${}^1V_7 \dots \dots$	$\begin{Bmatrix} {}^1V_1 = {}^0V_1 \\ {}^1V_3 = {}^0V_3 \end{Bmatrix}$	${}^1V_7 = bB \dots \dots x^1 \cos^1 x$

It will now be perceived that a general application may be made of the principle developed in the preceding example, to every species of process which it may be proposed to effect on series submitted to calculation. It is sufficient that the law of formation of the coefficients be known, and that this law be inscribed on the cards of the machine, which will then

²⁷ See Note E.

²⁸ For an explanation of the upper left-hand indices attached to the V 's in this and in the preceding Table, we must refer the reader to Note D, amongst those appended to the memoir. — Note by Translator.

of itself execute all the calculations requisite for arriving at the proposed result. If, for instance, a recurring series were proposed, the law of formation of the coefficients being here uniform, the same operations which must be performed for one of them will be repeated for all the others; there will merely be a change in the locality of the operation, that is it will be performed with different columns. Generally, since every analytical expression is susceptible of being expressed in a series ordered according to certain functions of the variable, we perceive that the machine will include all analytical calculations which can be definitively reduced to the formation of coefficients according to certain laws, and to the distribution of these with respect to the variables.

We may deduce the following important consequence from these explanations, viz. that since the cards only indicate the nature of the operations to be performed, and the columns of Variables with which they are to be executed, these cards will themselves possess all the generality of analysis, of which they are in fact merely a translation. We shall now further examine some of the difficulties which the machine must surmount, if its assimilation to analysis is to be complete. There are certain functions which necessarily change in nature when they pass through zero or infinity, or whose values cannot be admitted when they pass these limits. When such cases present themselves, the machine is able, by means of a bell, to give notice that the passage through zero or infinity is taking place, and it then stops until the attendant has again set it in action for whatever process it may next be desired that it shall perform. If this process has been foreseen, then the machine, instead of ringing, will so dispose itself as to present the new cards which have relation to the operation that is to succeed the passage through zero and infinity. These new cards may follow the first, but may only come into play contingently upon one or other of the two circumstances just mentioned taking place.

Let us consider a term of the form ab^n ; since the cards are but a translation of the analytical formula, their number in this particular case must be the same, whatever be the value of n ; that is to say, whatever be the number of multiplications required for elevating b to the n th power (we are supposing for the moment that n is a whole number). Now, since the exponent n indicates that b is to be multiplied n times by itself, and all these operations are of the same nature, it will be sufficient to employ one single operation card, viz. that which orders the multiplication.

But when n is given for the particular case to be calculated, it will be further requisite that the machine limit the number of its multiplications according to the given values. The process may be thus arranged. The three numbers a, b and n will be written on as many distinct columns of the store; we shall designate them V_0, V_1, V_2 ; the result ab^n will place itself on the column V_3 . When the number n has been introduced into the machine, a card will order a certain registering-apparatus to mark $(n - 1)$, and will at the same time execute the multiplication of b by b . When this is completed, it will be found that the registering-apparatus has effaced a unit, and that it only marks $(n - 2)$; while the machine will now again order the number b written on the column V_1 to multiply itself with the product b^2 written on the column V_3 , which will give b^3 . Another unit is then effaced from the registering-apparatus, and the same processes are continually repeated until it only marks zero. Thus the number b^n will be found inscribed on V_3 , when the machine, pursuing its

course of operations, will order the product of b^n by a ; and the required calculation will have been completed without there being any necessity that the number of operation-cards used should vary with the value of n . If n were negative, the cards, instead of ordering the multiplication of a by b^n , would order its division; this we can easily conceive, since every number, being inscribed with its respective sign, is consequently capable of reacting on the nature of the operations to be executed. Finally, if n were fractional, of the form $\frac{p}{q}$, an additional column would be used for the inscription of q , and the machine would bring into action two sets of processes, one for raising b to the power p , the other for extracting the q th root of the number so obtained.

Again, it may be required, for example, to multiply an expression of the form $ax^m + bx^n$ by another $Ax^p + Bx^q$, and then to reduce the product to the least number of terms, if any of the indices are equal. The two factors being ordered with respect to x , the general result of the multiplication would be $Aax^{m+p} + Abx^{n+p} + Bax^{m+q} + Bbx^{n+q}$. Up to this point the process presents no difficulties; but suppose that we have $m = p$ and $n = q$, and that we wish to reduce the two middle terms to a single one $(Ab + Ba)x^{m+q}$. For this purpose, the cards may order $m + q$ and $n + p$ to be transferred into the mill, and there subtracted one from the other; if the remainder is nothing, as would be the case on the present hypothesis, the mill will order other cards to bring to it the coefficients Ab and Ba , that it may add them together and give them in this state as a coefficient for the single term $x^{n+p} = x^{m+q}$.

This example illustrates how the cards are able to reproduce all the operations which intellect performs in order to attain a determinate result, if these operations are themselves capable of being precisely defined.

Let us now examine the following expression: —

$$2 \times \frac{2^2.4^2.6^2.8^2.10^2 \dots (2n)^2}{1^2.3^2.5^2.7^2.9^2 \dots (2n-1)^2.(2n+1)^2},$$

which we know becomes equal to the ratio of the circumference to the diameter, when n is infinite. We may require the machine not only to perform the calculaiton [sic] of this fractional expression, but further to give indication as soon as the value becomes identical with that of the ratio of the circumference to the diameter when n is infinite, a case in which the computation would be impossible. Observe that we should thus require of the machine to interpret a result not of itself evident, and that this is not amongst its attributes, since it is no thinking being. Nevertheless, when the $\cos$ of $n = \inf$ has been foreseen, a card may immediately order the substitution of the value of π (π being the ratio of the circumference to the diameter), without going through the series of calculations indicated. This would merely require that the machine contain a special card, whose office it should be to place the number π in a direct and independent manner on the column indicated to it. And here we should introduce the mention of a third species of cards, which may be called *cards of numbers*. There are certain numbers, such as those expressing the ratio of the circumference to the diameter, the Numbers of Bernoulli &c., which frequently present themselves in calculations. To avoid the necessity for computing them every time they have

to be used, certain cards may be combined specially in order to give these numbers ready made into the mill, whence they afterwards go and place themselves on those columns of the store that are destined for them. Through this means the machine will be susceptible of those simplifications afforded by the use of numerical tables. It would be equally possible to introduce, by means of these cards, the logarithms of numbers; but perhaps it might not be in this case either the shortest or the most appropriate method; for the machine might be able to perform the same calculations by other more expeditious combinations, founded on the rapidity with which it executes the four first operations of arithmetic. To give an idea of this rapidity, we need only mention that Mr. Babbage believes he can, by his engine, form the product of two numbers, each containing twenty figures, in *three minutes*.

Perhaps the immense number of cards required for the solution of any rather complicated problem may appear to be an obstacle; but this does not seem to be the case. There is no limit to the number of cards that can be used. Certain stuffs require for their fabrication not less than *twenty thousand* cards, and we may unquestionably far exceed even this quantity²⁹.

Resuming what we have explained concerning the Analytical Engine, we may conclude that it is based on two principles: the first, consisting in the fact that every arithmetical calculation ultimately depends on four principal operations – addition, subtraction, multiplication, and division; the second, in the possibility of reducing every analytical calculation to that of the coefficients for the several terms of a series. If this last principle be true, all the operations of analysis come within the domain of the engine. To take another point of view: the use of the cards offers a generality equal to that of algebraical formulæ, since such a formula simply indicates the nature and order of the operations requisite for arriving at a certain definite result, and similarly the cards merely command the engine to perform these same operations; but in order that the mechanisms may be able to act to any purpose, the numerical data of the problem must in every particular case be introduced. Thus the same series of cards will serve for all questions whose sameness of nature is such as to require nothing altered excepting the numerical data. In this light the cards are merely a translation of algebraical formulæ, or, to express it better, another form of analytical notation.

Since the engine has a mode of acting peculiar to itself, it will in every particular case be necessary to arrange the series of calculations conformably to the means which the machine possesses; for such or such a process which might be very easy for a calculator, may be long and complicated for the engine, and *vice versa*.

Considered under the most general point of view, the essential object of the machine being to calculate, according to the laws dictated to it, the values of numerical coefficients which it is then to distribute appropriately on the columns which represent the variables, it follows that the interpretation of formulæ and of results is beyond its province, unless indeed this very interpretation be itself susceptible of expression by means of the symbols which the machine employs. Thus, although it is not itself the being that reflects, it may yet

²⁹ See Note F.

be considered as the being which executes the conceptions of intelligence³⁰. The cards receive the impress of these conceptions, and transmit to the various trains of mechanism composing the engine the orders necessary for their action. When once the engine shall have been constructed, the difficulty will be reduced to the making out of the cards; but as these are merely the translation of algebraical formulæ, it will, by means of some simple notations, be easy to consign the execution of them to a workman. Thus the whole intellectual labour will be limited to the preparation of the formulæ, which must be adapted for calculation by the engine.

Now, admitting that such an engine can be constructed, it may be inquired: what will be its utility? To recapitulate; it will afford the following advantages: – First, rigid accuracy. We know that numerical calculations are generally the stumbling-block to the solution of problems, since errors easily creep into them, and it is by no means always easy to detect these errors. Now the engine, by the very nature of its mode of acting, which requires no human intervention during the course of its operations, presents every species of security under the head of correctness; besides, it carries with it its own check; for at the end of every operation it prints off, not only the results, but likewise the numerical data of the question; so that it is easy to verify whether the question has been correctly proposed. Secondly, economy of time: to convince ourselves of this, we need only recollect that the multiplication of two numbers, consisting each of twenty figures, requires at the very utmost three minutes. Likewise, when a long series of identical computations is to be performed, such as those required for the formation of numerical tables, the machine can be brought into play so as to give several results at the same time, which will greatly abridge the whole amount of the processes. Thirdly, economy of intelligence: a simple arithmetical computation requires to be performed by a person possessing some capacity; and when we pass to more complicated calculations, and wish to use algebraical formulæ in particular cases, knowledge must be possessed which pre-supposes preliminary mathematical studies of some extent. Now the engine, from its capability of performing by itself all these purely material operations, spares intellectual labour, which may be more profitably employed. Thus the engine may be considered as a real manufactory of figures, which will lend its aid to those many useful sciences and arts that depend on numbers. Again, who can foresee the consequences of such an invention? In truth, how many precious observations remain practically barren for the progress of the sciences, because there are not powers sufficient for computing the results! And what discouragement does the perspective of a long and arid computation cast into the mind of a man of genius, who demands time exclusively for meditation, and who beholds it snatched from him by the material routine of operations! Yet it is by the laborious route of analysis that he must reach truth; but he cannot pursue this unless guided by numbers; for without numbers it is not given us to raise the veil which envelopes the mysteries of nature. Thus the idea of constructing an apparatus capable of aiding human weakness in such researches, is a conception which, being realized, would mark a glorious epoch in the history of the sciences. The plans have been arranged for all the various parts, and for all the wheel-work, which compose this immense apparatus, and their action studied; but these have not yet been fully combined together in the drawings³¹

³⁰ See Note G.

³¹ This sentence has been slightly altered in the translation in order to express more exactly the present

and mechanical notation³². The confidence which the genius of Mr. Babbage must inspire, affords legitimate ground for hope that this enterprise will be crowned with success; and while we render homage to the intelligence which directs it, let us breathe aspirations for the accomplishment of such an undertaking.

state of the engine. – Note by Translator.

³² The notation here alluded to is a most interesting and important subject, and would have well deserved a separate and detailed Note upon it, amongst those appended to the Memoir. It has, however, been impossible within the space allotted, even to touch upon so wide a field. – Note by Translator.

NOTES BY THE TRANSLATOR.

Note A. – Page 108.

The particular function whose integral the Difference Engine was constructed to tabulate, is

$$\Delta^7 u_z = 0.$$

The purpose which that engine has been specially intended and adapted to fulfill, is the computation of nautical and astronomical tables. The integral of

$$\Delta^7 u_z = 0.$$

being

$$u_z = a + bx + cx^2 + dx^3 + ex^4 + fx^5 + gx^6,$$

the constants $a, b, c, \&c.$ are represented on the seven columns of discs, of which the engine consists. It can therefore tabulate *accurately* and to an *unlimited extent*, all series whose general term is comprised in the above formula; and it can also tabulate *approximatively* between *intervals of greater or less extent*, all other series which are capable of tabulation by the Method of Differences.

The Analytical Engine, on the contrary, is not merely adapted for *tabulating* the results of one particular function and of no other, but for *developing and tabulating* any function whatever. In fact the engine may be described as being the material expression of any indefinite function of any degree of generality and complexity, such as for instance,

$$F(x, y, z, \log x, \sin y, x^p, \&c.),$$

which is, it will be observed, a function of all other possible functions of any number of quantities.

In this, which we may call the *neutral* or *zero* state of the engine, it is ready to receive at any moment, by means of cards constituting a portion of its mechanism (and applied on the principle of those used in the Jacquard-loom), the impress of whatever *special* function we may desire to develop or to tabulate. These cards contain within themselves (in a manner explained in the Memoir itself, page 110) the law of development of the particular function that may be under consideration, and they compel the mechanism to act accordingly in a certain corresponding order. One of the simplest cases would be, for example, to suppose that

$$F(x, y, z, \&c. \&c.)$$

is the particular function

$$\Delta^n u_z = 0$$

which the Difference Engine tabulates for values of n up to 7. In this case the cards would order the mechanism to go through that succession of operations which would tabulate

$$u_z = a + bx + cx^2 + \dots mx^{n-1},$$

where n might be any number whatever.

These cards, however have nothing to do with the regulation of the particular *numerical* data. They merely determine the *operations*³³ to be effected, which operations may of course be performed on an infinite variety of particular numerical values, and do not bring out any definite numerical results unless the numerical data of the problem have been impressed on the requisite portions of the train of mechanism. In the above example, the first essential step towards an arithmetical result, would be the substitution of specific numbers for n , and for the other primitive quantities which enter into the function.

Again, let us suppose that for F we put two complete equations of the fourth degree between x and y . We must then express on the cards the law of elimination for such equations. The engine would follow out those laws, and would ultimately give the equation of one variable which results from such elimination. Various *modes* of elimination might be selected; and of course the cards must be made out accordingly. The following is one mode that might be adopted. The engine is able to multiply together any two functions of the form

$$a + bx + cx^2 + \dots px^n.$$

This granted, the two equations may be arranged according to the powers of y , and the coefficients of the powers of y may be arranged according to powers of x . The elimination of y will result from the successive multiplications and subtractions of several such functions. In this, and in all other instances, as was explained above, the particular *numerical* data and the *numerical* results are determined by means and by portions of the mechanism which act quite independently of those that regulate the *operations*.

In studying the action of the Analytical Engine, we find that the peculiar and independent nature of the considerations which in all mathematical analysis belong to *operations*, as distinguished from *the objects operated upon* and from the *results* of the operations performed upon those objects, is very strikingly defined and separated.

³³ We do not mean to imply that the *only* use made of the Jacquard cards is that of regulating the algebraical *operations*. But we mean to explain that *those* cards and portions of mechanism which regulate these *operations*, are wholly independent of those which are used for other purposes. M. Menabrea explains that there are three classes of cards used in the engine for three different sets of objects, viz. *Cards of the Operations*, *Cards of the Variables*, and *Cards of Numbers*. (See pages 110 and 118.)

It is well to draw attention to this point, not only because its full appreciation is essential to the attainment of any very just and adequate general comprehension of the powers and mode of action of the Analytical Engine, but also because it is one which is perhaps too little kept in view in the study of mathematical science in general. It is, however, impossible to confound it with other considerations, either when we trace the manner in which that engine attains its results, or when we prepare the data for its attainment of those results. It were much to be desired, that when mathematical processes pass through the human brain instead of through the medium of inanimate mechanism, it were equally a necessity of things that the reasonings connected with *operations* should hold the same just place as a clear and well-defined branch of the subject of analysis, a fundamental but yet independent ingredient in the science, which they must do in studying the engine. The confusion, the difficulties, the contradictions which, in consequence of a want of accurate distinctions in this particular, have up to even a recent period encumbered mathematics in all those branches involving the consideration of negative and impossible quantities, will at once occur to the reader who is at all versed in this science, and would alone suffice to justify dwelling somewhat on the point, in connexion with any subject so peculiarly fitted to give forcible illustration of it, as the Analytical Engine. It may be desirable to explain, that by the word *operation*, we mean *any process which alters the mutual relation of two or more things*, be this relation of what kind it may. This is the most general definition, and would include all subjects in the universe. In abstract mathematics, of course operations alter those particular relations which are involved in the considerations of number and space, and the *results* of operations are those peculiar results which correspond to the nature of the subjects of operation. But the science of operations, as derived from mathematics more especially, is a science of itself, and has its own abstract truth and value; just as logic has its own peculiar truth and value, independently of the subjects to which we may apply its reasonings and processes. Those who are accustomed to some of the more modern views of the above subject, will know that a few fundamental relations being true, certain other combinations of relations must of necessity follow; combinations unlimited in variety and extent if the deductions from the primary relations be carried on far enough. They will also be aware that one main reason why the separate nature of the science of operations has been little felt, and in general little dwelt on, is the *shifting* meaning of many of the symbols used in mathematical notation. First, the symbols of *operation* are frequently *also* the symbols of the *results* of operations. We may say that these symbols are apt to have both a *retrospective* and a *prospective* signification. They may signify either relations that are the consequence of a series of processes already performed, or relations that are yet to be effected through certain processes. Secondly, figures, the symbols of *numerical magnitude*, are frequently *also* the symbols of *operations*, as when they are the indices of powers. Wherever terms have a shifting meaning, independent sets of considerations are liable to become complicated together, and reasonings and results are frequently falsified. Now in the Analytical Engine the operations which come under the first of the above heads, are ordered and combined by means of a notation and of a train of mechanism which belong exclusively to themselves; and with respect to the second head, whenever numbers meaning *operations* and not *quantities* (such as the indices of powers), are inscribed on any column or set of columns, those columns immediately act in a wholly separate and independent manner, becoming connected with the *operating mechanism* exclusively, and re-acting upon

this. They never come into combination with numbers upon any other columns meaning *quantities*; though, of course, if there are numbers meaning *operations* upon n columns, those may *combine amongst each other*, and will often be required to do so, just as numbers meaning *quantities* combine with each other in any variety. It might have been arranged that all numbers meaning *operations* should have appeared on some separate portion of the engine from that which presents numerical *quantities*: but the present mode is in some cases more simple, and offers in reality quite as much distinctness when understood.

The operating mechanism can even be thrown into action independently of any object to operate upon (although of course no *result* could then be developed). Again, it might act upon other things besides *number*, were objects found whose mutual fundamental relations could be expressed by those of the abstract science of operations, and which should be also susceptible of adaptations to the action of the operating notation and mechanism of the engine. Supposing, for instance, that the fundamental relations of pitched sounds in the science of harmony and of musical composition were susceptible of such expression and adaptations, the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.

The Analytical Engine is an *embodying of the science of operations*, constructed with peculiar reference to abstract number as the subject of those operations. The Difference Engine is the embodying of *one particular and very limited set of operations*, which (see the notation used in note B) may be expressed thus, $(+, +, +, +, +, +)$, or thus, $6(+)$. Six repetitions of the one operation, $+$, is, in fact, the whole sum and object of that engine. It has seven columns, and a number on any column can add itself to a number on the next column to its *right-hand*. So that, beginning with the column furthest to the left, six additions can be effected, and the result appears on the seventh column, which is the last on the right-hand. The *operating* mechanism of this engine acts in as separate and independent a manner as that of the Analytical Engine; but being susceptible of only one unvarying and restricted combination, it has little force or interest in illustration of the distinct nature of the *science of operations*. The importance of regarding the Analytical Engine under this point of view will, we think, become more and more obvious, as the reader proceeds with M. Menabrea's clear and masterly article. The calculus of operations is likewise in itself a topic of so much interest, and has of late years been so much more written on and thought on than formerly, that any bearing which that engine, from its mode of constitution, may possess upon the illustration of this branch of mathematical science, should not be overlooked. Whether the inventor of this engine had any such views in his mind while working out the invention, or whether he may subsequently ever have regarded it under this phase, we do not know; but it is one that forcibly occurred to ourselves on becoming acquainted with the means through which analytical combinations are actually attained by the mechanism. We cannot forbear suggesting one practical result which it appears to us must be greatly facilitated by the independent manner in which the engine orders and combines its *operations*: we allude to the attainment of those combinations into which *imaginary quantities* enter. This is a branch of its processes into which we have not had the opportunity of inquiring, and our conjecture therefore as to the principle on which we conceive the accomplishment of such results may have been made to depend, is very probably not in accordance with the fact, and less subservient for the purpose than some other principles, or at least requiring

the cooperation of others, it seems to us obvious, however, that where operations are so independent in their mode of acting, it must be easy by means of a few simple provisions and additions in arranging the mechanism, to bring out a *double* set of *results*, viz. – 1st, the *numerical magnitudes* which are the results of operations performed on *numerical data*. (These results are the *primary* object of the engine). 2ndly, the symbolical results to be attached to those numerical results, which symbolical results are not less the necessary and logical consequences of operations performed upon *symbolical data*, than are numerical results when the data are numerical³⁴.

If we compare together the powers and the principles of construction of the Difference and of the Analytical Engines, we shall perceive that the capabilities of the latter are immeasurably more extensive than those of the former, and that they in fact hold to each other the same relationship as that of analysis to arithmetic. The Difference Engine can effect but one particular series of operations, viz. that required for tabulating the integral of the special function

$$\Delta^n u_z = 0;$$

and as it can only do this for values of n up to 7³⁵, it cannot be considered as being the most *general* expression even of *one particular* function, much less as being the expression of any and all possible functions of all degrees of generality. The Difference Engine can in reality (as has been already partly explained) do nothing but *add*; and any other processes, not excepting those of simple subtraction, multiplication and division, can be performed by it only just to that extent if, which it is possible, by judicious mathematical arrangement and artifices, to reduce them to a *series of additions*. The method of differences is, in fact, a method of additions; and as it includes within its means a larger number of results attainable by *addition* simply, than any other mathematical principle, it was very appropriately selected as the basis on which to construct an *Adding Machine*, so as to give to the powers of such a machine the widest possible range. The Analytical Engine, on the contrary, can either add, subtract, multiply or divide with equal facility; and performs each of these four operations in a direct manner, without the aid of any of the other three. This one fact implies everything; and it is scarcely necessary to point out, for instance,

³⁴ In fact such an extension as we allude to, would merely constitute a further and more perfected development of any system introduced for making the proper combinations of the signs *plus* and *minus*. How ably M. Menabrea has touched on this restricted case is pointed out in Note B.

³⁵ The machine might have been constructed so as to tabulate for a higher value of n than seven. Since, however, every unit added to the value of n increases the extent of the mechanism requisite, there would on this account be a limit beyond which it could not be practically carried. Seven is sufficiently high for the calculation of all ordinary tables.

The fact that, in the Analytical Engine, the same extent of mechanism suffices for the solution of $\Delta^n u_z = 0$, whether $n = 7$, $n = 100,000$, or $n =$ any number whatever, at once suggests how entirely distinct must be the *nature of the principles* through whose application matter has been enabled to become the working agent of abstract mental operations in each of these engines respectively; and it affords an equally obvious presumption, that in the case of the Analytical Engine, not only are those principles in themselves of a higher and more comprehensive description, but also such as must vastly extend the *practical* value of the engine whose basis they constitute.

that while the Difference Engine can merely *tabulate*, and is incapable of *developing*, the Analytical Engine can *either tabulate or develope*.

The former engine is in its nature strictly *arithmetical*, and the results it can arrive at lie within a very clearly defined and restricted range, while there is no finite line of demarcation which limits the powers of the Analytical Engine. These powers are co-extensive with our knowledge of the laws of analysis itself, and need be bounded only by our acquaintance with the latter. Indeed we may consider the engine as the *material and mechanical representative* of analysis, and that our actual working powers in this department of human study will be enabled more effectually than heretofore to keep pace with our theoretical knowledge of its principles and laws, through the complete control which the engine gives us over the *executive manipulation* of algebraical and numerical symbols.

Those who view mathematical science not merely as a vast body of abstract and immutable truths, whose intrinsic beauty, symmetry and logical completeness, when regarded in their connexion together as a whole, entitle them to a prominent place in the interest of all profound and logical minds, but as possessing a yet deeper interest for the human race, when it is remembered that this science constitutes the language through which alone we can adequately express the great facts of the natural world, and those unceasing changes of mutual relationship which, visibly or invisibly, consciously or unconsciously to our immediate physical perceptions, are interminably going on in the agencies of the creation we live amidst : those who thus think on mathematical truth as the instrument through which the weak mind of man can most effectually read his Creator's works, will regard with especial interest all that can tend to facilitate the translation of its principles into explicit practical forms.

The distinctive characteristic of the Analytical Engine, and that which has rendered it possible to endow mechanism with such extensive faculties as bid fair to make this engine the executive right-hand of abstract algebra, is the introduction into it of the principle which Jacquard devised for regulating, by means of punched cards, the most complicated patterns in the fabrication of brocaded stuffs. It is in this that the distinction between the two engines lies. Nothing of the sort exists in the Difference Engine. We may say most aptly that the Analytical Engine *weaves algebraical patterns* just as the Jacquard-loom weaves flowers and leaves. Here, it seems to us, resides much more of originality than the Difference Engine can be fairly entitled to claim. We do not wish to deny to this latter all such claims. We believe that it is the only proposal or attempt ever made to construct a calculating machine *founded on the principle of successive orders of differences*, and capable of *printing off its own results*; and that this engine surpasses its predecessors, both in the extent of the calculations which it can perform, in the facility, certainty and accuracy with which it can effect them, and in the absence of all necessity for the intervention of human intelligence *during the performance of its calculations*. Its nature is, however, limited to the strictly arithmetical, and it is far from being the first or only scheme for constructing *arithmetical* calculating machines with more or less of success.

The bounds of arithmetic were however outstepped the moment the idea of applying the cards had occurred; and the Analytical Engine does not occupy common ground with mere "calculating machines." It holds a position wholly its own; and the considerations it

suggests are most interesting in their nature. In enabling mechanism to combine together *general* symbols, in successions of unlimited variety and extent, a uniting link is established between the operations of matter and the abstract mental processes of the *most abstract* branch of mathematical science. A new, a vast, and a powerful language is developed for the future use of analysis, in which to wield its truths so that these may become of more speedy and accurate practical application for the purposes of mankind than the means hitherto in our possession have rendered possible. Thus not only the mental and the material, but the theoretical and the practical in the mathematical world, are brought into more intimate and effective connexion with each other. We are not aware of its being on record that anything partaking in the nature of what is so well designated the *Analytical Engine* has been hitherto proposed, or even thought of, as a practical possibility, any more than the idea of a thinking or of a reasoning machine.

We will touch on another point which constitutes an important distinction in the modes of operating of the Difference and Analytical Engines. In order to enable the former to do its business, it is necessary to put into its columns the series of numbers constituting the first terms of the several orders of differences for whatever is the particular table under consideration. The machine then works *upon* these as its data. But these data must themselves have been already computed through a series of calculations by a human head. Therefore that engine can only produce results depending on data which have been arrived at by the explicit and actual working out of processes that are in their nature different from any that come within the sphere of its own powers. In other words, an *analysing* process must have been gone through by a human mind in order to obtain the data upon which the engine then *synthetically* builds its results. The Difference Engine is in its character exclusively *synthetical*, while the Analytical Engine is equally capable of analysis or of synthesis.

It is true that the Difference Engine can calculate to a much greater extent with these few preliminary data, than the data themselves required for their own determination. The table of squares, for instance, can be calculated to any extent whatever, when the numbers *one* and *two* are furnished; and a very few differences computed at any part of a table of logarithms would enable the engine to calculate many hundreds or even thousands of logarithms. Still the circumstance of its requiring, as a previous condition, that any function whatever shall have been numerically worked out, makes it very inferior in its nature and advantages to an engine which, like the Analytical Engine, requires merely that we should know the *succession and distribution of the operations* to be performed; without there being any occasion³⁶, in order to obtain data on which it can work, for our ever having gone through either the same particular operations which it is itself to effect, or any others. Numerical data must of course be given it, but they are mere arbitrary ones; not data that could only be arrived at through a systematic and necessary series of previous numerical calculations, which is quite a different thing.

To this it may be replied that an analysing process must equally have been performed in order to furnish the Analytical Engine with the necessary operative data; and that herein may also lie a possible source of error. Granted that the actual mechanism is unerring in

³⁶ This subject is further noticed in Note F.

its processes, the *cards* may give it wrong orders. This is unquestionably the case; but there is much less chance of error, and likewise far less expenditure of time and labour, where operations only, and the distribution of these operations, have to be made out, than where explicit numerical results are to be attained. In the case of the Analytical Engine we have undoubtedly to lay out a certain capital of analytical labour in one particular line; but this is in order that the engine may bring us in a much larger return in another line. It should be remembered also that the cards when once made out for any formula, have all the generality of algebra, and include an infinite number of particular cases.

We have dwelt considerably on the distinctive peculiarities of each of these engines, because we think it essential to place their respective attributes in strong relief before the apprehension of the public; and to define with clearness and accuracy the wholly different nature of the principles on which each is based, so as to make it self-evident to the reader (the mathematical reader at least) in what manner and degree the powers of the Analytical Engine transcend those of an engine, which, like the Difference Engine, can only work out such results as may be derived from *one restricted and particular series of processes*, such as those included in $\Delta^n u_z = 0$. We think this of importance, because we know that there exists considerable vagueness and inaccuracy in the mind of persons in general on the subject. There is a misty notion amongst most of those who have attended at all to it, that *two* “calculating machines” have been successively invented by the same person within the last few years; while others again have never heard but of the one original “calculating machine,” and are not aware of there being any extension upon this. For either of these two classes of persons the above considerations are appropriate. While the latter require a knowledge of the fact that there *are two* such inventions, the former are not less in want of accurate and well-defined information on the subject. No very clear or correct ideas prevail as to the characteristics of each engine, or their respective advantages or disadvantages; and, in meeting with those incidental allusions, of a more or less direct kind, which occur in so many publications of the day, to these machines, it must frequently be matter of doubt *which* “calculating machine” is referred to, or whether both are included in the general allusion.

We are desirous likewise of removing two misapprehensions which we know obtain, to some extent, respecting these engines. In the first place it is very generally supposed that the Difference Engine, after it had been completed up to a certain point, *suggested* the idea of the Analytical Engine; and that the second is in fact the improved offspring of the first, and *grew out* of the existence of its predecessor, through some natural or else accidental combination of ideas suggested by this one. Such a supposition is in this instance contrary to the facts; although it seems to be almost an obvious inference, wherever two inventions, similar in their nature and objects, succeed each other closely in order of *time*, and strikingly in order of *value*; more especially when the same individual is the author of both. Nevertheless the ideas which led to the Analytical Engine occurred in a manner wholly independent of any that were connected with the Difference Engine. These ideas are indeed in their own intrinsic nature independent of the latter engine, and might equally have occurred had it never existed nor been even thought of at all.

The second of the misapprehensions above alluded to, relates to the well-known suspen-

sion, during some years past, of all progress in the construction of the Difference Engine. Respecting the circumstances which have interfered with the actual completion of either invention, we offer no opinion; and in fact are not possessed of the data for doing so, had we the inclination. But we know that some persons suppose these obstacles (be they what they may) to have arisen *in consequence* of the subsequent invention of the Analytical Engine while the former was in progress. We have ourselves heard it even *lamented* that an idea should ever have occurred at all, which had turned out to be merely the means of arresting what was already in a course of successful execution, without substituting the superior invention in its stead. This notion we can contradict in the most unqualified manner. The progress of the Difference Engine had long been suspended, before there were even the least crude glimmerings of any invention superior to it. Such glimmerings, therefore, and their subsequent development, were in no way the original *cause* of that suspension; although, where difficulties of some kind or other evidently already existed, it was not perhaps calculated to remove or lessen them that an invention should have been meanwhile thought of, which, while including all that the first was capable of, possesses powers so extended as to eclipse it altogether.

We leave it for the decision of each individual (*after he has possessed himself* of competent information as to the characteristics of each engine), to determine how far it ought to be matter of regret that such an accession has been made to the powers of human science, even if it *has* (which we greatly doubt) increased to a certain limited extent some already existing difficulties that had arisen in the way of completing a valuable but lesser work. We leave it for each to satisfy himself as to the wisdom of desiring the obliteration (were that now possible) of all records of the more perfect invention, in order that the comparatively limited one might be finished. The Difference Engine would doubtless fulfil all those practical objects which it was originally destined for. It would certainly calculate all the tables that are more directly necessary for the physical purposes of life, such as nautical and other computations. Those who incline to very strictly utilitarian views, may perhaps feel that the peculiar powers of the Analytical Engine bear upon questions of abstract and speculative science, rather than upon those involving every-day and ordinary human interests. These persons being likely to possess but little sympathy, or possibly acquaintance, with any branches of science which they do not find to be *useful* (according to their definition of that word), may conceive that the undertaking of that engine, now that the other one is already in progress, would be a barren and unproductive laying out of yet more money and labour; in fact, a work of supererogation. Even in the utilitarian aspect, however, we do not doubt that very valuable practical results would be developed by the extended faculties of the Analytical Engine; some of which results we think we could now hint at, had we the space; and others, which it may not yet be possible to foresee, but which would be brought forth by the daily increasing requirements of science, and by a more intimate practical acquaintance with the powers of the engine, were it in actual existence.

On general grounds, both of an *a priori* description as well as those founded on the scientific history and experience of mankind, we see strong presumptions that such would be the case. Nevertheless ail will probably concur in feeling that the completion of the Difference Engine would be far preferable to the non-completion of any calculating engine at all. With whomsoever or wheresoever may rest the present causes of difficulty that apparently exist

towards either the completion of the old engine, or the commencement of the new one, we trust they will not ultimately result in this generation's being acquainted with these inventions through the medium of pen, ink and paper merely; and still more do we hope, that for the honour of our country's reputation in the future pages of history, these causes will not lead to the completion of the undertaking by some *other* nation or government. This could not but be matter of just regret; and equally so, whether the obstacles may have originated in private interests and feelings, in considerations of a more public description, or in causes combining the nature of both such solutions.

We refer the reader to the "Edinburgh Review" of July 1834, for a very able account of the Difference Engine. The writer of the article we allude to, has selected as his prominent matter for exposition, a wholly different view of the subject from that which M. Menabrea has chosen. The former chiefly treats it under its mechanical aspect, entering but slightly into the mathematical principles of which that engine is the representative, but giving, in considerable length, many details of the mechanism and contrivances by means of which it translates the various orders of differences. M. Menabrea, on the contrary, exclusively develops the analytical view; taking it for granted that mechanism is able to perform certain processes, but without attempting to explain *how*; and devoting his whole attention to explanations and illustrations of the manner in which analytical laws can be so arranged and combined as to bring every branch of that vast subject within the grasp of the assumed powers of mechanism. It is obvious that, in the invention of a calculating engine, these two branches of the subject are equally essential fields of investigation, and that on their mutual adjustment, one to the other, must depend all success. They must be made to meet each other, so that the weak points in the powers of either department may be compensated by the strong points in those of the other. They are indissolubly connected, though so different in their intrinsic nature that perhaps the same mind might not be likely to prove equally profound or successful in both. We know those who doubt whether the powers of mechanism will in practice prove adequate in all respects to the demands made upon them in the working of such complicated trains of machinery as those of the above engines, and who apprehend that unforeseen practical difficulties and disturbances will arise in the way of accuracy and of facility of operation. The Difference Engine, however, appears to us to be in a great measure an answer to these doubts. It is complete as far as it goes, and it does work with all the anticipated success. The Analytical Engine, far from being more complicated, will in many respects be of simpler construction; and it is a remarkable circumstance attending it, that with very *simplified* means it is so much more powerful.

The article in the 'Edinburgh Review' was written some time previous to the occurrence of any ideas such as afterwards led to the invention of the Analytical Engine; and in the nature of the Difference Engine there is much less that would invite a writer to take exclusively, or even prominently, the mathematical view of it, than in that of the Analytical Engine; although mechanism has undoubtedly gone much further to meet mathematics, in the case of this engine, than of the former one. Some publication embracing the mechanical view of the Analytical Engine is a desideratum which we trust will be supplied before long.

Those who may have the patience to study a moderate quantity of rather dry details,

will find ample compensation, after perusing the article of 1834, in the clearness with which a succinct view will have been attained of the various practical steps through which mechanism can accomplish certain processes; and they will also find themselves still further capable of appreciating M. Menabrea's more comprehensive and generalized memoir. The very difference in the style and object of these two articles, makes them peculiarly valuable to each other; at least for the purposes of those who really desire something more than a merely superficial and popular comprehension of the subject of calculating engines.

A. A. L.

Note B. – Page 109

That portion of the Analytical Engine here alluded to is called the storehouse. It contains an indefinite number of the columns of discs described by M. Menabrea. The reader may picture to himself a pile of rather large draughtsmen heaped perpendicularly one above another to a considerable height, each counter having the digits from 0 to 9 inscribed on its *edge* at equal intervals; and if he then conceives that the counters do not actually lie one upon another so as to be in contact, but are fixed at small intervals of vertical distance on a common axis which passes perpendicularly through their centres, and around which each disc can *revolve horizontally* so that any required digit amongst those inscribed on its margin can be brought into view, he will have a good idea of one of these columns. The *lowest* of the discs on any column belongs to the units, the next above to the tens, the next above this to the hundreds, and so on. Thus, if we wished to inscribe 1345 on a column of the engine, it would stand thus: –

1
3
4
5

In the Difference Engine there are seven of these columns placed side by side in a row, and the working mechanism extends behind them; the general form of the whole mass of machinery is that of a quadrangular prism (more or less approaching to the cube); the results always appearing on that perpendicular face of the engine which contains the columns of discs, opposite to which face a spectator may place himself. In the Analytical Engine there would be many more of these columns, probably at least two hundred. The precise form and arrangement which the whole mass of its mechanism will assume is not yet finally determined.

We may conveniently represent the columns of discs on paper in a diagram like the following:

–

V_1	V_2	V_3	V_4	&c.
○	○	○	○	&c.
0	0	0	0	
0	0	0	0	
0	0	0	0	&c.
0	0	0	0	
□	□	□	□	&c.

The V's are for the purpose of convenient reference to any column, either in writing or speaking, and are consequently numbered. The reason why the letter V is chosen for this purpose in preference to any other letter, is because these columns are designated (as the reader will find in proceeding with the Memoir) the *Variables*, and sometimes the *Variable columns*, or the *columns of Variables*. The origin of this appellation is, that the values on the columns are destined to change, that is to *vary*, in every conceivable manner. But it is necessary to guard against the natural misapprehension that the columns are only intended to receive the values of the *variables* in an analytical formula, and not of the *constants*. The columns are called Variables on a ground wholly unconnected with the *analytical* distinction between constants and variables. In order to prevent the possibility of confusion, we have, both in the translation and in the notes, written Variable with a capital letter when we use the word to signify a *column of the engine*, and variable with a small letter when we mean the *variable of a formula*. Similarly, *Variable-cards* signify any cards that belong to a column of the engine.

To return to the explanation of the diagram: each circle at the top is intended to contain the algebraic sign + or −, either of which can be substituted³⁷ for the other, according as the number represented on the column below is positive or negative. In a similar manner any other purely *symbolical* results of algebraical processes might be made to appear in these circles. In Note A. the practicability of developing *symbolical* with no less ease than *numerical* results has been touched on.

The zeros beneath the *symbolic* circles represent each of them a disc, supposed to have the digit 0 presented in front. Only four tiers of zeros have been figured in the diagram, but these may be considered as representing thirty or forty, or any number of tiers of discs that may be required. Since each disc can present any digit, and each circle any sign, the discs of every column may be so adjusted³⁸ as to express any positive or negative number whatever

³⁷ A fuller account of the manner in which the signs are regulated, is given in Mons. Menabrea's Memoir, page 114. He himself expresses doubts (in a note of his own at the bottom of the page) as to his having been likely to hit on the precise methods really adopted; his explanation being merely a conjectural one. That it *does* accord precisely with the fact is a remarkable circumstance, and affords a convincing proof how completely Mons. Menabrea has been imbued with the true spirit of the invention. Indeed the whole of the above Memoir is a striking production, when we consider that Mons. Menabrea had had but very slight means for obtaining any adequate ideas respecting the Analytical Engine. It requires however a considerable acquaintance with the abstruse and complicated nature of such a subject, in order fully to appreciate the penetration of the writer who could take so just and comprehensive a view of it upon such limited opportunity.

³⁸ This adjustment is done by hand merely.

within the limits of the machine; which limits depend on the *perpendicular* extent of the mechanism, that is, on the number of discs to a column.

Each of the squares below the zeros is intended for the inscription of any *general* symbol or combination of symbols we please; it being understood that the number represented on the column immediately above, is the numerical value of that symbol, or combination of symbols. Let us, for instance, represent the three quantities a , n , x , and let us further suppose that $a = 5$, $n = 7$, $x = 98$. We should have ³⁹ –

V_1	V_2	V_3	V_4	&c.
+	+	+	+	
0	0	0	0	
0	0	0	0	&c.
0	0	9	0	
5	7	8	0	&c.
a	n	x		

We may now combine these symbols in a variety of ways, so as to form any required function or functions of them, and we may then inscribe each such function below brackets, every bracket uniting together those quantities (and those only) which enter into the function inscribed below it. We must also, when we have decided on the particular function whose numerical value we desire to calculate, assign another column to the right-hand for receiving the *results*, and must inscribe the function in the square below this column. In the above instance we might have any one of the following functions: –

$$ax^n, x^{an}, anx, \frac{a}{n}x, a + n + x, \&c. \&c.$$

Let us select the first. It would stand as follows, previous to calculation: –

V_1	V_2	V_3	V_4	&c.
+	+	+	+	
0	0	0	0	&c.
0	0	0	0	
0	0	9	0	
5	7	8	0	&c.
a	n	x	ax^n	&c.
ax_n				

The data being given, we must now put into the engine the cards proper for directing the operations in the case of the particular function chosen. These operations would in this

³⁹ It is convenient to omit the circles whenever the signs + or – can be actually represented.

instance be, —

Firstly, six multiplications in order to get x^n ($= 98^7$) for the above particular data).

Secondly, one multiplication in order then to get ax^n ($= 5 \times 98^7$).

In all, seven multiplications to complete the whole process. We may thus represent them :

—

$$(\times, \times, \times, \times, \times, \times, \times), \text{ or } 7(\times).$$

The multiplications would, however, at successive stages in the solution of the problem, operate on pairs of numbers, derived from *different* columns. In other words, the *same operation* would be performed on different *subjects of operation*. And here again is an illustration of the remarks made in the preceding Note on the independent manner in which the engine directs its *operations*. In determining the value of ax^n , the *operations* are *homogeneous*, but are distributed amongst different *subjects of operation*, at successive stages of the computation. It is by means of certain punched cards, belonging to the Variables themselves, that the action of the operations is so *distributed* as to suit each particular function. The *Operation-cards* merely determine the succession of operations in a general manner. They in fact throw all that portion of the mechanism included in the *mill*, into a series of different *states*, which we may call the *adding state*, or the *multiplying state*, &c. respectively. In each of these states the mechanism is ready to act in the way peculiar to that state, on any pair of numbers which may be permitted to come within its sphere of action. Only *one* of these operating states of the mill can exist at a time; and the nature of the mechanism is also such that only *one pair of numbers* can be received and acted on at a time. Now, in order to secure that the mill shall receive a constant supply of the proper pairs of numbers in succession, and that it shall also rightly locate the result of an operation performed upon any pair, each Variable has cards of its own belonging to it. It has, first, a class of cards whose business it is to *allow* the number on the Variable to pass into the mill, there to be operated upon. These cards may be called the *Supplying-cards*. They furnish the mill with its proper food. Each Variable has, secondly, another class of cards, whose office it is to allow the Variable to *receive* a number from the mill. These cards may be called the *Receiving-cards*. They regulate the location of results, whether temporary or ultimate results. The Variable-cards in general (including both the preceding classes) might, it appears to us, be even more appropriately designated the *Distributive-cards*, since it is through their means that the action of the operations, and the results of this action, are rightly *distributed*.

There are *two varieties* of the *Supplying* Variable-cards, respectively adapted for fulfilling two distinct subsidiary purposes: but as these modifications do not bear upon the present subject, we shall notice them in another place.

In the above case of ax^n , the *Operation-cards* merely order seven multiplications, that is, they order the mill to be in the *multiplying state* seven successive times (without any reference to the particular columns whose numbers are to be acted upon). The proper

Distributive Variable-cards step in at each successive multiplication, and cause the distributions requisite for the particular case.

For x^{an}	the operations would be	34 ($\times$)
... $a.n.x$		($\times, \times$), or 2 ($\times$)
... $\frac{a}{n}.x$		($\div, \times$)
... $a + n + x$		($+, +$), or 2 ($+$)

The engine might be made to calculate all these in succession. Having completed ax^n , the function x^{an} might be written under the brackets instead of ax^n , and a new calculation commenced (the appropriate Operation and Variable-cards for the new function of course coming into play). The results would then appear on V_5 . So on for any number of different functions of the quantities a, n, x . Each *result* might either permanently remain on its column during the succeeding calculations, so that when all the functions had been computed, their values would simultaneously exist on V_4, V_5, V_6 , &c.; or each result might (after being printed off, or used in any specified manner) be effaced, to make way for its successor. The square under V_4 , ought, for the latter arrangement, to have the functions ax^n, x^{an}, anx , &c. successively inscribed in it.

Let us now suppose that we have two expressions whose values have been computed by the engine independently of each other (each having its own group of columns for data and results). Let them be ax^n, bpy . They would then stand as follows on the columns: –

V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	V_9
+	+	+	+	+	+	+	+	+
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
a	n	x	ax^n	b	p	y	bpy	$\frac{ax^n}{bpy}$

We may now desire to combine together these two *results*, in any manner we please; in which case it would only be necessary to have an additional card or cards, which should order the requisite operations to be performed with the numbers on the two result-columns, V_4 , and V_8 , and the *result of these further operations* to appear on a new column, V_9 . Say that we wish to divide ax^n by bpy . The numerical value of this division would then appear on the column V_9 , beneath which we have inscribed $\frac{ax^n}{bpy}$. The whole series of operations from the beginning would be as follows (n being = 7) –

$$\{7(\times), 2(\times), \div\}, \text{ or } \{9(\times), \div\}.$$

This example is introduced merely to show that we may, if we please, retain separately and permanently any *intermediate* results (like ax^n, bpy), which occur in the course of processes having an ulterior and more complicated result as their chief and final object (like $\frac{ax^n}{bpy}$).

Any group of columns may be considered as representing a *general* function, until a *special* one has been implicitly impressed upon them through the introduction into the engine of the

Operation and Variable-cards made out for a *particular* function. Thus, in the preceding example, $V_1, V_2, V_3, V_5, V_6, V_7$ represent the *general* function $\phi(a, n, b, p, x, y)$ until the function $\frac{ax^n}{bpy}$ has been determined on, and *implicitly* expressed by the placing of the right cards in the engine. The actual working of the mechanism, as regulated by these cards, then *explicitly* develops the value of the function. The inscription of a function under the brackets, and in the square under the result-column, in no way influences the processes or the results, and is merely a memorandum for the observer, to remind him of what is going on. It is the Operation and the Variable-cards only, which in reality determine the function. Indeed it should be distinctly kept in mind that the inscriptions within *any* of the squares, are quite independent of the mechanism or workings of the engine, and are nothing but arbitrary memorandums placed there at pleasure to assist the spectator.

The further we analyse the manner in which such an engine performs its processes and attains its results, the more we perceive how distinctly it places in a true and just light the mutual relations and connexion of the various steps of mathematical analysis, how clearly it separates those things which are in reality distinct and independent, and unites those which are mutually dependent.

A. A. L.

Note C. – Page 110.

Those who may desire to study the principles of the Jacquard-loom in the most effectual manner, viz. that of practical observation, have only to step into the Adelaide Gallery or the Polytechnic Institution. In each of these valuable repositories of scientific *illustration*, a weaver is constantly working at a Jacquard-loom, and is ready to give any information that may be desired as to the construction and modes of acting of his apparatus. The volume on the manufacture of silk, in Lardner's Cyclopædia, contains a chapter on the Jacquard-loom, which may also be consulted with advantage.

The mode of application of the cards, as hitherto used in the art of weaving, was not found, however, to be sufficiently powerful for all the simplifications which it was desirable to attain in such varied and complicated processes as those required in order to fulfil the purposes of an Analytical Engine. A method was devised of what was technically designated *backing* the cards in certain groups according to certain laws. The object of this extension is to secure the possibility of bringing any particular card or set of cards into use *any number of times successively* in the solution of one problem. Whether this power shall be taken advantage of or not, in each particular instance, will depend on the nature of the operations which the problem under consideration may require. The process is alluded to by M. Menabrea in page 112, and it is a very important simplification. It has been proposed to use it for the reciprocal benefit of that art, which, while it has itself no apparent connexion with the domains of abstract science, has yet proved so valuable to the latter, in suggesting the principles which, in their new and singular field of application, seem likely to place *algebraical* combinations not less completely within the province of mechanism, than are all those varied intricacies of which *intersecting threads* are susceptible. By the introduction of the system of *backing* into the Jacquard-loom itself, patterns which should possess symmetry, and follow regular laws of any extent, might be woven by means of

comparatively few cards.

Those who understand the mechanism of this loom will perceive that the above improvement is easily effected in practice, by causing the prism over which the train of pattern-cards is suspended, to revolve *backwards* instead of *forwards*, at pleasure, under the requisite circumstances; until, by so doing, any particular card, or set of cards, that has done duty once, and passed on in the ordinary regular succession, is brought back to the position it occupied just before it was used the preceding time. The prism then resumes its *forward* rotation, and thus brings the card or set of cards in question into play a second time. This process may obviously be repeated any number of times.

A. A. L.

Note D. – Page 112.

We have represented the solution of these two equations, with every detail, in a diagram⁴⁰ similar to those used in Note B.; but additional explanations are requisite, partly in order to make this more complicated case perfectly clear, and partly for the comprehension of certain indications and notations not used in the preceding diagrams. Those who may wish to understand Note G. completely, are recommended to pay particular attention to the contents of the present Note, or they will not otherwise comprehend the similar notation and indications when applied to a much more complicated case.

In all calculations, the columns of Variables used may be divided into three classes: –

1st. Those on which the data are inscribed:

2ndly. Those intended to receive the final results:

3dly. Those intended to receive such intermediate and temporary combinations of the primitive data as are not to be permanently retained, but are merely needed for *working with*, in order to attain the ultimate results. Combinations of this kind might properly be called *secondary data*. They are in fact so many *successive stages* towards the final result. The columns which receive them are rightly named *Working-Variables*, for their office is in its nature purely *subsidiary* to other purposes. They develop an intermediate and transient class of results, which unite the original data with the final results.

The Result-Variables sometimes partake of the nature of Working-Variables. It frequently happens that a Variable destined to receive a final result is the recipient of one or more intermediate values successively, in the course of the processes. Similarly, the Variables for data often become Working-Variables, or Result-Variables, or even both in succession. It so happens, however, that in the case of the present equations the three sets of offices remain throughout perfectly separate and independent.

It will be observed, that in the squares below the *Working-Variables* nothing is inscribed. Any one of these Variables is in many cases destined to pass through various values successively during the performance of a calculation (although in these particular equations no instance of this occurs). Consequently no *one fixed* symbol, or combination of symbols, should be considered as properly belonging to a merely *Working-Variable*; and as a general rule their squares are left blank. Of course in this, as in all other cases where we mention a *general* rule, it is understood that many particular exceptions may be expedient.

In order that all the indications contained in the diagram may be completely understood, we shall now explain two or three points, not hitherto touched on. When the value on any Variable is called into use, one of two consequences may be made to result. Either the value may *return* to the Variable after it has been used, in which case it is ready for a second use if needed; or the Variable may be made zero. (We are of course not considering a third case, of not unfrequent occurrence, in which the same Variable is destined to receive the *result* of the very operation which it has just supplied with a number.) Now the ordinary rule is, that the value *returns* to the Variable; unless it has been foreseen that no use for that value

⁴⁰ See the diagram of page 139.

can recur, in which case zero is substituted. At the *end* of a calculation, therefore, every column ought as a general rule to be zero, excepting those for results. Thus it will be seen by the diagram, that when m , the value on V_0 , is used for the second time by Operation 5, V_0 becomes 0, since m is not again needed; that similarly, when $(mn' - m'n)$, on V_{12} , is used for the third time by Operation 11, V_{12} , becomes zero, since $(mn' - m'n)$ is not again needed. In order to provide for the one or the other of the courses above indicated, there are *two* varieties of the Supplying Variable-cards. One of these varieties has provisions which cause the number given off from any Variable to *return* to that Variable after doing its duty in the mill. The other variety has provisions which cause *zero* to be substituted on the Variable, for the number given off. These two varieties are distinguished, when needful, by the respective appellations of the *Retaining* Supply-cards and the *Zero* Supply-cards. We see that the *primary* office (see Note B.) of both these varieties of cards is the same; they only differ in their *secondary* office.

Every Variable thus has belonging to it one class of *Receiving* Variable-cards and *two* classes of *Supplying* Variable-cards. It is plain however that only the one or the other of these two latter classes can be used by any one Variable for *one* operation; never *both* simultaneously; their respective functions being mutually incompatible.

It should be understood that the Variable-cards are not placed in *immediate contiguity* with the columns. Each card is connected by means of wires with the column it is intended to act upon.

Our diagram ought in reality to be placed side by side with M. Menabrea's corresponding table, so as to be compared with it, line for line belonging to each operation. But it was unfortunately inconvenient to print them in this desirable form. The diagram is, in the main, merely another manner of indicating the various relations denoted in M. Menabrea's table. Each mode has some advantages and some disadvantages. Combined, they form a complete and accurate method of registering every step and sequence in all calculations performed by the engine.

No notice has yet been taken of the *upper* indices which are added to the left of each V in the diagram; an addition which we have also taken the liberty of making to the V 's in M. Menabrea's tables of pages 114, 116, since it does not *alter* anything therein represented by him, but merely *adds* something to the previous indications of those tables. The *lower* indices are obviously indices of *locality* only, and are wholly independent of the operations performed or of the results obtained, their value continuing unchanged during the performance of calculations. The *upper* indices, however, are of a different nature. Their office is to indicate any *alteration* in the value which a Variable represents; and they are of course liable to changes during the processes of a calculation. Whenever a Variable has only zeros upon it, it is called 0V ; the moment a value appears on it (whether that value be placed there arbitrarily, or appears in the natural course of a calculation), it becomes 1V . If this value gives place to another value, the Variable becomes 2V , and so forth. Whenever a *value* again gives place to *zero*, the Variable again becomes 0V , even if it have been nV , the moment before. If a *value* then again be substituted, the Variable becomes ^{n+1}V (as it would have done if it had not passed through the intermediate 0V); &c. &c. Just before any calculation is commenced, and after the data have been given, and everything adjusted

and prepared for setting the mechanism in action, the upper indices of the Variables for data are all unity, and those for the Working and Result-variables are all zero. in this state the diagram represents them⁴¹.

There are several advantages in having a set of indices of this nature; but these advantages are perhaps hardly of a kind to be immediately perceived, unless by a mind somewhat accustomed to trace the successive steps by means of which the engine accomplishes its purposes. We have only space to mention in a general way, that the whole notation of the tables is made more consistent by these indices, for they are able to mark a *difference* in certain cases, where there would otherwise be an apparent *identity* confusing in its tendency. In such a case as $V_n = V_p + V_n$ there is more clearness and more consistency with the usual laws of algebraical notation, in being able to write ${}^{m+1}V_n = {}^qV_p + V_n$. It is also obvious that the indices furnish a powerful means of tracing back the derivation of any result; and of registering various circumstances concerning that *series of successive substitutions*, of which every *result* is in fact merely the final consequence; circumstances that may in certain cases involve relations which it is important to observe, either for purely analytical reasons, or for practically adapting the workings of the engine to their occurrence. The series of substitutions which lead to the equations of the diagram are as follow: –

$$\begin{array}{llll}
 (1.)(2.) & & (3.)(4.) & \\
 {}^1V_{16}^* = \frac{{}^1V_{14}}{{}^1V_{12}} & = \frac{{}^1V_{10} - {}^1V_{11}}{{}^1V_6 - {}^1V_7} = \frac{{}^1V_0 {}^1V_5 - {}^1V_2 {}^1V_3}{{}^1V_0 {}^1V_4 - {}^1V_3 - {}^1V_1} & = \frac{d'm - dm'}{mn' - m'n} \\
 (1.)(2.) & & (3.)(4.) & \\
 {}^1V_{15} = \frac{{}^1V_{13}}{{}^1V_{12}} & = \frac{{}^1V_8 - {}^1V_9}{{}^1V_6 - {}^1V_7} = \frac{{}^1V_2 {}^1V_4 - {}^1V_5 {}^1V_1}{{}^1V_0 {}^1V_4 - {}^1V_3 {}^1V_1} & = \frac{dn' - d'n}{mn' - m'n}
 \end{array}$$

There are *three* successive substitutions for each of these equations. The formulæ(2.), (3.), and (4.) are implicitly contained in (1.), which latter we may consider as being in fact the *condensed* expression of any of the former. It will be observed that every succeeding substitution must contain *twice* as many *V*'s as its predecessor. So that if a problem require *n* substitutions, the successive series of numbers for the *V*'s in the whole of them will be 2, 4, 8, 16 . . . 2^n .

The substitutions in the preceding equations happen to be of little value towards illustrating the power and uses of the upper indices; for, owing to the nature of these particular equations the indices are all unity throughout. We wish we had space to enter more fully into the relations which these indices would in many cases enable us to trace.

M. Menabrea incloses the three centre columns of his table under the general title *Variable-cards*. The *V*'s however in reality all represent the actual *Variable-columns* of the engine,

⁴¹ We recommend the reader to trace the successive substitutions backwards from (1.) to (4.), in Mons. Menabrea's Table. This he will easily do by means of the upper and lower indices, and it is interesting to observe how each *V* successively ramifies (so to speak) into two other *V*'s in some other column of the Table; until at length the *V*'s of the original data are arrived at.

and not the cards that belong to them. Still the title is a very just one, since it is through the special action of certain Variable-cards (when *combined* with the more generalised agency of the Operation-cards) that every one of the particular relations he has indicated under that title is brought about.

Suppose we wish to ascertain how often any *one* quantity, or combination of quantities, is brought into use during a calculation. We easily ascertain *this*, from the inspection of any vertical column or columns of the diagram in which that quantity may appear. Thus, in the present case, we see that all the data, and all the intermediate results likewise, are used twice, excepting $(mn' - m'n)$, which is used three times.

The *order* in which it is possible to perform the operations for the present example, enables us to effect all the eleven operations of which it consists, with only *three Operation-cards*; because the problem is of such a nature that it admits of each *class* of operations being performed in a group together; all the multiplications one after another, all the subtractions one after another, &c. The operations are $\{6(\times), 3(-), 2(\div)\}$.

Since the very definition of an operation implies that there must be *two* numbers to act upon, there are of course *two Supplying* Variable cards necessarily brought into action for every operation, in order to furnish the two proper numbers. (See Note B.) Also, since every operation must produce a *result*, which must be placed *somewhere*, each operation entails the action of a *Receiving* Variable-card, to indicate the proper locality for the result. Therefore, at least three times as many Variable-cards as there are *operations* (not *Operation-cards*, for these, as we have just seen, are by no means always as numerous as the *operations*) are brought into use in every calculation. Indeed, under certain contingencies, a still larger proportion is requisite; such, for example, would probably be the case when the same result has to appear on more than one Variable simultaneously (which is not unfrequently a provision necessary for subsequent purposes in a calculation), and in some other cases which we shall not here specify. We see therefore that a great disproportion exists between the amount of *Variable* and of *Operation-cards* requisite for the working of even the simplest calculation.

All calculations do not admit, like this one, of the operations of the same nature being performed in groups together. Probably very few do so without exceptions occurring in one or other stage of the progress; and some would not admit it at all. The *order* in which the operations shall be performed in every particular case, is a very interesting and curious question, on which our space does not permit us fully to enter. In almost every computation a great *variety* of arrangements for the succession of the processes is possible, and various considerations must influence the selection amongst them for the purposes of a Calculating Engine. One essential object is to choose that arrangement which shall tend to reduce to a minimum the *time* necessary for completing the calculation.

It must be evident how multifarious and how mutually complicated are the considerations which the workings of such an engine involve. There are frequently several distinct *sets of effects* going on simultaneously; all in a manner independent of each other, and yet to a greater or less degree exercising a mutual influence. To adjust each to every other, and indeed even to perceive and trace them out with perfect correctness and success,

entails difficulties whose nature partakes to a certain extent of those involved in every question where *conditions* are very numerous and inter-complicated; such as for instance the estimation of the mutual relations amongst *statistical* phænomena, and of those involved in many other classes of facts.

A. A. L.

Note E. – Page 116.

This example has evidently been chosen on account of its brevity and simplicity, with a view merely to explain the *manner* in which the engine would proceed in the case of an *analytical calculation containing variables*, rather than to illustrate the *extent of its powers* to solve cases of a difficult and complex nature. The equations of page 111 are in fact a more complicated problem than the present one.

We have not subjoined any diagram of its development for this new example, as we did for the former one, because this is unnecessary after the full application already made of those diagrams to the illustration of M. Menabrea's excellent tables.

It may be remarked that a slight discrepancy exists between the formulæ

$$\frac{(a + Bx^1)}{(A + B \cos^1 x)}$$

given in the Memoir as the *data* for calculation, and the *results* of the calculation as developed in the last division of the table which accompanies it. To agree perfectly with this latter, the data should have been given as

$$\frac{(ax^0 + bx^1)}{(A \cos^0 x + B \cos^1 x)}.$$

The following is a more complicated example of the manner in which the engine would compute a trigonometrical function containing variables. To multiply

$$A + A_1 \cos \Theta + A_2 \cos 2\Theta + A_3 \cos 3\Theta + \dots$$

by

$$B + B_1 \cos \Theta.$$

Let the resulting products be represented under the general form

$$C_0 + C_1 \cos \Theta + C_2 \cos 2\Theta + C_3 \cos 3\Theta + \dots (1.)$$

This trigonometrical series is not only in itself very appropriate for illustrating the processes of the engine, but is likewise of much practical interest from its frequent use in astronomical computations. Before proceeding further with it, we shall point out that there are three very distinct classes of ways in which it may be desired to deduce numerical values from any analytical formula.

First. We may wish to find the collective, numerical value of the *whole formula*, without any reference to the quantities of which that formula is a function, or to the particular mode of their combination and distribution, of which the formula is the result and representative. Values of this kind are of a strictly arithmetical nature in the most limited sense of the term, and retain no trace whatever of the processes through which they have been deduced. In fact, any one such numerical value may have been attained from an *infinite variety* of data, or of problems. The values for x and y in the two equations (see Note D.), come under this class of numerical results.

Secondly. We may propose to compute the collective numerical value of *each term* of a formula, or of a series, and to keep these results separate. The engine must in such a case appropriate as many columns to *results* as there are terms to compute.

Thirdly. It may be desired to compute the numerical value of various *subdivisions of each term*, and to keep all these results separate. It may be required, for instance, to compute each coefficient separately from its variable, in which particular case the engine must appropriate *two* result-columns to *every term that contains both a variable and coefficient*.

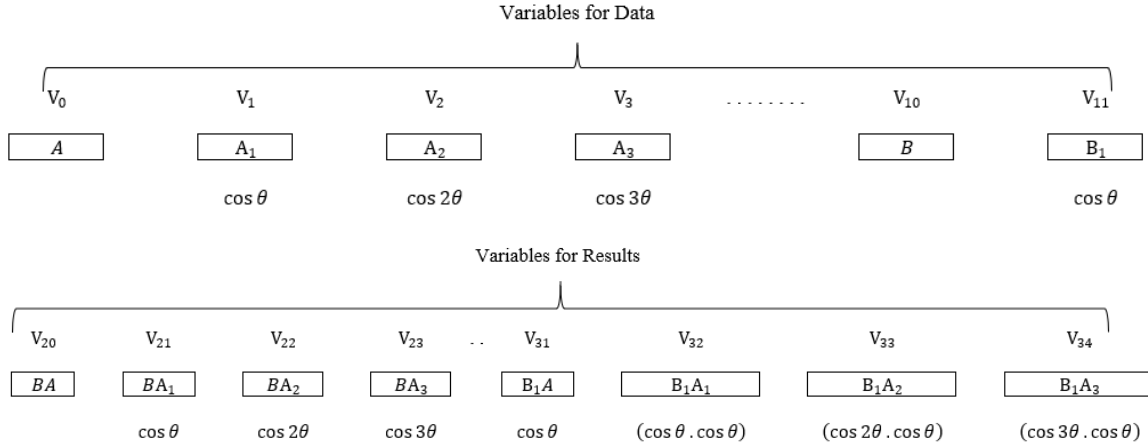
There are many ways in which it may be desired in special cases to distribute and keep separate the numerical values of different parts of an algebraical formula; and the power of effecting such distributions to any extent is essential to the *algebraical* character of the Analytical Engine. Many persons who are not conversant with mathematical studies, imagine that because the business of the engine is to give its results in *numerical notation*, the *nature of its processes* must consequently be *arithmetical* and *numerical*, rather than *algebraical* and *analytical*. This is an error. The engine can arrange and combine its numerical quantities exactly as if they were *letters* or any other *general* symbols; and in fact it might bring out its results in algebraical *notation*, were provisions made accordingly. It might develop three sets of results simultaneously, viz. *symbolic* results (as already alluded to in Notes A. and B.); *numerical* results (its chief and primary object); and *algebraical* results in *literal* notation. This latter however has not been deemed a necessary or desirable addition to its powers, partly because the necessary arrangements for effecting it would increase the complexity and extent of the mechanism to a degree that would not be commensurate with the advantages, where the main object of the invention is to translate into *numerical* language general formulæ of analysis already known to us, or whose laws of formation are known to us. But it would be a mistake to suppose that because its *results* are given in the *notation* of a more restricted science, its *processes* are therefore restricted to those of that science. The object of the engine is in fact to give the *utmost practical efficiency* to the resources of *numerical interpretations* of the higher science of analysis, while it uses the processes and combinations of this latter.

To return to the trigonometrical series. We shall only consider the four first terms of the

factor $(A + A_1 \cos \Theta + \&c.)$, since this will be sufficient to show the method. We propose to obtain separately the numerical value of *each coefficient* $C_0, C_1, \&c.$ of (1.). The direct multiplication of the two factors gives

$$BA + BA_1 \cos \Theta + BA_2 \cos 2\Theta + BA_3 \cos 3\Theta + \dots \\ + B_1 A \cos \Theta + B_1 A_1 \cos \Theta \cos \Theta + B_1 A_2 \cos 2\Theta \cos \Theta + B_1 A_3 \cos 3\Theta \cos \Theta. (2.)$$

a result which would stand thus on the engine: –



The variable belonging to each coefficient is written below it, as we have done in the diagram, by way of memorandum. The only further reduction which is at first apparently possible in the preceding result, would be the addition of V_{21} , to V_{31} , (in which case $B_1 A$ should be effaced from V_{31}). The whole operations from the beginning would then be –

<i>First series of Operations</i>	<i>Second Series of Operations</i>	<i>Third Series, which contains only one (final) operation</i>
${}^1V_{10} \times {}^1V_0 = {}^1V_{20}$	${}^1V_{11} \times {}^1V_0 = {}^1V_{31}$	${}^1V_{21} + {}^1V_{31} = {}^2V_{21}$, and
${}^1V_{10} \times {}^1V_1 = {}^1V_{21}$	${}^1V_{11} \times {}^1V_1 = {}^1V_{32}$	V_{31} becomes = 0.
${}^1V_{10} \times {}^1V_2 = {}^1V_{22}$	${}^1V_{11} \times {}^1V_2 = {}^1V_{33}$	
${}^1V_{10} \times {}^1V_3 = {}^1V_{23}$	${}^1V_{11} \times {}^1V_3 = {}^1V_{34}$	

We do not enter into the same detail of *every* step of the processes as in the examples of Notes D. and G., thinking it unnecessary and tedious to do so. The reader will remember the meaning and use of the upper and lower indices, &c., as before explained.

To proceed: we know that

$$\cos n\Theta \cos \Theta = \frac{1}{2} \cos \overline{n+1}\Theta + \frac{1}{2} \overline{n-1}\Theta \dots (3.)$$

Consequently, a slight examination of the second line of (2.) will show that by making the proper substitutions, (2.) will become

$$\begin{array}{c|c|c|c|c}
BA & +BA_1 \cdot \cos 2\Theta & +BA_2 \cdot \cos 2\Theta & +BA_3 \cdot \cos 3\Theta & \\
+\frac{1}{2}B_1A_1 & B_1A \cdot \cos \Theta & +\frac{1}{2}B_1A_1 \cdot \cos 2\Theta & +\frac{1}{2}B_1A_2 \cdot \cos 3\Theta & \\
C_0 & +\frac{1}{2}B_1A_2 \cdot \cos \Theta & +\frac{1}{2}B_1A_3 \cdot \cos 2\Theta & +\frac{1}{2}B_1A_3 \cdot \cos 4\Theta & \\
& C_1 & C_2 & C_3 & C_4
\end{array}$$

These coefficients should respectively appear on

$$V_{20} \quad V_{21} \quad V_{22} \quad V_{23} \quad V_{24}.$$

We shall perceive, if we inspect the particular arrangement of the results in (2.) on the Result-columns as represented in the diagram, that, in order to effect this transformation, each successive coefficient upon $V_{32}, V_{33}, \&c.$ (beginning with V_{32}), must through means of proper cards be divided by *two*⁴²; and that one of the halves thus obtained must be added to the coefficient on the Variable which precedes it by ten columns, and the other half to the coefficient on the Variable which precedes it by twelve columns; $V_{32}, V_{33}, \&c.$ themselves becoming zeros during the process.

This series of operations may be thus expressed: –

Fourth Series

$$\begin{cases}
{}^1V_{32} \div 2 + {}^1V_{22} = {}^2V_{22} = BA_2 + \frac{1}{2}B_1A_1 \\
{}^1V_{32} \div 2 + {}^1V_{20} = {}^2V_{20} = BA + \frac{1}{2}B_1A_1 \dots = C_0
\end{cases}$$

$$\begin{cases}
{}^1V_{33} \div 2 + {}^1V_{23} = {}^2V_{23} = BA_3 + \frac{1}{2}B_1A_2 \dots = C_3 \\
{}^1V_{33} \div 2 + {}^2V_{21} = {}^3V_{21} = BA_1 + B_1A + \frac{1}{2}B_1A_2 = C_1
\end{cases}$$

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$$\begin{cases}
{}^1V_{34} \div 2 + {}^0V_{24} = \frac{1}{2}B_1A_3 \dots = C_4 \\
{}^1V_{34} \div 2 + V_{22} = {}^3V_{22} = BA_2 + \frac{1}{2}B_1A_1 + \frac{1}{2}B_1A_3 = C_2.
\end{cases}$$

The calculation of the coefficients $C_0, C_1, \&c.$ of (1.), would now be completed, and they would stand ranged in order on $V_{20}, V_{21}, \&c.$ It will be remarked, that from the moment the fourth series of operations is ordered, the Variables $V_{31}, V_{32}, \&c.$ cease to be *Result-Variables*, and become mere *Working-Variables*.

The substitution made by the engine of the processes in the second side of (3.) for those in the first side, is an excellent illustration of the manner in which we may arbitrarily order it

⁴² This division would be managed by ordering the number two to appear on any separate new column which should be conveniently situated for the purpose, and then directing this column (which is in the strictest sense a *Working-Variable*) to divide itself successively with $V_{32}, V_{33}, \&c.$

⁴³It should be observed, that there were the rest of the factor $(A + A \cos \Theta + \&c.)$ taken into account, instead of *four* terms only, C_3 would have the additional term $\frac{1}{2}B_1A_4$; and C_4 the two additional terms, $BA_4, \frac{1}{2}B_1A_5$. This would have been the case had even *six* terms been multiplied.

to substitute any function, number, or process, at pleasure, for any other function, number or process, on the occurrence of a specified contingency.

We will now suppose that we desire to go a step further, and to obtain the numerical value of each *complete* term of the product (1.); that is, of each *coefficient and variable united*, which for the $(n + 1)$ th term would be $C_n \cos n\Theta$.

We must for this purpose place the variables themselves on another set of columns, V_{41} , V_{42} , &c., and then order their successive multiplication by V_{21} , V_{22} , &c., each for each. There would thus be a final series of operations as follows: –

Fifth and Final Series of Operations

$${}^2V_{20} \times {}^0V_{40} = {}^1V_{40}$$

$${}^3V_{21} \times {}^0V_{41} = {}^1V_{41}$$

$${}^3V_{22} \times {}^0V_{42} = {}^1V_{42}$$

$${}^2V_{23} \times {}^0V_{43} = {}^1V_{43}$$

$${}^1V_{24} \times {}^0V_{44} = {}^1V_{44}$$

(N.B. that V_{40} being intended to receive the coefficient on V_{20} , which has *no* variable, will only have $\cos 0\Theta (= 1)$ inscribed on it, preparatory to commencing the fifth series of operations.)

From the moment that the fifth and final series of operations is ordered, the Variables V_{20} , V_{21} , &c. then in their turn cease to be *Result-Variables* and become mere *Working-Variables*; V_{40} , V_{41} , &c. being now the recipients of the ultimate results.

We should observe, that if the variables $\cos \Theta$, $\cos 2\Theta$, $\cos 3\Theta$, &c. are furnished, they would be placed directly upon V_{41} , V_{42} , &c., like any other data. If not, a separate computation might be entered upon in a separate part of the engine, in order to calculate them, and place them on V_{41} , &c.

We have now explained how the engine might compute (1.) in the most direct manner, supposing we knew nothing about the *general* term of the resulting series. But the engine would in reality set to work very differently, whenever (as in this case) we *do* know the law for the general term.

The first two terms of (1.) are

$$(BA + \frac{1}{2}B_1A_1) + \overline{(BA_1 + B_1A + \frac{1}{2}B_1A_2 \cos \Theta)}(4.)$$

and the general term for all after these is

$$(BA_n + \frac{1}{2}B_1\overline{A_{n-1} + A_{n+2}}) \cos n\Theta(5.)$$

which is the coefficient of the $(n + 1)$ th term. The engine would calculate the two first terms by means of a separate set of suitable Operation-cards, and would then need another set for the third term; which last set of Operation-cards would calculate all the succeeding terms *ad infinitum*: merely requiring certain new Variable-cards for each term to direct the operations to act on the proper columns. The following would be the successive sets of operations for computing the coefficients of $(n + 2)$ terms: –

$$(\times, \times, \div, +), (\times, \times, \times, \div, +, +), n(\times, +, \times, \div, +)$$

Or we might represent them as follows, according to the numerical order of the operations:

–

$$(1, 2 \dots 4), (5, 6 \dots 10), n(11, 12 \dots 15).$$

The brackets, it should be understood, point out the relation in which the operations may be *grouped*, while the comma marks *succession*. The symbol $+$ might be used for this latter purpose, but this would be liable to produce confusion, as $+$ is also necessarily used to represent one class of the actual operations which are the subject of that succession. In accordance with this meaning attached to the comma, care must be taken when any one group of operations recurs more than once, as is represented above by $n(11 \dots 15)$, not to insert a comma after the number or letter prefixed to that group. $n, (11 \dots 15)$ would stand for *an operation n , followed by the group of operations $(11 \dots 15)$* ; instead of denoting *the number of groups which are to follow each other*.

Wherever a *general term* exists, there will be a *recurring group* of operations, as in the above example. Both for brevity and for distinctness, a *recurring group* is called a *cycle*. A *cycle* of operations, then, must be understood to signify any *set of operations* which is repeated *more than once*. It is equally a *cycle*, whether it be repeated *twice* only, or an indefinite number of times; for it is the fact of a *repetition occurring at all* that constitutes it such. In many cases of analysis there is a *recurring group* of one or more *cycles*; that is, a *cycle of a cycle*, or a *cycle of cycles*. For instance: suppose we wish to divide a series by a series,

$$(1.) \frac{a + bx + cx^2 + \dots}{a' + b'x + c'x^2 + \dots},$$

it being required that the result shall be developed, like the dividend and the divisor, in successive powers of x . A little consideration of (1.), and of the steps through which algebraical division is effected, will show that (if the denominator be supposed to consist of p terms) the first partial quotient will be completed by the following operations: –

$$(2.) \quad \{(\div), p(\times, -)\} \text{ or } \{(1), p(2, 3)\}$$

that the second partial quotient will be completed by an exactly similar set of operations, which acts on the remainder obtained by the first set, instead of on the original dividend.

The whole of the processes therefore that have been gone through, by the time the *second* partial quotient has been obtained, will be, –

$$(3.) \quad 2 \{(\div), p(\times, -)\} \text{ or } 2 \{(1), p(2, 3)\}$$

which is a cycle that includes a cycle, or a cycle of the second order. The operations for the *complete* division, supposing we propose to obtain n terms of the series constituting the quotient, will be, –

$$(4.) \quad n \{(\div), p(\times, -)\} \text{ or } n \{(1), p(2, 3)\}$$

It is of course to be remembered that the process of algebraical division in reality continues *ad infinitum*, except in the few exceptional cases which admit of an exact quotient being obtained. The number n in the formula (4.), is always that of the number of terms we propose to ourselves to obtain; and the n th partial quotient is the coefficient of the $(n-1)$ th power of x .

There are some cases which entail *cycles of cycles of cycles*, to an indefinite extent. Such cases are usually very complicated, and they are of extreme interest when considered with reference to the engine. The algebraical development in a series, of the n th function of any given function, is of this nature. Let it be proposed to obtain the n th function of

$$(5.) \quad \phi(a, b, c, \dots, x), x \text{ being the variable.}$$

We should premise that we suppose the reader to understand what is meant by an n th function. We suppose him likewise to comprehend distinctly the difference between developing *an n th function algebraically*, and merely *calculating an n th function arithmetically*. If he does not, the following will be by no means very intelligible; but we have not space to give any preliminary explanations. To proceed: the law, according to which the successive functions of (5.) are to be developed, must of course first be fixed on. This law may be of very various kinds. We may propose to obtain our results in successive *powers* of x , in which case the general form would be

$$C + C_1x + C_2x^2 + \&c.$$

or in successive powers of n itself, the index of the function we are ultimately to obtain, in which case the general form would be

$$C + C_1n + C_2n^2 + \&c.$$

and x would only enter in the coefficients. Again, other functions of x or of n instead of *powers*, might be selected. It might be in addition proposed, that the coefficients themselves should be arranged according to given functions of a certain quantity. Another mode would

be to make equations arbitrarily amongst the coefficients only, in which case the several functions, according to either of which it might be possible to develop the n th function of (5.), would have to be determined from the combined consideration of these equations and of (5.) itself.

The *algebraical* nature of the engine (so strongly insisted on in a previous part of this Note) would enable it to follow out any of these various modes indifferently; just as we recently showed that it can distribute and separate the numerical results of any one presented series of processes, in a perfectly arbitrary manner. Were it otherwise, the engine could merely *compute the arithmetical n th function*, a result which, like any other purely arithmetical results, would be simply a collective number, bearing no traces of the data or the processes which had led to it.

Secondly, the *law* of development for the n th function being selected, the next step would obviously be to develop (5.) itself, according to this law. This result would be the first function, and would be obtained by a determinate series of processes. These in most cases would include amongst them one or more *cycles* of operations.

The third step (which would consist of the various processes necessary for effecting the actual substitution of the series constituting the *first function*, for the *variable* itself) might proceed in either of two ways. It might make the substitution either wherever x occurs in the original (5.), or it might similarly make it wherever x occurs in the first function itself which is the equivalent of (5.). In some cases the former mode might be best, and in others the latter.

Whichever is adopted, it must be understood that the result is to appear arranged in a series following the law originally prescribed for the development of the n th function. This result constitutes the second function; with which we are to proceed exactly as we did with the first function, in order to obtain the third function; and so on, $n - 1$ times, to obtain the n th function. We easily perceive that since every successive function is arranged in a series *following the same law*, there would (after the *first function* is obtained) be a *cycle*, *of a cycle*, *of a cycle*, &c. of operations⁴⁴, one, two, three, up to $n - 1$ times, in order to get the n th function. We say, after the first function is obtained, because (for reasons on which we cannot here enter) the first function might in many cases be developed through a set of processes peculiar to itself, and not recurring for the remaining functions.

We have given but a very slight sketch, of the principal *general* steps which would be requisite for obtaining an n th function of such a formula as (5.). The question is so exceedingly complicated, that perhaps few persons can be expected to follow, to their own satisfaction, so brief and general a statement as we are here restricted to on this subject. Still it is a very important case as regards the engine, and suggests ideas peculiar to itself, which we should regret to pass wholly without allusion. Nothing could be more interesting than to

⁴⁴A cycle that includes n other cycles, successively *contained one within another*, is called a cycle of the $n + 1$ th order. A cycle may simply *include* many other cycles, and yet only be of the second order. If a series follows a certain law for a certain number of terms, and then another law for another number of terms, there will be a cycle of operations for every new law; but these cycles will not be *contained one within another*, – they merely *follow each other*. Therefore their number may be infinite without influencing the order of a cycle that includes a repetition of such a series.

follow out, in every detail, the solution by the engine of such a case as the above; but the time, space and labour this would necessitate, could only suit a very extensive work.

To return to the subject of *cycles* of operations: some of the notation of the integral calculus lends itself very aptly to express them: (2.) might be thus written : –

$$(6.) \quad (\div), \sum (+1)^p (\times, -), \text{ or } (1), \sum (+1)^p (2, 3),$$

where p stands for the variable; $(+1)^p$ for the function of the variable, that is, for ϕp ; and the limits are from 1 to p , or from 0 to $p-1$, each increment being equal to unity. Similarly, (4.) would be, –

$$(7.) \quad \sum (+1)^n \left\{ (\div), \sum (+1)^p (\times, -) \right\}$$

the limits of n being from 1 to n , or from 0 to $n-1$,

$$(8.) \quad \text{or } \sum (+1)^n \left\{ (1), \sum (+1)^p (2, 3) \right\}$$

Perhaps it may be thought that this notation is merely a circuitous way of expressing what was more simply and as effectually expressed before; and, in the above example, there may be some truth in this. But there is another description of cycles which *can* only effectually be expressed, in a condensed form, by the preceding notation. We shall call them *varying cycles*. They are of frequent occurrence, and include successive cycles of operations of the following nature: –

$$(9.) \quad p(1, 2, \dots m), \overline{p-1}(1, 2 \dots m), \overline{p-2}(1, 2 \dots m) \dots \overline{p-n}(1, 2 \dots m),$$

where each cycle contains the same group of operations, but in which the number of repetitions of the group varies according to a fixed rate, with every cycle. (9.) can be well expressed as follows: –

$$(10.) \quad \sum p(1, 2 \dots m), \text{ the limits of } p \text{ being from } p-n \text{ to } p.$$

Independent of the intrinsic advantages which we thus perceive to result in certain cases from this use of the notation of the integral calculus, there are likewise considerations which make it interesting, from the connections and relations involved in this new application. It has been observed in some of the former Notes, that the processes used in analysis form a logical system of much higher generality than the applications to number merely. Thus, when we read over any algebraical formula, considering it exclusively with reference to the processes of the engine, and putting aside for the moment its abstract signification as to the relations of quantity, the symbols $+$, $\times$, &c., in reality represent (as their immediate and proximate effect, when the formula is applied to the engine) that a certain prism which is a part of the mechanism (see Note C.), turns a new face, and thus presents a new card to act

on the bundles of levers of the engine; the new card being perforated with holes, which are arranged according to the peculiarities of the operation of addition, or of multiplication, &c. Again, the *numbers* in the preceding formula (8.), each of them really represents one of these very pieces of card that are hung over the prism.

Now in the use made in the formulæ(7.), (8.) and (10.), of the notation of the integral calculus, we have glimpses of a similar new application of the language of the *higher* mathematics. $\sum$, in reality, here indicates that when a certain number of cards have acted in succession, the prism over which they revolve must *rotate backwards*, so as to bring those cards into their former position; and the limits 1 to n , 1 to p , &c., regulate how often this backward rotation is to be repeated.

A. A. L.

Note F. – Page 119

There is in existence a beautiful woven portrait of Jacquard, in the fabrication of which 24,000 cards were required.

The power of *repeating* the cards, alluded to by M. Menabrea in page 112, and more fully explained in Note C., reduces to an immense extent the number of cards required. It is obvious that this mechanical improvement is especially applicable wherever *cycles* occur in the mathematical operations, and that, in preparing data for calculations by the engine, it is desirable to arrange the order and combination of the processes with a view to obtain them as much as possible *symmetrically* and in cycles, in order that the mechanical advantages of the *backing* system may be applied to the utmost. It is here interesting to observe the manner in which the value of an *analytical* resource is *met* and *enhanced* by an ingenious *mechanical* contrivance. We see in it an instance of one of those mutual *adjustments* between the purely mathematical and the mechanical departments, mentioned in Note A. as being a main and essential condition of success in the invention of a calculating engine. The nature of the resources afforded by such adjustments would be of two principal kinds. In some cases, a difficulty (perhaps in itself insurmountable) in the one department, would be overcome by facilities in the other; and sometimes (as in the present case) a strong point in the one, would be rendered still stronger and more available, by combination with a corresponding strong point in the other.

As a mere example of the degree to which the combined systems of cycles and of backing can diminish the *number* of cards requisite, we shall choose a case which places it in strong evidence, and which has likewise the advantage of being a perfectly different *kind* of problem from those that are mentioned in any of the other Notes. Suppose it be required to eliminate nine variables from ten simple equations of the form –

$$ax_0 + bx_1 + cx_2 + dx_3 + \dots = p \quad (1.)$$

$$a^1x_0 + b^1x_1 + c^1x_2 + d^1x_3 + \dots = p' \quad (2.)$$

$$\&c. \quad \&c. \quad \&c. \quad \&c.$$

We should explain, before proceeding, that it is not our object to consider this problem with reference to the actual arrangement of the data on the Variables of the engine, but simply as an abstract question of the *nature* and *number* of the *operations* required to be performed during its complete solution.

The first step would be the elimination of the first unknown quantity x_0 , between the two first equations. This would be obtained by the form –

$$(a^1a - aa^1)x_0 + (a^1b - ab^1)x_1 + (a^1c - ac^1)x_2 \quad ++ (a^1d0ad^1)x_3 + \dots = a^1p - ap^1.$$

for which the operations $10(\times, \times, -)$ would be needed. The second step would be the elimination of x_0 , between the second and third equations, for which the operations would be precisely the same. We should then have had altogether the following operations: –

$$10(\times, \times, -), 10(\times, \times, -) = 20(\times, \times, -).$$

Continuing in the same manner, the total number of operations for the complete elimination of x_0 between all the successive pairs of equations would be –

$$9 \cdot 10(\times, \times, -) = 90(\times, \times, -).$$

We should then be left with nine simple equations of nine variables from which to eliminate the next variable x_1 , for which the total of the process would be –

$$8 \cdot 9(\times, \times, -) = 72(\times, \times, -).$$

We should then be left with eight simple equations of eight variables from which to eliminate x_2 , for which the process would be –

$$7 \cdot 8(\times, \times, -) = 56(\times, \times, -).$$

and so on. The total operations for the elimination of all the variables would thus be –

$$9 \cdot 10 + 8 \cdot 9 + 7 \cdot 8 + 6 \cdot 7 + 5 \cdot 6 + 4 \cdot 5 + 3 \cdot 4 + 2 \cdot 3 + 1 \cdot 2 = 330.$$

So that *three* Operation-cards would perform the office of 330 such cards.

If we take n simple equations containing $n-l$ variables, n being a number unlimited in magnitude, the case becomes still more obvious, as the same three cards might then take the place of thousands or millions of cards.

We shall now draw further attention to the fact, already noticed, of its being by no means necessary that a formula proposed for solution should ever have been actually worked out, as a condition for enabling the engine to solve it. Provided we know the *series of operations* to be gone through, that is sufficient. In the foregoing instance this will be obvious enough on a slight consideration. And it is a circumstance which deserves particular notice, since herein may reside a latent value of such an engine almost incalculable in its possible ultimate results. We already know that there are functions whose numerical value it is of importance for the purposes both of abstract and of practical science to ascertain, but whose determination requires processes so lengthy and so complicated, that, although it is possible to arrive at them through great expenditure of time, labour and honey, it is yet on these accounts practically almost unattainable; and we can conceive there being some results which it may be *absolutely impossible* in practice to attain with any accuracy, and whose precise determination it may prove highly important for some of the future wants of science in its manifold, complicated and rapidly-developing fields of inquiry, to arrive at.

Without, however, stepping into the region of conjecture, we will mention a particular problem which occurs to us at this moment as being an apt illustration of the use to which such an engine may be turned for determining that which human brains find it difficult or impossible to work out unerringly. In the solution of the famous problem of the Three Bodies, there are, out of about 295 coefficients of lunar perturbations given by M. Clausen (*Astroee. Nachrichten*, No. 406) as the result of the calculations by Burg, of two by Damoiseau, and of one by Burckhardt, fourteen coefficients that differ in the nature of their algebraic sign; and out of the remainder there are only 101 (or about one-third) that agree precisely both in signs and in amount. These discordances, which are generally small in individual magnitude, may arise either from an erroneous determination of the abstract coefficients in the development of the problem, or from discrepancies in the data deduced from observation, or from both causes combined. The former is the most ordinary source of error in astronomical computations, and this the engine would entirely obviate.

We might even invent laws for series or formula in an arbitrary manner, and set the engine to work upon them, and thus deduce numerical results which we might not otherwise have thought of obtaining. But this would hardly perhaps in any instance be productive of any great practical utility, or calculated to rank higher than as a kind of philosophical amusement.

A. A. L.

Note G. – Page 120

It is desirable to guard against the possibility of exaggerated ideas that might arise as to the powers of the Analytical Engine. In considering any new subject, there is frequently a tendency, first, to *overrate* what we find to be already interesting or remarkable; and, secondly, by a sort of natural reaction, to *undervalue* the true state of the case, when we do discover that our notions have surpassed those that were really tenable.

The Analytical Engine has no pretensions whatever to *originate* anything. It can do whatever we *know how to order it* to perform. It can *follow* analysis; but it has no power

of *anticipating* any analytical relations or truths. Its province is to assist us in making *available* what we are already acquainted with. This it is calculated to effect primarily and chiefly of course, through its executive faculties; but it is likely to exert an *indirect* and reciprocal influence on science itself in another manner. For, in so distributing and combining the truths and the formulæ of analysis, that they may become most easily and rapidly amenable to the mechanical combinations of the engine, the relations and the nature of many subjects in that science are necessarily thrown into new lights, and more profoundly investigated. This is a decidedly indirect, and a somewhat *speculative*, consequence of such an invention. It is however pretty evident, on general principles, that in devising for mathematical truths a new form in which to record and throw themselves out for actual use, views are likely to be induced. which should again react on the more theoretical phase of the subject. There are in all extensions of human power, or additions to human knowledge, various *collateral* influences, besides the main and primary object attained.

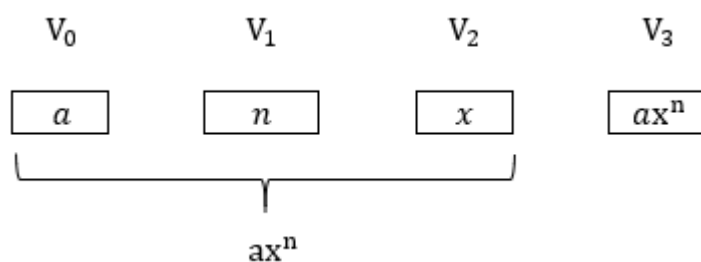
To return to the executive faculties of this engine: the question must arise in every mind, are they *really* even able to *follow* analysis in its whole extent? No reply, entirely satisfactory to all minds, can be given to this query, excepting the actual existence of the engine, and actual experience of its practical results. We will however sum up for each reader's consideration the chief elements with which the engine works:—

1. It performs the four operations of simple arithmetic upon any numbers whatever.
2. By means of certain artifices and arrangements (upon which we cannot enter within the restricted space which such a publication as the present may admit of), there is no limit either to the *magnitude* of the *numbers* used, or to the *number of quantities* (either variables or constants) that may be employed.
3. It can combine these numbers and these quantities either algebraically or arithmetically, in relations unlimited as to variety, extent, or complexity.
4. It uses algebraic *signs* according to their proper laws, and developes the logical consequences of these laws.
5. It can arbitrarily substitute any formula for any other; effacing the first from the columns on which it is represented, and making the second appear in its stead.
6. It can provide for singular values. Its power of doing this is referred to in M. Menabrea's memoir, page 117, where he mentions the passage of values through zero and infinity. The practicability of causing it arbitrarily to change its processes at any moment, on the occurrence of any specified contingency (of which its substitution of $\frac{1}{2} \cos n + 1\Theta + \frac{1}{2} \cos n - 1\Theta$ for $(\cos n\Theta \cos n\Theta)$ explained in Note E., is in some degree an illustration), at once secures this point.

The subject of integration and of differentiation demands some notice. The engine can effect these processes in either of two ways:—

First. We may order it, by means of the Operation and of the Variable-cards, to go through the various steps by which the required *limit* can be worked out for whatever function is under consideration.

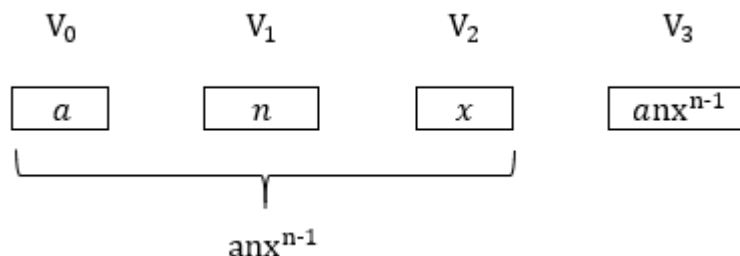
Secondly. It may (if we know the form of the limit for the function in question) effect the integration or differentiation by direct⁴⁵ substitution. We remarked in Note B., that any *set* of columns on which numbers are inscribed, represents merely a *general* function of the several quantities, until the special function have been impressed by means of the Operation and Variable-cards. Consequently, if instead of requiring the value of the function, we require that of its integral, or of its differential coefficient, we have merely to order whatever particular combination of the ingredient quantities may constitute that integral or that coefficient. In a ax^n , for instance, instead of the quantities



being ordered to appear on V , in the combination ax^n , they would be ordered to appear in that of

$$anx^{n-1}$$

They would then stand thus: –



Similarly, we might have $\frac{a}{n+1}x^{n+1}$, the integral of ax^n .

An interesting example for following out the process of the engine would be such a form as

⁴⁵ The engine cannot of course compute limits for perfectly *simple* and *uncompounded* functions, except in this manner. It is obvious that it has no power of representing or of manipulating with any but *finite* increments or decrements; and consequently that wherever the computation of limits (or of any other functions) depends upon the *direct* introduction of quantities which either increase or decrease *indefinitely*, we are absolutely beyond the sphere of its powers. Its nature and arrangements are remarkably adapted for taking into account all *finite* increments or decrements (however small or large), and for developing the true and logical modifications of form or value dependent upon differences of this nature. The engine may indeed be considered as including the whole Calculus of Finite Differences; many of whose theorems would be especially, and beautifully fitted for development by its processes, and would offer peculiarly interesting considerations. We may mention, as an example, the calculation of the Numbers of Bernoulli by means of the *Differences of Nothing*.

$$\int \frac{x^n dx}{\sqrt{a^2 - x^2}}$$

or any other cases of integration by successive reductions, where an integral which contains an operation repeated n times can be made to depend upon another which contains the same $n-1$ or $n-2$ times, and so on until by continued reduction we arrive at a certain *ultimate* form, whose value has then to be determined.

The methods in Arbogast's *Calcul des Dérivations* are peculiarly fitted for the notation and the processes of the engine. Likewise the whole of the Combinatorial Analysis, which consists first in a purely numerical calculation of indices, and secondly in the distribution and combination of the quantities according to laws prescribed by those indices.

We will terminate these Notes by following up in detail the steps through which the engine could compute the Numbers of Bernoulli, this being (in the form in which we shall deduce it) a rather complicated example of its powers. The simplest manner of computing these numbers would be from the direct expansion of

$$\frac{x}{e^x - 1} = \frac{1}{1 + \frac{x}{2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{2 \cdot 3 \cdot 4} + \&c.} \dots \quad (1.)$$

which is in fact a particular case of the development of

$$\frac{a + bx + cx^2 + \&c.}{a' + b'x + c'x^2 + \&c.}$$

mentioned in Note E. Or again, we might compute them from the well-known form

$$B_{2n-1} = 2 \cdot \frac{1 \cdot 2 \cdot 3 \dots 2n}{(2\pi)^{2n}} \left\{ 1 + \frac{1}{2^{2n}} + \frac{1}{3^{2n}} + \dots \right\} \dots \quad (2.)$$

or from the form

$$B_{2n-1} = \frac{\pm 2^n}{(2^{2n} - 1)2^{n-1}} \left\{ \begin{array}{l} \frac{1}{2} n^{2n-1} \\ -(n-1)^{2n-1} \left\{ 1 + \frac{1}{2} \cdot \frac{2n}{1} \right\} \\ +(n-2)^{2n-1} \left\{ 1 + \frac{2n}{1} + \frac{1}{2} \cdot \frac{2n \cdot (2n-1)}{1 \cdot 2} \right\} \\ -(n-3)^{2n-1} \left\{ 1 + \frac{2n}{1} + \frac{2n \cdot 2n-1}{1 \cdot 2} + \frac{1}{2} \cdot \frac{2n \cdot (2n-1) \cdot (2n-2)}{1 \cdot 2 \cdot 3} \right\} \\ + \dots \dots \end{array} \right\} \quad (3.)$$

or from many others. As however our object is not simplicity or facility of computation, but the illustration of the powers of the engine, we prefer selecting the formula below, marked (8.). This is derived in the following manner: –

If in the equation

$$\frac{x}{e^x - 1} = 1 - \frac{x}{2} + B_1 \frac{x^2}{2} + B_3 \frac{x^4}{2 \cdot 3 \cdot 4} + B_5 \frac{x^6}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + \dots \quad (4.)$$

(in which $B_1, B_3, \dots$, &c. are the Numbers of Bernoulli), we expand the denominator of the first side in powers of x , and then divide both numerator and denominator by x , we shall derive

$$1 = \left(1 - \frac{x}{2} + B_1 \frac{x^2}{2} + B_3 \frac{x^4}{2 \cdot 3 \cdot 4} + \dots\right) \left(1 + \frac{x}{2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{2 \cdot 3 \cdot 4} \dots\right) \quad (5.)$$

If this latter multiplication be actually performed, we shall have a series of the general form

$$1 + D_1 x + D_2 x^2 + D_3 x^3 + \dots \quad (6.)$$

in which we see, first, that all the coefficients of the powers of x are severally equal to zero; and secondly, that the general form for D_{2n} , the coefficient of the $2n + 1$ th *term* (that is of x^{2n} any *even* power of x), is the following: –

$$\left. \begin{aligned} & \frac{1}{2 \cdot 3 \dots 2n + 1} - \frac{1}{2} \cdot \frac{1}{2 \cdot 3 \dots 2n} + \frac{B_1}{2} \cdot \frac{1}{2 \cdot 3 \dots 2n - 1} + \frac{B_3}{2 \cdot 3 \cdot 4} \cdot \frac{1}{2 \cdot 3 \dots 2n - 3} + \\ & + \frac{B_5}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} \cdot \frac{1}{2 \cdot 3 \dots 2n - 5} + \dots + \frac{B_{2n-1}}{2 \cdot 3 \dots 2n} \cdot 1 = 0 \end{aligned} \right\} \quad (7.)$$

Multiplying every term by $(2 \cdot 3 \dots 2n)$, we have

$$\left. \begin{aligned} 0 = & -\frac{1}{2} \cdot \frac{2n-1}{2n+1} + B_1 \left(\frac{2n}{2}\right) + B_3 \left(\frac{2n \cdot 2n - 1 \cdot 2n - 2}{2 \cdot 3 \cdot 4}\right) + \\ & + B_5 \left(\frac{2n \cdot 2n - 1 \dots 2n - 4}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6}\right) + \dots + B_{2n-1} \end{aligned} \right\} \quad (8.)$$

which it may be convenient to write under the general form: –

$$0 = A_0 + A_1 B_1 + A_3 B_3 + A_5 B_5 + \dots + B_{2n-1}$$

A_1, A_3 , &c. being those functions of n which respectively belong to B_1, B_3 , &c.

We might have derived a form nearly similar to (8.), from D_{2n-1} the coefficient of any *odd* power of x in (6.); but the general form is a little different for the coefficients of the *odd* powers, and not quite so convenient.

On examining (7.) and (8.), we perceive that, when these formulæ are isolated from (6.) whence they are derived, and considered in themselves separately and independently, n may be any whole number whatever; although when (7.) occurs *as one of the D's* in (6.), it is obvious that n is then not arbitrary, but is always a certain function of the *distance of that D from the beginning*. If that distance be $= d$, then

$$2n + 1 = d, \quad \text{and} \quad n = \frac{d-1}{2} \text{ (for any even power of } x \text{)}$$

$$2n = d, \quad \text{and} \quad n = \frac{d}{2} \text{ (for any odd power of } x \text{)}.$$

It is with the *independent* formula (8.) that we have to do. Therefore it must be remembered that the conditions for the value of n are now modified, and that n is a perfectly *arbitrary* whole number. This circumstance, combined with the fact (which we may easily perceive) that whatever n is, every term of (8.) after the $(n+1)$ th is $= 0$, and that the $(n+1)$ th term itself is always $= B_{2n-1} \cdot \frac{1}{1} = B_{2n-1}$, enables us to find the value (either numerical or algebraical) of any n th Number of Bernoulli B_{2n-1} , *in terms of all the preceding ones*, if we but know the values of $B_1, B_3 \dots B_{2n-3}$. We append to this Note a Diagram and Table, containing the details of the computation for B_7 , (B_1, B_3, B_5 being supposed given).

On attentively considering (8.), we shall likewise perceive that we may derive from it the numerical value of *every* Number of Bernoulli in succession, from the very beginning, *ad infinitum*, by the following series of computations: —

1st Series. — Let $n = 1$, and calculate (8.) for this value of n . The result is B_1 .

2nd Series. — Let $n = 2$. Calculate (8.) for this value of n , substituting the value of B_1 , just obtained. The result is B_3 .

3rd Series — Let $n = 3$. Calculate (8.) for this value of n , substituting the values of B_1, B_3 , before obtained. The result is B_5 . And so on, to any extent.

The diagram⁴⁶ represents the columns of the engine when just prepared for computing B_{2n-1} , (in the case of $n = 4$); while the table beneath them presents a complete simultaneous view of all the successive changes which these columns then severally pass through in order to perform the computation. (The reader is referred to Note D, for explanations respecting the nature and notation of such tables.)

Six numerical *data* are in this case necessary for making the requisite combinations. These data are 1, 2, $n(= 4)$, B_1, B_3, B_5 . Were $n = 5$, the additional datum B_7 would be needed. Were $n = 6$, the datum B_9 would be needed; and so on. Thus the actual *number of data* needed will always be $n + 2$, for $n = n$; and out of these $n + 2$ data, $(\overline{n+2}-3)$ of

⁴⁶See the diagram at the end of these Notes.

them are successive Numbers of Bernoulli. The reason why the Bernoulli Numbers used as data, are nevertheless placed on *Result* columns in the diagram, is because they may properly be supposed to have been previously computed in succession by the *engine* itself; under which circumstances each *B* will appear as a *result*, previous to being used as a *datum* for computing the succeeding *B*. Here then is an instance (of the kind alluded to in Note D.) of the same Variables filling more than one office in turn. It is true that if we consider our computation of B_7 as a perfectly *isolated* calculation, we may conclude B_1, B_3, B_5 , to have been arbitrarily placed on the columns; and it would then perhaps be more consistent to put them on V_4, V_5, V_6 as data and not results. But we are not taking this view. On the contrary, we suppose the engine to be *in the course of computing* the Numbers to an indefinite extent, from the very beginning; and that we merely single out, by way of example, *one amongst* the successive but distinct series of computations it is thus performing. Where the *B*'s are fractional, it must be understood that they are computed and appear in the notation of *decimal* fractions. Indeed this is a circumstance that should be noticed with reference to all calculations. In any of the examples already given in the translation and in the Notes, some of the *data*, or of the temporary or permanent results, might be fractional, quite as probably as whole numbers. But the arrangements are so made, that the nature of the processes would be the same as for whole numbers.

In the above table and diagram we are not considering the *signs* of any of the *B*'s, merely their numerical magnitude. The engine would bring out the sign for each of them correctly of course, but we cannot enter on *every* additional detail of this kind, as we might wish to do. The circles for the signs are therefore intentionally left blank in the diagram.

Operation-cards 1, 2, 3, 4, 5, 6 prepare $-\frac{1}{2} \cdot \frac{2n-1}{2n+1}$. Thus, Card 1 multiplies *two* into n , and the three *Receiving* Variable-cards belonging respectively to V_4, V_5, V_6 , allow the result $2n$ to be placed on each of these latter columns (this being a case in which a triple receipt of the result is needed for subsequent purposes); we see that the upper indices of the two Variables used, during Operation 1, remain unaltered.

We shall not go through the details of every operation singly, since the table and diagram sufficiently indicate them; we shall merely notice some few peculiar cases.

By Operation 6, a *positive* quantity is turned into a *negative* quantity, by simply subtracting the quantity from a column which has only zero upon it. (The sign at the top of V_8 , would become $-$ during this process.)

Operation 7 will be unintelligible, unless it be remembered that if we were calculating for $n = 1$ instead of $n = 4$, Operation 6 would have completed the computation of B_1 itself; in which case the engine, instead of continuing its processes, would have to put B_1 on V_{21} ; and then either to stop altogether, or to begin Operations 1, 2...7 all over again for value of $n (= 2)$, in order to enter on the computation of B_3 ; (having however taken care, previous to this recommencement, to make the number on V_3 equal to *two*, by the addition of unity to the former $n = 1$ on that column). Now Operation 7 must either bring out a result equal to zero (if $n = 1$); or a result *greater* than zero, as in the present case; and the engine follows the one or the other of the two courses just explained, contingently on the one or the other result of Operation 7. In order fully to perceive the necessity of this *experimental*

operation, it is important to keep in mind what was pointed out, that we are not treating a perfectly isolated and independent computation, but one out of a series of antecedent and prospective computations.

Cards 8, 9, 10 produce $-\frac{1}{2} \cdot \frac{2n-1}{2n+1} + B_1 \frac{2n}{2}$. In Operation 9 we see an example of an upper index which again becomes a value after having passed from preceding values to zero. V_{11} has successively been ${}^0V_{11}$, ${}^1V_{11}$, ${}^2V_{11}$, ${}^0V_{11}$, ${}^3V_{11}$; and, from the nature of the office which V_{11} performs in the calculation, its index will continue to go through further changes of the same description, which, if examined, will be found to be regular and periodic.

Card 12 has to perform the same office as Card 7 did in the preceding section; since, if n had been $= 2$, the 11th operation would have completed the computation of B_3 .

Cards 13 to 20 make A_3 . Since A_{2n-1} always consists of $2n-1$ factors, A_3 , has three factors; and it will be seen that Cards 13, 14, 15, 16 make the second of these factors, and then multiply it with the first; and that 17, 18, 19, 20 make the third factor, and then multiply this with the product of the two former factors.

Card 23 has the office of Cards 11 and 7 to perform, since if n were $= 3$, the 21st and 22nd operations would complete the computation of B_5 . As our case is B_7 , the computation will continue one more stage; and we must now direct attention to the fact, that it, order to compute A_7 it is merely necessary precisely to repeat the group of Operations 13 to 20; and then, in order to complete the computation of B_7 , to repeat Operations 21, 22.

It will be perceived that every unit added to n in B_{2n-1} , entails an additional repetition of operations (13 ... 23) for the computation of B_{2n-1} . Not only are all the *operations* precisely the same however for every such repetition, but they require to be respectively supplied with numbers from the very *same pairs of columns*; with only the one exception of Operation 21, which will of course need B_5 (from V_{23}) instead of B_3 , (from V_{22}). This identity in the *columns* which supply the requisite numbers, must not be confounded with identity in the *values* those columns have upon them and give out to the mill. Most of th^{ose} values undergo alterations during a performance of the operations (13 ... 23), and consequently the columns present a new set of values for the *next* performance of (13 ... 23) to work on.

At the termination of the *repetition* of operations (13 ... 23) in computing B_7 , the alterations in the values on the Variables are, that

$$\begin{array}{ll} V_6 = 2n - 4 \text{ instead of} & 2n - 2. \\ V_7 = 6 \dots & 4. \\ V_{10} = 0 \dots & 1. \\ V_{13} = A_0 + A_1B_1 + A_3B_3 + A_5B_5 \text{ instead of} & A_0 + A_1B_1 + A_3B_3. \end{array}$$

In this state the only remaining processes are first: to transfer the value which is on V_{13} to V_{24} ; and secondly to reduce V_6 , V_7 , V_{13} to zero, and to add⁴⁷ *one* to V_3 , in order that the

⁴⁷ It is interesting to observe, that so complicated a case as this calculation of the Bernoullian Numbers, nevertheless, presents a remarkable simplicity in one respect; viz., that during the processes for the

engine may be ready to commence computing B_9 . Operations 24 and 25 accomplish these purposes. It may be thought anomalous that Operation 25 is represented as leaving the upper index of V_3 still = unity. But it must be remembered that these indices always begin anew for a separate calculation, and that Operation 25 places upon V_3 the *first* value for the new calculation.

It should be remarked, that when the group (13 ... 23) is *repeated*, changes occur in some of the *upper* indices during the course of the repetition: for example, 3V_6 would become 4V_6 and 5V_6 .

We thus see that when $n = 1$, nine Operation-cards are used; that when $n = 2$, fourteen Operation-cards are used; and that when $n > 2$, twenty-five Operation-cards are used; but that no *more* are needed, however great n may be; and not only this, but that these same twenty-five cards suffice for the successive computation of all the Numbers from B_1 , to B_{2n-1} inclusive. With respect to the number of *Variable*-cards, it will be remembered, from the explanations in previous Notes, that an average of three such cards to each *operation* (not however to each Operation-card) is the estimate. According to this the computation of B_1 will require twenty-seven Variable-cards; B_3 forty-two such cards; B_5 seventy-five ; and for every succeeding B after B_5 , there would be thirty-three additional Variable-cards (since each repetition of the group (13 ... 23) adds eleven to the number of operations required for computing the previous B). But we must now explain, that whenever there is a *cycle of operations*, and if these merely require to be supplied with numbers from the *same pairs of columns*, and likewise each operation to place its *result* on the *same* column for every repetition of the whole group, the process then admits of a *cycle of Variable-cards* for effecting its purposes. There is obviously much more symmetry and simplicity in the arrangements, when cases do admit of repeating the Variable as well as the Operation-cards. Our present example is of this nature. The only exception to a *perfect identity* in *all* the processes and columns used, for every repetition of Operations (13 ... 23) is, that Operation 21 always requires one of its factors from a new column, and Operation 24 always puts its result on a new column. But as these variations follow the same law at each repetition (Operation 21 always requiring its factor from a column *one* in advance of that which it used the previous time, and Operation 24 always putting its result on the column *one* in advance of that which received the previous result), they are easily provided for in arranging the recurring group (or cycle) of Variable-cards.

We may here remark that the average estimate of three Variable-cards coming into use to each operation, is not to be taken as an absolutely and literally correct amount for all cases and circumstances. Many special circumstances, either in the nature of a problem, or in the arrangements of the engine under certain contingencies, influence and modify this average to a greater or less extent. But it is a very safe and correct *general* rule to go upon. In the preceding case it will give us seventy-five Variable-cards as the total number which will be necessary for computing any B after B_3 . This is very nearly the precise amount really used, but we cannot here enter into the minutiae of the few particular circumstances which occur

computation of *millions* of these Numbers, no other arbitrary modification would be requisite in the arrangements, excepting the above simple and uniform provision for causing one of the data periodically to receive the finite increment unity.

in this example (as indeed at some one stage or other of probably most computations) to modify slightly this number.

It will be obvious that the very *same* seventy-five Variable-cards may be repeated for the computation of every succeeding Number, just on the same principle as admits of the repetition of the thirty-three Variable-cards of Operations (13 ... 23) in the computation of any *one* Number. Thus there will be a *cycle of a cycle* of Variable-cards.

If we now apply the notation for cycles, as explained in Note E, we may express the operations for computing the Numbers of Bernoulli in the following manner: –

$$\begin{array}{ll}
 (1 \dots 7), (24, 25) \dots \dots \dots \text{ gives } B_1 & = \text{1st number; } (n \text{ being } = 1). \\
 (1 \dots 7), (8 \dots 12), (24, 25) \dots \dots \dots B_3 & = \text{2nd } \dots \dots \dots ; (n \dots = 2). \\
 (1 \dots 7), (8 \dots 12), (13 \dots 23), \\
 (24, 25) \dots \dots \dots B_5 & = \text{3rd } \dots \dots \dots ; (n \dots = 3). \\
 \dots \dots \dots & \dots \dots \dots \\
 \dots \dots \dots & \dots \dots \dots \\
 (1 \dots 7), (8 \dots 12), \Sigma (+1)^{n-2} \\
 (13 \dots 23), (24, 25) \dots \dots \dots B_{2n-1} & = \text{nth } \dots \dots \dots ; (n \dots = n).
 \end{array}$$

Again,

$$(1 \dots 7), (24, 25), \sum_{\text{limits 1 to } n} (+1)^n \left\{ (1 \dots 7), (8 \dots 12), \sum_{\text{limits 0 to } (n-2)} (n+2)(13 \dots 23), (24, 25) \right\}$$

represents the total operations for computing every number in succession, from B_1 to B_{2n-1} inclusive.

In this formula we see a *varying cycle* of the *first* order, and an ordinary cycle of the *second* order. The latter cycle in this case includes in it the varying cycle.

On inspecting the ten Working-Variables of the diagram, it will be perceived, that although the *value* on any one of them (excepting V_4 and V_5) goes through a series of changes, the *office* which each performs is in this calculation *fixed* and *invariable*. Thus V_6 always prepares the *numerators* of the factors of any A ; V_7 the *denominators*. V_8 always receives the $(2n-3)$ th factor of A_{2n-1} , and V_9 the $(2n-1)$ th. V_{10} always decides which of two courses the succeeding processes are to follow, by feeling for the value of n through means of a subtraction; and so on; but we shall not enumerate further. It is desirable in all calculations, so to arrange the processes, that the *offices* performed by the Variables may be as uniform and fixed as possible.

Supposing that it was desired not only to tabulate B_1, B_3 &c., but A_0, A_1 , &c.; we have only then to appoint another series of Variables, V_{41}, V_{42} , &c., for receiving these latter results as they are successively produced upon V_{11} . Or again, we may, instead of this, or in addition to this second series of results, wish to tabulate the value of each successive total term of the series (8.), viz: $A_0, A_1 B_1, A_3 B_3$, &c. We have then merely to multiply each B with each corresponding A , as produced; and to place these successive products on Result-columns appointed for the purpose.

The formula (8.) is interesting in another point of view. It is one particular case of the general Integral of the following Equation of Mixed Differences: –

$$\frac{d^2}{dx^2} (z_{n+1}x^{2n+2}) = (2n+1)(2n+2)z^n x^{2n}$$

for certain special suppositions respecting z , x and n .

The *general* integral itself is of the form,

$$z_n = f(n) \cdot x + f_1(n) + f_2(n) \cdot x^{-1} + f_3(n) \cdot x^{-3} + \dots$$

and it is worthy to remark, that the engine might (in a manner more or less similar to the preceding) calculate the value of this formula upon most *other* hypotheses for the functions in the integral with as much, or (in many cases) with more ease than it can in formula (8.).

A. A. L.

Diagram for the computation by the Engine of the Numbers of Bernoulli. See Note G. (page 44 *et seq.*)

Number of Operation.	Nature of Operation.	Variables receiving results.	Indication of change in the value on any Variable.	Statement of Results.	Data.											Working Variables.													Result Variables.			
					W_1	W_2	W_3	W_4	W_5	W_6	W_7	W_8	W_9	W_{10}	W_{11}	W_{12}	W_{13}	W_{14}	W_{15}	W_{16}	W_{17}	W_{18}	W_{19}	W_{20}	W_{21}	W_{22}	W_{23}	W_{24}	W_{25}	W_{26}	W_{27}	
1	$\times$	$W_5 \times W_3$	$\{W_5 = W_3\}$	$= 2n$	1	2	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2	$-$	$W_6 - W_1$	$\{W_6 = W_1\}$	$= 2n - 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3	$+$	$W_6 + W_1$	$\{W_6 = W_1\}$	$= 2n + 1$	1	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4	$+$	$W_6 + W_4$	$\{W_6 = W_4\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
5	$+$	$W_6 + W_5$	$\{W_6 = W_5\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
6	$-$	$W_6 - W_1$	$\{W_6 = W_1\}$	$= 2n - 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7	$-$	$W_6 - W_1$	$\{W_6 = W_1\}$	$= 2n - 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
8	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
10	$\times$	$W_6 \times W_1$	$\{W_6 = W_1\}$	$= 2n$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
11	$+$	$W_6 + W_1$	$\{W_6 = W_1\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12	$-$	$W_6 - W_1$	$\{W_6 = W_1\}$	$= 2n - 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
13	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
14	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
15	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
16	$\times$	$W_6 \times W_1$	$\{W_6 = W_1\}$	$= 2n$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
17	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
18	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
19	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
20	$\times$	$W_6 \times W_1$	$\{W_6 = W_1\}$	$= 2n$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
21	$\times$	$W_6 \times W_1$	$\{W_6 = W_1\}$	$= 2n$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
22	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
23	$-$	$W_6 - W_1$	$\{W_6 = W_1\}$	$= 2n - 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
24	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
25	$+$	$W_6 + W_7$	$\{W_6 = W_7\}$	$= 2n + 1$	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

Here follows a repetition of Operations thirteen to twenty-three.

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Appendix A

List of abbreviations and titles

Below is a list of abbreviations and honorary titles which you might encounter while reading the materials for this game.

Baronet A rank in the United Kingdom's peerage system of hereditary titles. A baronet is a member of the lowest hereditary titled British order, still a commoner, but allowed to use the prefix "Sir." A 1st Baronet would be the first commoner in a family elevated to this status. A 2nd Baronet would be the eldest son of the 1st Baronet, and so on.

Countess A rank in the United Kingdom's peerage system of hereditary titles. A countess is the female spouse of an Earl, or, less frequently, a woman who holds an earldom in her own right. A countess would properly be referred to as "Lady" So, Ada Lovelace's proper title is Lady King-Noel, Countess of Lovelace.

DD Doctor of Divinity, an advanced or honorary academic degree in divinity. In the United Kingdom, D.D. is awarded to religious scholars for distinctive works beyond the doctorate (Ph.D.) level.

Duke A rank in the United Kingdom's peerage system of hereditary titles. A duke is a member of the highest titled British order, above marquesses and earls, and is the highest rank short of the monarch him or herself.

Earl A rank in the United Kingdom's peerage system of hereditary titles. An earl is a member of the middle titled British order, above viscounts and baronets, but below dukes and marquesses. These titles would often be associated with locations, e.g., Charles Jenkinson was elevated to the title of Earl of Liverpool by the king in 1796. The titles were passed down from father to first-born son, and so to continue the example, Charles' son Robert was known as Sir Robert Jenkinson, 2nd Earl of Liverpool. It would also be proper to refer to this person as Lord Jenkinson.

FRS Fellow of the Royal Society. A post-nominal title used by members of the Royal Society of London, the academy of sciences for the United Kingdom. Fellows are elected to the society by existing fellows. In the time period of our game, election to "The Royal" had as much to do with who you were as with what scientific achievements

you made. Also, during the time of our game, all fellows of the Royal Society really were in fact “fellows;” no women were allowed in the society until Kathleen Lonsdale and Marjory Stephenson were elected in 1945.

FRSA Fellow of the Royal Society of Arts. A post-nominal title awarded to individuals who have made outstanding achievements in the arts, manufacturing, and / or commerce. The Royal Society for the Encouragement of Arts, Manufactures and Commerce (RSA) is a London-based, British organization committed to finding practical solutions to social challenges.

GCB Knight Grand Cross of the Most Honourable Order of the Bath. This order was expanded in 1815 by Prince George Augustus Frederick, five years before he became King George IV in 1820. The restructured order was to honor those high-ranking officers who served in the Napoleonic Wars, and “Knight Grand Cross” is the highest rank in this order.

GCH Knight Grand Cross of the Royal Guelphic Order, a.k.a. the Hanoverian Guelphic Order, a chivalric order of the Kingdom of Hanover. Hanover was a Germanic kingdom created after the Napoleonic Wars, which was jointly ruled by its own royalty, Great Britain, and Ireland. The order was established in 1815 by Prince George Augustus Frederick, five years before he became King George IV in 1820. Since this was a “foreign” order, holders of this honorific title were not entitled to be called “sir.” “Knight Grand Cross” is the highest rank in this order.

GCMG Knight Grand Cross of the Most Distinguished Order of Saint Michael and Saint George. This order was created in 1818 by Prince George Augustus Frederick, two years before he became King George IV in 1820. The order was intended to honor those with high positions in the Mediterranean territories acquired during the Napoleonic Wars. “Knight Grand Cross” is the highest rank in this order.

GS Game Session. One of the class sessions for the game. In this game, there are seven or eight game sessions, GS 1, GS 2, . . . , GS 7, and three optional lab sessions.

IP Influence Point. IPs are used to exert influence on the PM when he makes a decision regarding Babbage’s machines. IPs can be earned in a variety of ways as the game progresses.

KCB Knight Commander of the Most Honorable Order of the Bath. Knight Commander is the mid-level rank (between Knight Grand Cross and Companion) of the chivalric Order of Bath. The Order of Bath is used to honor senior civil servants.

KG Knight of the Most Noble Order of the Garter, the highest order of chivalry in the United Kingdom. This title is given by the king or queen to subjects at his or her pleasure, but there are only allowed to be 24 members of the order (other than members of the royal family) at any one time.

KH Knight of the Royal Guelphic Order, a.k.a. the Hanoverian Guelphic Order, a chivalric order of the Kingdom of Hanover. Hanover was a Germanic kingdom created after the Napoleonic Wars, which was jointly ruled by its own royalty, Great Britain, and

Ireland. The order was established in 1815 by Prince George Augustus Frederick, five years before he became King George IV in 1820. Since this was a “foreign” order, holders of this honorific title were not entitled to be called “sir.” “Knight” is the lowest rank in this order.

KT Knight Companion of the Order of the Thistle. A chivalric order of Scotland, established in 1687 by King James II. This order is limited to the sovereign and sixteen Knights and Ladies, who are selected for the order by the sovereign.

MOS Men of Science. This abbreviation stands for the collection of men (and some women) game characters who are engineers, mathematicians and / or men of science (a.k.a. natural philosophers). During our game, it is inappropriate to call men of science “scientists,” as that term was not coined until 1833 by William Whewell, and did not come into full use until late in the nineteenth century.

MP Member of Parliament. A person elected by the general public to represent their interests in the House of Commons, one of the two houses of the British Parliament. At the time of our game, all MPs are male; the first woman elected to the House of Commons was Constance Markievicz, who won her seat in 1918. However, Markievicz was a member of Sinn Féin, the Irish political party advocating for Irish independence, and so she did not take her seat. A year later, Nancy Astor was elected and took her seat in 1919.

PC Member of the Privy Council. The Privy Council is an advisory group serving the monarch (first King William IV and then Queen Victoria in our game). PC is a post-nominal title used by current or past members of the Privy Council.

PM Prime Minister. The head of the UK government and is ultimately responsible for the policy and decisions of the government. The PM oversees the operation of the Civil Service and government agencies, appoints members of the government, and is the principal government figure in the House of Commons.

PRS President of the Royal Society. A post-nominal title used by fellows of the Royal Society of London who have served as president of the society. The governing council and president of the Royal Society are elected by the fellows of the society.

Appendix B

Currency conversion

Table B.1 gives a rough equivalent between British Pounds Sterling in the year 1819 with pounds and U.S. dollars in 2016.

Table B.1: Currency conversion

1819 Pounds	2016 Pounds	2016 U.S. Dollars
£10.00	£804.14	\$1,093.11
£20.00	£1,608.28	\$2,186.23
£30.00	£2,412.42	\$3,279.34
£40.00	£3,216.56	\$4,372.45
£50.00	£4,020.70	\$5,465.57
£60.00	£4,824.84	\$6,558.68
£70.00	£5,628.98	\$7,651.79
£80.00	£6,433.12	\$8,744.91
£90.00	£7,237.26	\$9,838.02
£100.00	£8,041.40	\$10,931.13
£200.00	£16,082.80	\$21,862.27
£300.00	£24,124.20	\$32,793.40
£400.00	£32,165.60	\$43,724.53
£500.00	£40,207.00	\$54,655.67
£600.00	£48,248.40	\$65,586.80
£700.00	£56,289.80	\$76,517.93
£800.00	£64,331.20	\$87,449.07
£900.00	£72,372.60	\$98,380.20
£1,000.00	£80,414.00	\$109,311.33
£2,000.00	£160,828.00	\$218,622.67
£3,000.00	£241,242.00	\$327,934.00
£4,000.00	£321,656.00	\$437,245.34
£5,000.00	£402,070.00	\$546,556.67
£6,000.00	£482,484.00	\$655,868.00
£7,000.00	£562,898.00	\$765,179.34
£8,000.00	£643,312.00	\$874,490.67
£9,000.00	£723,726.00	\$983,802.00
£10,000.00	£804,140.00	\$1,093,113.34
£20,000.00	£1,608,280.00	\$2,186,226.68
£30,000.00	£2,412,420.00	\$3,279,340.01
£40,000.00	£3,216,560.00	\$4,372,453.35
£50,000.00	£4,020,700.00	\$5,465,566.69
£60,000.00	£4,824,840.00	\$6,558,680.03
£70,000.00	£5,628,980.00	\$7,651,793.37
£80,000.00	£6,433,120.00	\$8,744,906.70
£90,000.00	£7,237,260.00	\$9,838,020.04
£100,000.00	£8,041,400.00	\$10,931,133.38

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